

SECOND-LARGEST AND MINIMUM DISTANCE MULTIPLICITIES IN PLANAR POINT SETS: AN UPPER BOUND OF $4/3$, AND RELATED RESULTS ON ERDŐS PROBLEM #132

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ABSTRACT. For a finite set $X \subset \mathbb{R}^2$ let Δ_2 and δ be the second-largest and the smallest distance determined by X , and let $\mu(d)$ be the number of pairs of X at distance d . Clemen, Dumitrescu and Liu asked for $L = \limsup_{n \rightarrow \infty} \max_{|X|=n} \min\{\mu(\Delta_2), \mu(\delta)\}/n$; the known bounds were $9/7 \leq L \leq 3/2$, the upper bound coming from Vesztergombi's inequality $\mu(\Delta_2) \leq 3n/2$. We prove that $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{4}{3}n + C_0$ for an absolute constant C_0 , so $L \leq 4/3$. The new ingredient is an averaged estimate: if R is the set of points of X with no partner at distance Δ_2 , then $\sum_{y \in R} (\deg y + m_y) \leq 8|R| + O(1)$, where $\deg y$ is the number of points at distance δ from y and m_y the number of those outside R . Its key step is that every locally extremal configuration pins $(\Delta - \Delta_2)/\delta$ to one of two short intervals. When $\tau(X) \leq \sqrt{3}/2 - 0.02$ or $\tau(X) \geq \sqrt{3}/2 + 0.03$, where $\tau(X)$ is an explicit scale-invariant measure of the excess of Δ over Δ_2 in units of δ , a refinement of the method gives $\frac{21}{16}n + C_1$, and when $\tau(X) > 1$ a global Euler-type estimate gives $\frac{9}{7}n + O(n^{2/3})$, which matches the known lower bound $L \geq \frac{9}{7}$. A weaker bound, $\frac{15}{11}n + C_0$, obtained by pointwise estimates, is formalised in Lean 4 with Mathlib using only the standard axioms; the bounds $\frac{4}{3}$, $\frac{21}{16}$ and $\frac{9}{7}$ are not formalised.

We also give three results on Erdős Problem #132, which asks whether some distance other than the diameter must occur at most n times. First, if no vertex of the convex hull has four points of X at distance Δ_2 (for instance, if no four points of X are concyclic), then $\mu(\Delta_2) \leq \frac{3}{2}|L_1| + |L_2|$, where L_1, L_2 are the first two convex layers; so, under this hypothesis, the answer is yes whenever at least half as many points lie at depth ≥ 3 as on the hull. This rests on a layer bound whose inputs from Vesztergombi's structure theory we prove directly. Second, for n points in convex position there are at least three distances occurring between 1 and n times when n is even and at least 6, and when $6 \leq n \leq 18$. The cases $n = 11, 13, 15, 17$ are computer-assisted: they reduce to the unsatisfiability of finite ordinal problems, certified by DRAT (and for $n \leq 13$ LRAT) proof checkers, and for $n = 7, 11, 13$ also by a Lean 4 formalisation. Third, if all but w of the n points in convex position lie on one circle and $n + w \geq 91$, $n \geq 3w$, then at least $(n - 3w + 1)/4$ distances are rare; the proof combines Kneser's theorem with a bound of Beukers and Smyth on torsion points of plane curves.

AI DISCLOSURE

The author used an AI assistant (Claude, Anthropic) for exploration, computation, formalisation and drafting. The checks performed on the results are described in [Appendix A](#).

1. INTRODUCTION

Let $X \subset \mathbb{R}^2$ be a set of n points. Write $\Delta = \Delta(X)$ for its diameter, Δ_2 for its second-largest distance and δ for its minimum distance, and write $\mu(d) = \mu_X(d)$ for the number of unordered pairs of points of X at distance d . Hopf and Pannwitz [16] proved that $\mu(\Delta) \leq n$.

Erdős Problem #132. Erdős and Pach asked whether there must be a *second* distance, besides the diameter, that occurs at least once and at most n times, and whether the number of such distances must tend to infinity (perhaps being at least $n^{1-o(1)}$). This is Problem #132 on the Erdős problems website [6], where it is recorded as open and where Erdős’s offer of \$100 for “any nontrivial result” is recorded (reference [Er97e] there). The statement fails for $n = 4$ (two equilateral triangles glued along an edge) and holds for $n = 5, 6$ by Erdős and Fishburn [11]. We call a distance d *rare* if $1 \leq \mu(d) \leq n$.

Clemen, Dumitrescu and Liu [7] proved that the answer to the first question is yes for sets in convex position (their Theorem 1.2), and that $\mu(\Delta_2) \leq n$ whenever

$$(1) \quad \min\left\{\frac{3}{2}(|L_1| + |L_2|), \frac{4}{3}|L_1| + 2|L_2|, 2|L_1| + |L_2|\right\} \leq n,$$

where L_1, L_2 are the first two convex layers (their Theorem 1.3). They showed that neither Δ_2 nor δ is always rare: they constructed sets with $\min\{\mu(\Delta_2), \mu(\delta)\} \geq 9n/8 - o(n)$ (their Proposition 1.5). They asked (their Problem 1.6) for the value of

$$L := \limsup_{n \rightarrow \infty} \max_{|X|=n} \frac{\min\{\mu(\Delta_2), \mu(\delta)\}}{n}.$$

Vesztergombi [27] proved that $\mu(\Delta_2) \leq \frac{3}{2}n$ for every n -point planar set, which gives $L \leq 3/2$. On the Erdős problems forum, the user ienjoymath [17] gave a construction showing $L \geq 9/7$ and conjectured $L = 9/7$. As far as we know, $9/7 \leq L \leq 3/2$ were the best bounds before this paper.

Results. Our main result is an upper bound strictly below $3/2$. For a set with at least two distances put

$$\tau(X) := \frac{\Delta^2 - \Delta_2^2 - \delta^2}{2\Delta_2\delta},$$

a scale-invariant measure of how much the diameter exceeds Δ_2 (in the scaling $\delta = 1$ used in the proof, $\tau = t - (1 - t^2)/(2\Delta_2)$ with $t = \Delta - \Delta_2$; see Section 2).

Theorem 1. *Let $X \subset \mathbb{R}^2$ be a set of n points.*

- (a) $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{4}{3}n + C_0$, where C_0 is an absolute constant.
- (b) If $\tau(X) \leq \sqrt{3}/2 - 1/50$ or $\tau(X) \geq \sqrt{3}/2 + 3/100$, then $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{21}{16}n + C_1$, where C_1 is an absolute constant.
- (c) If $\tau(X) > 1$, then $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{9}{7}n + C_2 n^{2/3}$, where C_2 is an absolute constant.

Consequently $L \leq 4/3 \approx 1.3333$.

(Here X is assumed to determine at least two distances; otherwise $n \leq 3$.)

Part (c) matches the lower bound $L \geq \frac{9}{7}$: sets with $\tau(X) > 1$ cannot do better than the construction of [17]. (That construction has $\tau(X) \rightarrow 0$, so part (c) does not show that $\frac{9}{7}$ is attained with $\tau(X) > 1$.) For $\tau(X) > 1$ part (c) supersedes part (b). The constants are explicit: the proof gives $C_0 = 3N_0 + 2 \cdot 10^8$ with $N_0 = 1.51 \cdot 10^9$ (Section 9), $C_1 = 2 \cdot 10^{32}$ (Sections 10 and 11) and $C_2 = 3 \cdot 10^4$ (Section 12). We have made no attempt to optimise them. Part (a) improves the bound $\frac{15}{11}n + C_0$ of the previous version of this paper, which we keep as Theorem 8.1: its proof (Sections 2 to 8) is the backbone of the proof of part (a), and it is the part of this paper that is formalised.

Status of the proofs. Theorem 8.1(a), the bound $\frac{15}{11}n + C_0$, is formalised in Lean 4 with Mathlib: the formal proof uses only the standard axioms (no `sorry`, no `native_decide`) and is checked by the Lean kernel and replayed by the independent checker `leanchecker`; see Section 14.2. Theorem 1 is *not* formalised; a formalisation of part (a) is in progress. Part (a) adds to Theorem 8.1 one new estimate, Theorem 9.1 in Section 9; its proof has been through two

independent referee passes, which found no mathematical error. Part (b) is a finer version of the same argument. For $\tau(X) \leq \sqrt{3}/2 - 1/50$ and for $\tau(X) > 1$ (Section 10), two independent referee passes found it correct, but found gaps in an earlier write-up (one case of a load count, a step in a packing lemma, and several numerical slips); the text below incorporates the repairs. For $\sqrt{3}/2 + 3/100 \leq \tau(X) \leq 1$ (Section 11) part (b) needs two further inputs, a sharper count of rungs and the average weight 7; each of them has been through *one* independent referee pass, whose corrections are incorporated, and the text of Section 11 itself has not been refereed. Part (c) rests on a global estimate, Theorem 12.1 in Section 12, for the δ -edges inside K . Two independent referee passes found gaps in an earlier write-up of it (all at the level of constants or exposition); the complete corrected proof given here was checked in a third independent pass, which found no errors or gaps. Parts (b) and (c) are included because they show how far the method goes; the reader who wants only $L \leq \frac{4}{3}$ can skip Sections 10 to 12. A weaker bound, $\frac{54}{37}n + C_0$ with $C_0 = 3N_0 + 10^8$, obtained by a different argument in Regions II and III and with the trivial bound $\mu(\delta) \leq e(S) + 6|R|$ in place of Section 6, was formalised first; that argument is written out in Appendix B.

Theorem 13.6 and Corollary 13.7 give a new class of sets for which the answer to the first question of Problem #132 is yes. Let $m_3 := n - |L_1| - |L_2|$ be the number of points of depth at least 3. Condition (1) contains the case $2|L_1| + |L_2| \leq n$, i.e. $|L_1| \leq m_3$. We show that $|L_1| \leq 2m_3$ suffices under a genericity hypothesis that holds, for instance, when no four points of X are concyclic. The inputs we need from Vesztergombi's structure theory [28] are proved here from scratch (Proposition 13.2 and Theorem 13.5).

Section 14 treats the second question of Problem #132 in convex position. Theorem 14.1 states that a set of n points in convex position has at least three rare distances for every n with $6 \leq n \leq 18$. Some of these cases follow from the literature, some follow from a parity-sharp version of the counting argument of [7], and the cases $n = 11, 13, 15, 17$ are new and computer-assisted. Theorem 14.14 shows that a set in convex position with all but w points on one circle has at least $(n - 3w + 1)/4$ rare distances once $n + w \geq 91$ (and $n \geq 3w$); so the second question of Problem #132 has a positive answer for such sets when $w = o(n)$. In Section 15 we list open problems.

Idea of the proof of Theorem 1. Let S be the set of points that have a partner at distance Δ_2 , let $Q \subseteq S$ be the set of points with exactly one such partner, and put $s = |S|$, $p = |Q|$, $r = n - s$. Vesztergombi's bound applied to $S \setminus Q$ gives $\mu(\Delta_2) \leq \frac{3}{2}s - \frac{1}{2}p$. On the δ side we use the exact identity $\mu(\delta) = e(S) + \sum_{y \in R} \frac{1}{2}(\deg y + m_y)$, where $e(S)$ is the number of δ -pairs inside S , $R := X \setminus S$, $\deg y$ is the number of δ -neighbours of y and m_y the number of those in S . We prove two estimates: $e(S) \leq s + p + O(1)$ (Section 5), and $\sum_{y \in R} (\deg y + m_y) \leq 8r + O(1)$ (Theorem 9.1). The combination with weights $\frac{2}{3}$ and $\frac{1}{3}$ gives $\frac{4}{3}n + O(1)$ (Lemma 3.7). The pointwise bound $\deg y + m_y \leq 10$ for all but $O(1)$ points of R (Theorem 6.1), with weights $\frac{8}{11}$ and $\frac{3}{11}$, gives the formalised bound $\frac{15}{11}$.

For the first estimate: almost all points of S lie on the boundary of a convex body, either the “ball polygon” $K = \bigcap_{w \in X} B(w, \Delta_2)$ or the convex hull, and each has a well-defined inward direction. All but $O(1)$ of the δ -edges inside S join points with almost equal inward directions. Such edges are either tangential edges, of which there are at most s , or *rungs* joining the two boundaries, whose geometry is fixed by an exact identity (Lemma 2.2(c)). A corner lemma shows that the inner end of a rung has only one Δ_2 -partner, up to $O(1)$ exceptions, unless a second partner sits in one exactly determined reflected position; in that case the rung forces a missing tangential edge elsewhere. So every rung is paid for, either on the Δ_2 side (through p) or by a missing tangential edge.

For the second estimate: a point of R that is not a diametral endpoint lies in the interior of K , while its neighbours in S do not; where ∂K is nearly flat on a long stretch, this confines those neighbours to an open arc of less than 240° , so there are at most 4 of them, and the points y near which ∂K is not flat are $O(1)$ in number. This gives $\deg y + m_y \leq 10$. To get the average down to 8 we classify the points with $\deg y + m_y \geq 9$ (*heavy* points): they come in four local shapes. Each shape contains a neighbour k on ∂K and a diametral endpoint z whose heights above the boundary differ by about $\frac{\sqrt{3}}{2}$ or about $\frac{1}{2}$, and the exact identity of Lemma 2.2(b) then forces $t = \Delta - \Delta_2$ to be within 0.05 of that height difference (Lemma 9.8). So the shape of the heavy points is determined by t , and each heavy point has one or two neighbours deeper inside K that are light, not shared with another heavy point, and able to absorb its excess. The regime where the diameter is much larger than Δ_2 is handled separately by an incidence argument.

Part (b) repeats the classification with much smaller tolerances: in Region I and for $\tau > 1$ only one heavy shape survives (a “cap”, which needs $t = \frac{1}{2}$), all other points have weight at most 8, and a discharging along unit edges, with one forwarding step, brings every weight down to 7. In Region II the same discharging works, and in addition the rung count is sharpened to $e(S) \leq s + \frac{1}{2}p + O(1)$, which the weights $\frac{5}{8}, \frac{3}{8}$ can absorb. Part (c) is different in kind. When $\tau > 1$ no δ -edge joins D to K , and we show that the δ -graph inside K has at most $3|V| - 2|S \setminus D| + O(n^{2/3})$ edges, by Euler’s formula applied after removing a shell at depth about $n^{1/3}$; the points of S on ∂K are paid for by the boundary walk of the outer face together with a second closed walk, either around the hole that contains the deep core of K or, if there is no such hole, the return path forced by a winding-number argument.

Notation. $B(c, r)$ and $C(c, r)$ are the closed disc and the circle with centre c and radius r . For unit vectors u, u' , $\angle(u, u') \in [0, \pi]$ is the angle between them; recall $|u - u'| \leq \angle(u, u')$. Angles are in radians unless marked with a degree sign.

2. SET-UP FOR THE PROOF OF THEOREM 1

We scale so that $\delta = 1$. Throughout Sections 2 to 10 and 12 and Appendix B, X is an n -point set with $n \geq 4$ and at least two distinct distances, so $\Delta > \Delta_2 \geq \delta = 1$. We put

$$t := \Delta - \Delta_2 > 0, \quad \tau := t - \frac{1 - t^2}{2\Delta_2}, \quad \beta := \frac{1}{100}.$$

Then $\tau = (\Delta^2 - \Delta_2^2 - 1)/(2\Delta_2)$, which is the quantity $\tau(X)$ of the introduction, and $\tau - t = (t^2 - 1)/(2\Delta_2)$. Hence $\tau \leq t$ when $t \leq 1$, and $\tau > 1$ if and only if $t > 1$.^[C2] A crucial feature of X is that it has *no distance in the open interval* (Δ_2, Δ) .

We use the following objects.

- D is the set of endpoints of diametral pairs. Each point of D is a vertex of $P := \text{conv } X$.
- S is the set of points of X that have a partner at distance Δ_2 ; $R := X \setminus S$ and $r := |R|$, $s := |S|$.
- $Q \subseteq S$ is the set of points with exactly one partner at distance Δ_2 (Δ_2 -degree 1), and $p := |Q|$.
- $K := \bigcap_{w \in X} B(w, \Delta_2)$, a compact convex set (a “ball polygon”).
- $e(Y)$ is the number of δ -pairs inside $Y \subseteq X$. For $y \in X$, $\deg y$ is the number of δ -neighbours of y , and for $y \in R$, m_y is the number of δ -neighbours of y that lie in S .

The *degenerate regime* is $\Delta > 1.94 \Delta_2$ (Section 7); the *nondegenerate regime* is $\Delta \leq 1.94 \Delta_2$ (Sections 3 to 6, 9, 10 and 12).

Lemma 2.1 (ball polygon). (a) $X \setminus D \subseteq K$.
(b) $S \setminus D \subseteq \partial K$.

(c) $\text{dist}(z, K) \geq t$ for every $z \in D$.

Proof. (a) A point $y \notin D$ has all its distances at most Δ_2 , since there are none in (Δ_2, Δ) . (b) If $v \in S \setminus D$ and $|vw| = \Delta_2$, then $v \in K \cap C(w, \Delta_2) \subseteq \partial K$. (c) If $z \in D$ and $|zx| = \Delta$, then $K \subseteq B(x, \Delta_2)$, while $|zx| = \Delta_2 + t$. $\square$

Direction field. For each $p \in S$ we fix a unit vector u_p .

- If $p \in S \setminus D$, choose a Δ_2 -partner w_p and put $u_p := (w_p - p)/\Delta_2$. Since $K \subseteq B(w_p, \Delta_2)$, u_p is an inward normal of K at p .
- If $p \in S \cap D$, choose a diametral partner x_p and put $u_p := (x_p - p)/\Delta$. This is an inward normal of P at p .

Lemma 2.2 (exact identities). *Let $y \in X$, $y \neq p$.*

- (a) *If $p \in S \cap D$, then $(y - p) \cdot u_p \geq |y - p|^2/(2\Delta) > 0$.*
- (b) *If $p \in S \setminus D$, then either $(y - p) \cdot u_p \geq |y - p|^2/(2\Delta_2)$ (we say y is normal for p), or $|y - w_p| = \Delta$, so that $y \in D$ (we say y is an exception for p).*
- (c) *In the exception case with $|y - p| = 1$ we have exactly $(y - p) \cdot u_p = -\tau$. Hence $y - p = -\tau u_p \pm \sigma u_p^\perp$ with $\sigma := \sqrt{1 - \tau^2}$. In particular exceptions at distance 1 exist only when $\tau \leq 1$.*

Proof. (a) $|y - x_p|^2 = |y - p|^2 - 2\Delta(y - p) \cdot u_p + \Delta^2 \leq \Delta^2$. (b) $|y - w_p|^2 = |y - p|^2 - 2\Delta_2(y - p) \cdot u_p + \Delta_2^2$, and $|y - w_p| \in [0, \Delta_2] \cup \{\Delta\}$. (c) With $|y - p| = 1$ and $|y - w_p| = \Delta = \Delta_2 + t$ the same identity gives $(y - p) \cdot u_p = (1 + \Delta_2^2 - \Delta^2)/(2\Delta_2) = (1 - 2\Delta_2 t - t^2)/(2\Delta_2) = -\tau$.^[C2] $\square$

The proof uses only $|p - x_p| = \Delta$, resp. $|p - w_p| = \Delta_2$. So (a) holds for every $p \in D$ (also $p \in R \cap D$) and every diametral partner x of p in place of x_p , and (b), (c) hold for every Δ_2 -partner w of p in place of w_p , with u_p replaced by $(w - p)/\Delta_2$. We use these variants below.

Lemma 2.3 (no common normal). *If $p \neq q$ both lie in $S \setminus D$, then $u_p \neq u_q$.*

Proof. Neither point lies in D , so each is normal for the other: $(q - p) \cdot u_p > 0$ and $(p - q) \cdot u_q > 0$. These cannot both hold if $u_p = u_q$. $\square$

Good edges, T-edges and rungs. A δ -edge pq inside S is *good* if $\angle(u_p, u_q) \leq \beta$, and *bad* otherwise. For a good edge put $e := q - p$, $a := e \cdot u_p$ and $b := -e \cdot u_q$. Then $|a + b| = |e \cdot (u_p - u_q)| \leq \beta$.

Lemma 2.4 (dichotomy). *Assume $\tau > 2\beta$. Each good edge is of exactly one of the following two kinds.*

- T-edge: $a, b \in (0, \beta)$.
- Rung: *one end $v \in S \setminus D$ sees the other end $z \in S \cap D$ as an exception. Then $(z - v) \cdot u_v = -\tau$ and $(v - z) \cdot u_z \in [\tau - \beta, \tau + \beta]$.*

Proof. By Lemma 2.2, from each end the other end is either normal (then the corresponding one of a, b is positive) or an exception (then it equals $-\tau$). An exception target lies in D , and points of D see everything as normal, so both ends cannot be exceptions. If neither is, then $a, b > 0$ and $a + b \leq \beta$. If one is, say $a = -\tau$, then $b \in [\tau - \beta, \tau + \beta]$, and $b > 0$ shows that b is not an exception value. $\square$

We say a T-neighbour q of p is on the *right* (resp. *left*) of p if $(q - p) \cdot u_p^\perp > 0$ (resp. < 0), where $u^\perp$ denotes u rotated by -90° .

(T1) Each p has at most one T-neighbour on each side. Indeed, two T-directions on one side both make an angle in $(0, \arcsin \beta)$ with $\pm u_p^\perp$, so they are less than 60° apart, whereas two points at distance 1 from p and at distance ≥ 1 from each other are at least 60° apart as seen from p .

(T2) If q is the right T-neighbour of p then p is the left T-neighbour of q , since $\angle(u_p, u_q) \leq \beta$.

Hence the T-edges form vertex-disjoint directed paths and cycles, and every cycle has length at least 3 (a vertex cannot be both the left and the right T-neighbour of another). We say each point of S has a *left slot* and a *right slot*; a T-edge from p to its right T-neighbour q uses the right slot of p and the left slot of q . So the number e_T of T-edges is half the number of used slots, and $e_T \leq s$. T-edges with both ends in $S \setminus D$ or both in $S \cap D$ are *pure*; the others are *mixed*. Note that (T1) and (T2) apply to mixed T-edges too.

Turning. Let γ be a closed arc of the boundary of a compact convex set with nonempty interior. Its *turning* $\text{turn}(\gamma)$ is the length of the set $N(\gamma) \subseteq S^1$, the union of the normal cones at *all* points of γ , including the full normal cones at its endpoints.

Lemma 2.5 (multiplicity). *Let $\gamma_1, \dots, \gamma_m$ be closed boundary arcs of one convex body, and suppose every boundary point lies in at most M of them. Then $\sum_i \text{turn}(\gamma_i) \leq 2\pi M$.*

Proof. For all but countably many directions $\nu \in S^1$ the set of boundary points with outer (or inner) normal ν is a single point y_ν , and $\nu \in N(\gamma_i)$ if and only if $y_\nu \in \gamma_i$. Integrate over ν . $\square$

Points of $S \setminus D$ lie on ∂K and points of $S \cap D$ lie on ∂P ; in each case distinct points are at Euclidean, hence arc-length, distance at least 1. So a boundary arc of length ℓ contains at most $\ell + 1$ of them, and a boundary of perimeter ℓ contains at most ℓ of them.

3. REDUCTIONS

Lemma 3.1 ((VS)). *For every $S' \subseteq S$, the number of Δ_2 -pairs inside S' is at most $\frac{3}{2}|S'|$. In particular $\mu(\Delta_2) \leq \frac{3}{2}s$.*

Proof. Every Δ_2 -pair of X lies in S , and every distance within S' is a distance of X . If S' contains no Δ_2 -pair there is nothing to prove. If S' contains a diametral pair of X , then $\text{diam } S' = \Delta$ and, because X has no distances in (Δ_2, Δ) , Δ_2 is the second-largest distance of S' ; Vesztergombi's theorem [27] gives at most $\frac{3}{2}|S'|$ such pairs. Otherwise every distance in S' is at most Δ_2 , so $\Delta_2 = \text{diam } S'$ and Hopf–Pannwitz [16] gives at most $|S'|$. $\square$

We use Vesztergombi's theorem in exactly the form of its main theorem: for any set of m points in the plane, the second-largest distance occurs at most $\frac{3}{2}m$ times [27, Theorem]. There is no threshold on m and no additive constant.

Lemma 3.2 (weighted Δ_2 bound). $\mu(\Delta_2) \leq p + \frac{3}{2}(s - p) = \frac{3}{2}s - \frac{1}{2}p$.

Proof. At most p Δ_2 -pairs touch Q , since each point of Q lies in exactly one Δ_2 -pair. Every other Δ_2 -pair lies inside $S \setminus Q$, and there are at most $\frac{3}{2}(s - p)$ of them by Lemma 3.1. $\square$

Lemma 3.3 (exact identity). $\mu(\delta) = e(S) + \sum_{y \in R} \frac{1}{2}(\deg y + m_y)$. Moreover $\deg y \leq 6$ and $m_y \leq \deg y$, so $\deg y + m_y \leq 12$ for every $y \in R$.

Proof. The δ -pairs inside S number $e(S)$. The pairs with one end in S and one in R number $\sum_{y \in R} m_y$. The pairs inside R are counted twice in $\sum_{y \in R} (\deg y - m_y)$. Adding, $\mu(\delta) = e(S) + \sum_y m_y + \frac{1}{2} \sum_y (\deg y - m_y)$, which is the identity. Each point has at most 6 points at distance δ , since these are pairwise at least δ apart. $\square$

Corollary 3.4 ((E)). $\mu(\delta) \leq e(S) + 6r$. $\square$

Lemma 3.5 (linear combination). *Suppose $e(S) \leq s + p + C$, and that $\deg y + m_y \leq 10$ for all but at most C' points $y \in R$. Then*

$$\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{15}{11}n + C + C'.$$

If instead $\deg y + m_y \leq 8$ for all but at most C' points $y \in R$, then $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{4}{3}n + C + C'$.

Proof. For an exceptional y we still have $\deg y + m_y \leq 12$ (Lemma 3.3). So in the first case $\mu(\delta) \leq s + p + 5r + C + C'$, and with Lemma 3.2

$$\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{8}{11}(\frac{3}{2}s - \frac{1}{2}p) + \frac{3}{11}(s + p + 5r + C + C') = \frac{15}{11}(s + r) - \frac{1}{11}p + \frac{3}{11}(C + C').$$

In the second case $\mu(\delta) \leq s + p + 4r + C + 2C'$, and the weights $\frac{2}{3}, \frac{1}{3}$ give $\frac{4}{3}(s + r) + \frac{1}{3}(C + 2C')$. $\square$

Remark 3.6. If $\deg y + m_y \leq 2c$ for all but $O(1)$ points of R , the best weights give $\max\{\frac{4}{3}, \frac{3c}{2c+1}\}n + O(1)$ under the hypothesis $e(S) \leq s + p + O(1)$, and $\frac{3c}{2c+1}n + O(1)$ under the stronger hypothesis $e(S) \leq s + O(1)$; the weight on $\mu(\Delta_2)$ must be at least $\frac{2}{3}$ to cancel the p -term.^[LP] For $2c = 12, 10, 8, 6$ these values are $\frac{18}{13}, \frac{15}{11}, \frac{4}{3}, \frac{4}{3}$, respectively $\frac{18}{13}, \frac{15}{11}, \frac{4}{3}, \frac{9}{7}$. In particular the trivial bound (E) together with $e(S) \leq s + p + O(1)$ already gives $\frac{18}{13}n + O(1)$, with weights $\frac{10}{13}, \frac{3}{13}$. Since $\mu(\delta)$ depends only on $\sum_{y \in R}(\deg y + m_y)$, all these values remain valid if the pointwise hypothesis is replaced by $\sum_{y \in R}(\deg y + m_y) \leq 2cr + O(1)$.

Lemma 3.7 (averaged linear combination). (a) *If $e(S) \leq s + p + C$ and $\sum_{y \in R}(\deg y + m_y) \leq 8r + C''$, then $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{4}{3}n + \frac{1}{3}(C + \frac{1}{2}C'')$.*

(b) *If $e(S) \leq s + C$ and $\sum_{y \in R}(\deg y + m_y) \leq 7r + C''$, then $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{21}{16}n + \frac{3}{8}(C + \frac{1}{2}C'')$.*

Proof. (a) By Lemma 3.3, $\mu(\delta) \leq s + p + 4r + C + \frac{1}{2}C''$, and with Lemma 3.2, $\frac{2}{3}(\frac{3}{2}s - \frac{1}{2}p) + \frac{1}{3}(s + p + 4r + C + \frac{1}{2}C'') = \frac{4}{3}(s + r) + \frac{1}{3}(C + \frac{1}{2}C'')$.^[N3] (b) Now $\mu(\delta) \leq s + \frac{7}{2}r + C + \frac{1}{2}C''$, and with Lemma 3.1, $\frac{5}{8} \cdot \frac{3}{2}s + \frac{3}{8}(s + \frac{7}{2}r + C + \frac{1}{2}C'') = \frac{21}{16}(s + r) + \frac{3}{8}(C + \frac{1}{2}C'')$. $\square$

The value $\frac{4}{3}$ in (a) is the limit of this method in the presence of the p -term (Remark 3.6), and (b) is the value $\frac{3c}{2c+1}$ of Remark 3.6 with $2c = 7$.

So in the nondegenerate regime it suffices, for Theorem 1(a), to show $e(S) \leq s + p + O(1)$ in every τ -regime (Section 5) and $\sum_{y \in R}(\deg y + m_y) \leq 8r + O(1)$ (Section 9). The pointwise bounds $\deg y + m_y \leq 10$ (resp. ≤ 8 in Region II) for all but $O(1)$ points $y \in R$ (Section 6) give the weaker Theorem 8.1 through Lemma 3.5.

Small cases. Suppose we are in the nondegenerate regime and $\Delta_2 < 10^4$. Then $\Delta < 1.94 \cdot 10^4$, so X lies in a disc of radius Δ about any of its points, and since the discs of radius $\frac{1}{2}$ about points of X are disjoint, $n \leq (2\Delta + 1)^2 < N_0 := 1.51 \cdot 10^9$.^[C11] Since $\mu(\delta) \leq 3n$, Theorem 1 holds for such X once $C_0 \geq 3N_0$. From now on, in the nondegenerate regime, $\Delta_2 \geq 10^4$; in particular $2/\Delta_2 \leq \beta/2$.^[C3]

4. BAD EDGES ARE $O(1)$ IN THE NONDEGENERATE REGIME

4.1. A lens lemma.

Lemma 4.1 (lens). *Let $\Lambda = B(c_1, \rho) \cap B(c_2, \rho)$ with $0 < |c_1 c_2| = d < 2\rho$, and put $\cos \varphi_0 := d/(2\rho)$. Assume*

$$(2) \quad \rho \geq \frac{2}{\cos(\varphi_0/2) - \cos \varphi_0}.$$

Let p lie on the arc of $\partial\Lambda$ belonging to $C(c_1, \rho)$, let q lie on the arc belonging to $C(c_2, \rho)$, and suppose $|pq| \leq 1$. Then there is a corner c of Λ such that the boundary arcs from p to c and from c to q each have length at most $\rho_0 := 1/\sin(\varphi_0/2)$.

Proof. Take $c_1 = (-d/2, 0)$, $c_2 = (d/2, 0)$, $p = c_1 + \rho(\cos \varphi, \sin \varphi)$ and $q = c_2 + \rho(-\cos \psi, \sin \psi)$ with $|\varphi|, |\psi| \leq \varphi_0$. *Step 1.* $p_x - q_x = \rho(\cos \varphi - \cos \varphi_0) + \rho(\cos \psi - \cos \varphi_0)$ with both terms ≥ 0 ; since $|p_x - q_x| \leq 1$, each term is at most 1. By (2), $\cos \varphi < \cos(\varphi_0/2)$, i.e. $|\varphi| > \varphi_0/2$; likewise $|\psi| > \varphi_0/2$. *Step 2.* By the mean value theorem, $\rho(\varphi_0 - |\varphi|) = \rho(\cos \varphi - \cos \varphi_0)/\sin \xi \leq 1/\sin \xi$ for some $\xi \in (\varphi_0/2, \varphi_0) \subseteq (0, \pi/2)$, so $\rho(\varphi_0 - |\varphi|) \leq \rho_0$; likewise for ψ . *Step 3.* If φ, ψ had opposite signs, then $|p_y - q_y| \geq 2\rho \sin(\varphi_0/2)$. Now (2) gives $\rho \geq 2/(1 - \cos \varphi_0) = 1/\sin^2(\varphi_0/2)$, so $2\rho \sin(\varphi_0/2) \geq 2/\sin(\varphi_0/2) \geq 2 > 1$, a contradiction. *Step 4.* So the corner $c = (0, \text{sign}(\varphi)\rho \sin \varphi_0)$ works. $\square$

Lemma 4.2 (lens-corner arc length). *In the situation of Lemma 4.1, suppose moreover that $p, q \in \partial K'$ for a compact convex $K' \subseteq \Lambda$, and $|pq| = 1$. Let H be the closed half-plane bounded by the line pq that contains c , and let $\gamma := \partial K' \cap H$ be the boundary arc of K' from p to q on c 's side. Then $\text{length}(\gamma) \leq 2\rho_0$.*

Proof. $\partial\Lambda$ contains no segment, so the line pq meets it only in p and q , and $\partial\Lambda \cap H$ is exactly the arc $p \rightarrow c \rightarrow q$, of length at most $2\rho_0$. Since $K' \cap H \subseteq \Lambda \cap H$ are convex, monotonicity of perimeter gives $\text{length}(\gamma) + 1 \leq 2\rho_0 + 1$. (If $K' \cap H$ is the segment $[p, q]$, then γ is that segment, of length $1 \leq 2\rho_0$.) $\square$

Constants.^[C7] For *KK lenses* ($\rho = \Delta_2$, $d \leq \Delta \leq 1.94\Delta_2$) we have $\cos \varphi_0 \leq 0.97$, so $\rho_0^2 = 2/(1 - \cos \varphi_0) \leq 66.7 < 8.2^2$. The function $c \mapsto \cos(\varphi_0/2) - c = \sqrt{(1+c)/2} - c$ is decreasing on $[0, 1)$, so (2) holds as soon as $\rho \geq 89.01$, in particular for $\rho = \Delta_2 \geq 10^4$. For *DD lenses* ($\rho = \Delta$, $d \leq \Delta$) we have $\cos \varphi_0 \leq \frac{1}{2}$, $\rho_0 \leq 2$, and (2) needs only $\rho \geq 5.5$.

4.2. The count. We split bad edges pq by type: *KK* (both ends in $S \setminus D$), *DD* (both in $S \cap D$) and *cross* (one end in each).

If the perimeter of K is below 1202, then $|S \setminus D| < 1202$ and at most $6 \cdot 1202$ edges touch $S \setminus D$; similarly for P and $S \cap D$. We may therefore assume both perimeters are at least 1202 when counting the bad edges touching the corresponding row; these small cases contribute a constant below $1.5 \cdot 10^4$ in total.

KK edges. Let $p, q \in S \setminus D$ form a bad edge. If $w_p = w_q$, then $\angle(u_p, u_q) \leq 2 \arcsin(1/(2\Delta_2)) < \beta$,^[C3] so the edge would be good; hence $w_p \neq w_q$. Then $K \subseteq \Lambda := B(w_p, \Delta_2) \cap B(w_q, \Delta_2)$, p lies on the $C(w_p, \Delta_2)$ -arc and q on the $C(w_q, \Delta_2)$ -arc of $\partial\Lambda$, and $d = |w_p w_q| \leq \Delta \leq 1.94\Delta_2$. Lemmas 4.1 and 4.2 give an arc $\gamma_{pq} \subseteq \partial K$ from p to q of length at most $2\rho_0$. The union of the normal cones along the closed arc γ_{pq} is connected and contains u_p and u_q , so $\text{turn}(\gamma_{pq}) \geq \angle(u_p, u_q) > \beta$. If $y \in \gamma_{pq}$ then p lies within arc distance $2\rho_0$ of y ; there are at most $4\rho_0 + 1$ such p , each of degree at most 6, so every point lies on at most $M = 6(4\rho_0 + 1)$ of the arcs. By Lemma 2.5,

$$\#\text{KK-bad} \leq 12\pi(4\rho_0 + 1)/\beta < 1.28 \cdot 10^5. \text{[C9]}$$

DD edges. The same argument, on ∂P with the lens $B(x_p, \Delta) \cap B(x_q, \Delta)$ and $\rho_0 \leq 2$, gives $\#\text{DD-bad} \leq 12\pi \cdot 9/\beta < 3.4 \cdot 10^4$. (If $x_p = x_q$ the angle is less than $2/\Delta < \beta$ and the edge is good.)

Cross edges. Let zv be a bad edge with $z \in S \cap D$ and $v \in S \setminus D$. Since $z \in X$, $|zw_v| \in [0, \Delta_2] \cup \{\Delta\}$.

(c1) $|zw_v| = \Delta$. By Lemma 2.2(a), $(w_v - z)/\Delta$ is a second inward normal of P at z . Its angle to u_v is the angle at w_v of the triangle zw_v ; that angle is opposite the side of length 1, hence acute, and by the law of sines it is at most $\arcsin(1/\Delta) < 2/\Delta_2 \leq \beta/2$.^[C3] As $\angle(u_z, u_v) > \beta$, the normal cone of P at z has angle greater than $\beta/2$. Normal cones at distinct vertices have disjoint interiors, so fewer than $4\pi/\beta$ points z qualify, each of degree at most 6: $\#(c1) < 24\pi/\beta < 7600$.

(c2) $|zw_v| \leq \Delta_2$. Let g be the unit vector from v towards x_z , and $\alpha := \angle(u_v, g)$.

- (i) *Depth.* $v \notin D$ gives $|vx_z| \leq \Delta_2$, while $|zx_z| = \Delta$ and $|zv| = 1$ give $|vx_z| \geq \Delta - 1$. So $v \in B(x_z, \Delta_2)$ at depth $d_0 := \Delta_2 - |vx_z| \in [0, 1 - t]$; in particular this case needs $t \leq 1$. Also $K \subseteq B(x_z, \Delta_2)$.
- (ii) α is not small. $\angle(u_z, g)$ is the angle at x_z of the triangle zvx_z , which is at most $\arcsin(1/\Delta) < \beta/2$; hence $\alpha \geq \angle(u_v, u_z) - \angle(u_z, g) > \beta/2$.
- (iii) α is not close to π . Since $w_v, x_z \in X$, $|w_v x_z| \leq \Delta \leq 1.94\Delta_2$. The law of cosines in the triangle $vw_v x_z$ (sides Δ_2 and $\Delta_2 - d_0$ at v , enclosing the angle α) gives $\cos \alpha \geq (1 + \varrho^2 - 1.94^2)/(2\varrho)$, where $\varrho := (\Delta_2 - d_0)/\Delta_2 \in [1 - 10^{-4}, 1]$. The right side is increasing in ϱ , so $\cos \alpha \geq -0.8821$, i.e. $\alpha \leq 151.9^\circ$. By concavity of $\sin$ on $[0, \pi]$, $\sin \alpha \geq \min\{\sin(\beta/2), 0.471\} = \sin(\beta/2)$.^[C8]
- (iv) *The walk.* Let $e_0 \perp u_v$ be the unit vector with $e_0 \cdot g = -\sin \alpha < 0$. Follow ∂K from v in the direction whose forward one-sided tangent is closest to e_0 , parametrised by arc length, $\gamma(0) = v$. Suppose the turning of $\gamma([0, \sigma])$ is at most $\beta/4$. This turning includes the whole normal cone $N_K(v)$. Then every normal along the arc is within $\beta/4$ of u_v , so every forward tangent γ' is within $\beta/4$ of e_0 (at a corner v , the forward tangent is perpendicular to an extreme normal of $N_K(v)$, which is also within $\beta/4$ of u_v). Hence $\gamma' \cdot g \leq -\sin \alpha + 2\sin(\beta/8) \leq -\sin(\beta/2) + 2\sin(\beta/8) < -\beta/5$.^[C8] Integrating, $(\gamma(\sigma) - v) \cdot g \leq -\sigma\beta/5$, and so

$$|\gamma(\sigma) - x_z|^2 \geq (\Delta_2 - d_0)^2 + \frac{2}{5}(\Delta_2 - d_0)\sigma\beta.$$

For $\sigma = L := 6/\beta = 600$ the right side exceeds Δ_2^2 ,^[C8] which puts $\gamma(L)$ outside $B(x_z, \Delta_2) \supseteq K$, a contradiction. Hence the closed arc J_v of length L from v in the chosen direction has turning greater than $\beta/4$.

- (v) *Count.* A point y lies in J_v only if v is within arc distance L of y , which allows at most $2L + 1$ points v . By Lemma 2.5, $\#\{v\} \leq 2\pi(2L + 1)/(\beta/4)$. Each such v has degree at most 6, so $\#(c2) \leq 48\pi(2L + 1)/\beta < 1.82 \cdot 10^7$.^[C9]

Proposition 4.3. *In the nondegenerate regime the number C_{bad} of bad edges satisfies $C_{\text{bad}} < 1.83 \cdot 10^7$.^[C9]*

5. THE NONDEGENERATE REGIME: BOUNDING $e(S)$

In this section we prove $e(S) \leq s + p + C$ with an absolute constant C in every τ -regime of the nondegenerate regime. Rungs are counted with the bad edges excluded: $e(S) \leq e_T + \rho + C_{\text{bad}}$, where ρ is the number of good rungs. (When $\tau \leq 2\beta$, Lemma 2.4 is not used.)

5.1. Region I: $\tau \leq \sqrt{3}/2 - 2\beta$. Let $p \in S \setminus D$ and let q be a good neighbour of p ; the *elevation* of $q - p$ is $\arcsin((q - p) \cdot u_p)$. If q is normal for p , then $b > 0$ (by Lemma 2.2(a) if $q \in D$, by (b) if $q \notin D$), so $a < \beta$. Otherwise q is an exception and $a = -\tau$.

- If $\tau \geq 0$, every good neighbour on one side of $\mathbb{R}u_p$ has elevation in $[-\arcsin \tau, \arcsin \beta)$, a window of width less than $\arcsin(\sqrt{3}/2 - 2\beta) + \arcsin \beta \approx 58.35^\circ < 60^\circ$.^[C4]
- If $\tau < 0$, then $|\tau| \leq 1/(2\Delta_2)$ and all elevations lie in $(0, \arcsin \beta) \cup \{\arcsin |\tau|\}$; the window is again less than 60° .^[C4]

Let $p \in S \cap D$. Then $a > 0$ always, and either $a < \beta$, or p is the exception of q and $a \in [\tau - \beta, \tau + \beta]$; the window is at most $\arcsin(\sqrt{3}/2 - \beta) \approx 58.87^\circ < 60^\circ$.^[C4]

Since two neighbours of p are at least 60° apart, p has at most one good neighbour on each side. Hence

$$e(S) \leq s + C_{\text{bad}} \quad \text{in Region I.}$$

5.2. Common facts for $\tau > \sqrt{3}/2 - 2\beta$. Here $\tau > 2\beta$,^[C5,C6] so [Lemma 2.4](#) applies: every good edge is a T-edge or a rung. *Region II* is $\tau \geq \sqrt{3}/2 + 3\beta$; *Region III* is $\sqrt{3}/2 - 2\beta < \tau < \sqrt{3}/2 + 3\beta$. If $\tau > 1$ then $t > 1$, and by [Lemma 2.2\(c\)](#) there are no rungs; so $e(S) \leq e_T + C_{\text{bad}} \leq s + C_{\text{bad}}$ by (T1). *From now on in this section assume $\tau \leq 1$; then $t \leq 1$ and $t \geq \tau$ ([Section 2](#)).* Put

$$\theta_0 := \arccos \tau \in [0, \pi/2), \quad \sigma = \sqrt{1 - \tau^2} = \sin \theta_0, \quad o_{\pm}(u) := -\tau u \pm \sigma u^{\perp}.$$

By [Lemma 2.2\(c\)](#), a rung (v, z) with $v \in S \setminus D$ has $z = v + o_+(u_v)$ or $z = v + o_-(u_v)$; in particular a K -vertex has at most two rungs. The map $u \mapsto o_{\pm}(u)$ is a rotation, so $|o_{\pm}(u) - o_{\pm}(u')| = |u - u'| \leq \angle(u, u')$, and $|o_+(u) - o_-(u)| = 2\sigma$. The K -row is $S \setminus D$ and the D -row is $S \cap D$; rungs join the two rows.

T-steps. If q is the right T-neighbour of p , then $q - p = \cos \varepsilon u_p^{\perp} + \sin \varepsilon u_p$ with $\varepsilon \in (0, \arcsin \beta)$, since $a = (q - p) \cdot u_p \in (0, \beta)$.

Lemma 5.1 (mixed T-edges). *Assume $\sqrt{3}/2 - 2\beta < \tau \leq 1$. Let $v \in S \setminus D$ and $z \in S \cap D$ be joined by a T-edge. Then the closed arc of ∂K of length 2 starting at v on z 's side has turning at least $t - \arcsin \beta$. Consequently there are fewer than $2\pi \cdot 10/(t - \arcsin \beta)$ mixed T-edges: fewer than 71 in Region II and fewer than 76 in Region III.*

Proof. Here $t \geq \tau > \sqrt{3}/2 - 2\beta > 0.846$. Suppose the turning is $\Theta < t - \arcsin \beta$; note $t - \arcsin \beta < 1 < \pi/3$. Let e_0 be the unit tangent direction at v perpendicular to u_v pointing to z 's side. Every forward tangent along the arc is within Θ of e_0 , so $|\gamma(\sigma') - v| \geq \sigma' \cos \Theta \geq \sigma'/2$ for the arc-length parametrisation γ , and some $\sigma^* \leq 2$ has $|\gamma(\sigma^*) - v| = 1$. The unit vector $\gamma(\sigma^*) - v$ lies within Θ of e_0 , and the unit vector $z - v$ lies within $\arcsin \beta$ of e_0 . Hence $|\gamma(\sigma^*) - z| \leq \Theta + \arcsin \beta < t \leq \text{dist}(z, K)$ by [Lemma 2.1\(c\)](#). But $\gamma(\sigma^*) \in K$, a contradiction.

For the count: each v has at most two T-edges, and v must lie within arc distance 2 of a point y of its arc, so each boundary point lies on at most $M \leq 10$ of these arcs. By [Lemma 2.5](#) the number of mixed T-edges is less than $2\pi \cdot 10/(t - \arcsin \beta)$. In Region II, $t - \arcsin \beta \geq 0.886$ and the bound is below 71;^[C5] in Region III, $t - \arcsin \beta > 0.836$ and the bound is below 76.^[III] $\square$

Lemma 5.2 (corner lemma). *Let $v \in S \setminus D$ have a rung partner z and a second Δ_2 -partner $w' \neq w_v$, and put $u' := (w' - v)/\Delta_2$ and $c := z - v$. Then exactly one of the following holds.*

- (i) $|zw'| \leq \Delta_2$. Then $c \cdot u' \geq 1/(2\Delta_2) > 0$ and $\angle(u_v, u') > \pi/2 - \theta_0$.
- (ii) $|zw'| = \Delta$. Then $u' = u^*$, the reflection of u_v in the line $\mathbb{R}c$, and $\angle(u_v, u') = 2\theta_0 > 0$.

In both cases u_v and u' are inward normals of K at v , and the normal cone $N_K(v)$ has angle at least $\angle(u_v, u')$.

Proof. Since $z \in X$, $|zw'| \in [0, \Delta_2] \cup \{\Delta\}$, so exactly one case occurs. We have $c \cdot u_v = -\tau$ and $|c| = 1$, so $\angle(c, u_v) = \pi - \theta_0$. (i) By [Lemma 2.2\(b\)](#) applied with w' in place of w_v , z is normal for v with respect to w' : $c \cdot u' \geq |c|^2/(2\Delta_2) > 0$. So $\angle(c, u') < \pi/2$ and $\angle(u_v, u') \geq \angle(c, u_v) - \angle(c, u') > \pi/2 - \theta_0$. (ii) By [Lemma 2.2\(c\)](#) applied with w' , $c \cdot u' = -\tau$. The unit vectors u with $c \cdot u = -\tau$ are the two vectors at angle θ_0 from $-c$, namely u_v and its reflection u^* in $\mathbb{R}c$. Since $w' \neq w_v$ and both are at distance Δ_2 from v , $u' \neq u_v$; so $u' = u^*$, $\theta_0 > 0$, and $\angle(u_v, u') = 2\theta_0$. Both u_v and u' are inward normals of K at v , because $K \subseteq B(w_v, \Delta_2) \cap B(w', \Delta_2)$. Moreover $\angle(u_v, u') < \pi$, because $|w_v w'| \leq \Delta < 2\Delta_2$; so the (convex) normal cone contains the shorter arc between them. $\square$

Normal cones of K at distinct boundary points have disjoint interiors. We say that v is *of type* (i) if some second Δ_2 -partner of v is as in (i), and *of type* (ii) if v has Δ_2 -degree at least 2 and is not of type (i). A rung K-end of type (ii) has Δ_2 -degree exactly 2, since only the directions u_v and u^* are available.

5.3. Region II: $\tau \geq \sqrt{3}/2 + 3\beta$. We keep $\tau \leq 1$. Then $\theta_0 \leq \arccos(\sqrt{3}/2 + 3\beta) < 26.4^\circ < 30^\circ$ and $\sigma \leq \sigma_{\max} \approx 0.4440$.^[C5]

(R1) *Each vertex has at most one rung.* At $v \in S \setminus D$: the two candidate positions $v + o_{\pm}(u_v)$ are $2\sigma < 1$ apart. At $z \in S \cap D$: if z had rungs to $v \neq v'$, both rungs are good, so $\angle(u_v, u_{v'}) \leq 2\beta$ and $|v - v'| = |o_{\pm}(u_v) - o_{\pm'}(u_{v'})| \leq 2\sigma + 2\beta \approx 0.908 < 1$,^[C5] a contradiction.

So rungs form a matching. By Lemma 5.2, a type-(i) vertex has a normal cone of angle greater than $\pi/2 - \theta_0 > \pi/3$, so *there are at most 5 type-(i) vertices*. Every other rung K-end either lies in Q or is of type (ii).

Lemma 5.3 (far partners persist). *Let $z \in D$ and $w \in X$ with $|zw| = \Delta$, put $\hat{w} := (w - z)/\Delta$, and let $y \in X$ with $|zy| = 1$ and $\eta := (y - z) \cdot \hat{w}$. Then $|yw|^2 = \Delta^2 + 1 - 2\Delta\eta$. If*

$$\eta < \frac{\Delta^2 - \Delta_2^2 + 1}{2\Delta} = t + \frac{1 - t^2}{2\Delta},$$

then $|yw| > \Delta_2$, hence $|yw| = \Delta$. When $t \leq 1$ the threshold is at least t .

Proof. The identity is $|y - w|^2 = |y - z|^2 - 2(y - z) \cdot (w - z) + |w - z|^2$. The inequality $\Delta^2 + 1 - 2\Delta\eta > \Delta_2^2$ is the displayed condition, and $\Delta^2 - \Delta_2^2 = 2\Delta t - t^2$ gives the second form of the threshold. There are no distances in (Δ_2, Δ) .^[II] $\square$

Lemma 5.4. *Let (v, z) be a rung whose K-end v is of type (ii). Then z has no T-neighbour at all, pure or mixed.*

Proof. Let $w := w_v$ and let w^* be the second Δ_2 -partner of v , in direction u^* from v . Since the rung is an exception for v with respect to both partners (Lemma 5.2(ii)), $|zw| = |zw^*| = \Delta$. Put $\hat{w} := (w - z)/\Delta$ and $\hat{w}^* := (w^* - z)/\Delta$. The angle between $\hat{w}$ and u_v is the angle at w of the triangle vzw ; it is opposite the side vz of length 1, hence acute, and by the law of sines it is at most $\arcsin(1/\Delta)$. Likewise $\angle(\hat{w}^*, u^*) \leq \arcsin(1/\Delta)$. Also $\angle(u^*, u_v) = 2\theta_0$, and $\angle(u_v, u_z) \leq \beta$ because the rung is good.

Let z_1 be a T-neighbour of z and $e := z_1 - z$, a unit vector with $e \cdot u_z \in (0, \beta)$, so $\angle(e, u_z) \geq \pi/2 - \arcsin \beta$. Put $\varepsilon := \beta + \arcsin \beta + \arcsin(1/\Delta)$. By the triangle inequality for angles, $\angle(e, \hat{w}) \geq \pi/2 - \varepsilon$ and $\angle(e, \hat{w}^*) \geq \pi/2 - (2\theta_0 + \varepsilon)$, so

$$\eta := e \cdot \hat{w} \leq \sin \varepsilon, \quad \eta^* := e \cdot \hat{w}^* \leq \sin(2\theta_0 + \varepsilon).$$

Both are less than t : since $t \geq \tau = \cos \theta_0 = \sin(\pi/2 - \theta_0)$ and $0 \leq \varepsilon \leq 2\theta_0 + \varepsilon < \pi/2 - \theta_0$, it suffices that $3\theta_0 + \varepsilon < \pi/2$, and indeed $\theta_0 < 26.4^\circ$ and $\varepsilon < 1.2^\circ$ give $3\theta_0 + \varepsilon < 80.4^\circ$. By Lemma 5.3 (with $t \leq 1$), $|z_1 w| = |z_1 w^*| = \Delta$. Now $w \neq w^*$, and z, z_1 are two distinct points of $C(w, \Delta) \cap C(w^*, \Delta)$. These two points are $2\sqrt{\Delta^2 - |ww^*|^2}/4 \geq \sqrt{3}\Delta > 1$ apart, because $|ww^*| \leq \Delta$. This contradicts $|zz_1| = 1$. $\square$

The count in Region II. By (T1) every point of S has T-degree at most 2. By Lemma 5.4 the D-ends of rungs with a type-(ii) K-end have T-degree 0, and they are distinct by (R1). Let d be the number of these rungs. Summing T-degrees, $2e_T \leq 2(s - d)$. Every other rung has a K-end in Q or of type (i), and each K-vertex has at most one rung (R1), so $\rho \leq d + p + 5$. Hence

$$e(S) \leq e_T + \rho + C_{\text{bad}} \leq (s - d) + (d + p + 5) + C_{\text{bad}} = s + p + 5 + C_{\text{bad}} \quad \text{in Region II.}$$

Mixed T-edges need no separate count here.

5.4. Region III: $\sqrt{3}/2 - 2\beta < \tau < \sqrt{3}/2 + 3\beta$. Here $\theta_0 \in (26.36^\circ, 32.22^\circ)$ and $t \geq \tau > 0.846$. By [Lemma 5.2](#), a rung K-vertex of Δ_2 -degree at least 2 has a normal cone of angle at least $\min\{2\theta_0, \pi/2 - \theta_0\} > 2\pi/7$, because $\tau < \sqrt{3}/2 + 3\beta < \cos(\pi/7)$ and $\tau > \sqrt{3}/2 - 2\beta > \cos(3\pi/14)$; ^[C6] so there are at most 6 such vertices.

Let q_1 and q_2 be the numbers of K-vertices with exactly one and exactly two good rungs. Then $\rho \leq q_1 + 2q_2$, and $p \geq q_1 + q_2 - 6$. It therefore suffices to prove

$$(3) \quad e_T \leq s - q_2 + C,$$

since then $e(S) \leq e_T + \rho + C_{\text{bad}} \leq s + q_1 + q_2 + C + C_{\text{bad}} \leq s + p + C + C_{\text{bad}} + 6$. A K-vertex with two rungs is called a *two-rung vertex*; its rungs are $z = v + o_+(u_v)$ and $z' = v + o_-(u_v)$, and $z - z' = 2\sigma u_v^\perp$.

(a) *The case $\tau > \sqrt{3}/2$.* The two positions $v + o_\pm(u_v)$ are $2\sigma < 1$ apart, so $q_2 = 0$, and (3) holds with $C = 0$ by (T1).

(b) *The case $\tau < \sqrt{3}/2$: slot killing.* Here $2\sigma > 1$; more precisely $0 < 2\sigma - 1 < 0.067$ and $\sigma < 0.5332$. ^[III]

Lemma 5.5. *Let v be a two-rung vertex with rungs $z = v + o_+(u_v)$ and $z' = v + o_-(u_v)$. Then z' has no T-neighbour (pure or mixed) on its right, and z has no T-neighbour on its left.*

Proof. Let y be the right T-neighbour of z' . Then $y - z' = \cos \varepsilon u_{z'}^\perp + \sin \varepsilon u_{z'}$ with $\varepsilon \in (0, \arcsin \beta)$, and $\angle(u_v, u_{z'}) \leq \beta$ (good rung). So the angle γ' between $y - z'$ and $z - z' = 2\sigma u_v^\perp$ is at most $\gamma := \beta + \arcsin \beta < 0.0201$. Hence

$$|y - z|^2 = 1 + 4\sigma^2 - 4\sigma \cos \gamma' \leq (2\sigma - 1)^2 + 4\sigma(1 - \cos \gamma) < 0.005 < 1. \supset [III]$$

Also $y \neq z$, since $|yz'| = 1 \neq 2\sigma = |zz'|$. This contradicts $\delta = 1$. The statement for z is symmetric. $\square$

Injectivity. Suppose $v \neq \bar{v}$ are two-rung vertices with the same $z' = v + o_-(u_v) = \bar{v} + o_-(u_{\bar{v}})$. Put $\bar{z} := \bar{v} + o_+(u_{\bar{v}}) = z' + 2\sigma u_{\bar{v}}^\perp$. Since $\angle(u_v, u_{\bar{v}}) \leq 2\beta$ (both within β of $u_{z'}$), $|\bar{z} - z| \leq 2\sigma \cdot 2\beta < 1$, so $\bar{z} = z$. ^[III] Then $v, \bar{v}$ are the two points of $C(z, 1) \cap C(z', 1)$, mirror images in the line zz' , which has direction $u_v^\perp$. As $v - z' = \tau u_v + \sigma u_v^\perp$, we get $\bar{v} - z' = -\tau u_v + \sigma u_v^\perp$, and $(\bar{v} - z') \cdot u_{z'} \leq -\tau \cos \beta + \sigma \sin \beta < 0$, ^[III] contradicting [Lemma 2.2\(a\)](#) at $z' \in D$. The same holds for the z -ends. So the right slots killed at the z' -ends are pairwise distinct, and so are the left slots killed at the z -ends: [Lemma 5.5](#) kills $2q_2$ distinct slots. As e_T is half the number of used slots, $e_T \leq \frac{1}{2}(2s - 2q_2) = s - q_2$, which is (3).

(c) *The case $\tau = \sqrt{3}/2$.* Now $\sigma = \frac{1}{2}$ and $\theta_0 = 30^\circ$. For a two-rung vertex v with $w := w_v$, $z - z' = u_v^\perp$ has length 1, so $vz'z$ is an equilateral triangle of side 1, the *triangle at v* ; its base $z'z$ is a δ -edge inside $S \cap D$. By definition of rungs, $|zw| = |z'w| = \Delta$. (Property (R1) at D-vertices can fail here, since $2\sigma + 2\beta > 1$; the argument below does not use it.) We prove (3) with $C = C_{\text{bad}} + 75$.

Base determines apex. The other point of $C(z, 1) \cap C(z', 1)$ is $\bar{v} = z' - \tau u_v + \frac{1}{2}u_v^\perp$, and $(\bar{v} - z') \cdot u_{z'} \leq -\tau \cos \beta + \frac{1}{2} \sin \beta < 0$, contradicting [Lemma 2.2\(a\)](#). So a δ -edge is the base of at most one triangle, and at most C_{bad} triangles have a bad base. A good base joins two points of D , so by [Lemma 2.4](#) it is a pure D-row T-edge; since $z - z' = u_v^\perp$ is within β of $u_{z'}^\perp$, z is the right T-neighbour of z' .

Small rows. If the K-row has at most 2 points then $q_2 \leq 2$; if the D-row has at most 2 points, then there is at most one base and $q_2 \leq 1$. In both cases (3) holds with $C = 2$ since $e_T \leq s$. So assume both rows have at least 3 points.

Lemma 5.6 (rows are cyclic orders). *If v' is the right pure T -neighbour of v in the K -row, then v' is the successor of v in the counterclockwise cyclic order of $S \setminus D$ on ∂K . The same holds in the D -row, on ∂P .*

Proof. Take coordinates with $v = (0, 0)$, $v' = (1, 0)$. By the T -step form, $u_v = (\sin \varepsilon, \cos \varepsilon)$ and $u_{v'} = (-\sin \varepsilon', \cos \varepsilon')$ with $\varepsilon, \varepsilon' \in (0, \arcsin \beta)$. These are inward normals of K at v and v' , so the part of K with $y \leq 0$ lies in the triangle $\{y \leq 0, y \geq -x \tan \varepsilon, y \geq (x - 1) \tan \varepsilon'\}$. That triangle has $x \in [0, 1]$ and depth at most $\tan(\arcsin \beta)$, so each of its points is within $\sqrt{1/4 + \tan^2(\arcsin \beta)} < 1$ of v or v' .^[III] Hence the boundary arc of K from v to v' in $\{y \leq 0\}$ contains no other point of the row. Traversed from v to v' with K on the side of u_v (above), this arc runs counterclockwise, because $u^\perp$ is u rotated by -90° . For the D -row, u_z is an inward normal of P by Lemma 2.2(a), and the same argument applies on ∂P . $\square$

Let $N_K := |S \setminus D|$ and $N_D := |S \cap D|$, and let B_K (resp. B_D) be the number of pairs $(x, \text{succ } x)$ of cyclically consecutive points of the K -row (resp. D -row) that are not joined by a pure T -edge. By Lemma 5.6 the pure T -edges of a row are among its N consecutive pairs, so there are $N_K - B_K$ and $N_D - B_D$ of them. With Lemma 5.1,

$$(4) \quad e_T \leq s - B_K - B_D + 75.$$

If a point x has no pure right T -neighbour, then $(x, \text{succ } x)$ is counted in B_K (resp. B_D): otherwise x would be the right T -neighbour of $\text{succ } x$, and then $x = \text{succ}(\text{succ } x)$ by Lemma 5.6, so the row would have only 2 points.

Rotational sense. In both rows “right” means counterclockwise along the boundary (Lemma 5.6). Moreover, if $z_k \in S \cap D$ and its right T -neighbour z_{k+1} both lie on $C(w, \Delta)$, then z_{k+1} is counterclockwise from z_k about w . Indeed, put $\hat{w}_k := (w - z_k)/\Delta$ and $e := z_{k+1} - z_k$. Then $e \cdot \hat{w}_k = 1/(2\Delta)$, so $e = \frac{1}{2\Delta} \hat{w}_k \pm \sqrt{1 - \frac{1}{4\Delta^2}} \hat{w}_k^\perp$. Since x_{z_k} and w are both diametral partners of z_k and $|x_{z_k} w| \leq \Delta$, we have $\angle(u_{z_k}, \hat{w}_k) \leq \pi/3$; and $\angle(e, u_{z_k}^\perp) < \arcsin \beta$ by the T -step form. Hence $\angle(e, \hat{w}_k^\perp) < \pi/3 + \arcsin \beta < \pi/2$, the sign is $+$, and moving along $\hat{w}_k^\perp$ ($\hat{w}_k$ rotated by -90°) is moving counterclockwise about w .

Lemma 5.7 (a D -row component lies on one circle). *Let Z be the component of the graph of pure D -row T -edges that contains a good base $z'z$ of the triangle at v , and let $w := w_v$. Then $Z \subseteq C(w, \Delta)$, consecutive points of Z are at angle exactly $\alpha_1 := 2 \arcsin(1/(2\Delta))$ about w , all in the counterclockwise sense, and Z spans an angle at most $\pi/3$ about w . In particular Z is a path.*

Proof. We know $|zw| = |z'w| = \Delta$. Let $z_k \in Z$ have both a left T -neighbour z_{k-1} and a right T -neighbour z_{k+1} , and put $u := u_{z_k}$. By the T -step form, $z_{k+1} - z_k = \cos \varepsilon_1 u^\perp + \sin \varepsilon_1 u$ with $\varepsilon_1 \in (0, \arcsin \beta)$, and $z_{k-1} - z_k = -\cos \varepsilon_2 u^\perp + \sin \varepsilon_2 u$ where $\sin \varepsilon_2 = (z_{k-1} - z_k) \cdot u_{z_k}$ is the value b of the incoming T -edge, so $\varepsilon_2 \in (0, \arcsin \beta)$. Let $\nu = \cos \varphi u + \sin \varphi u^\perp$ be any inward normal of P at z_k . Then $\nu \cdot (z_{k+1} - z_k) = \sin(\varepsilon_1 + \varphi) \geq 0$ and $\nu \cdot (z_{k-1} - z_k) = \sin(\varepsilon_2 - \varphi) \geq 0$, so $\varphi \in [-\varepsilon_1, \varepsilon_2]$.

Suppose $|z_k w| = \Delta$. Then w is a diametral partner of z_k , and $\hat{w}_k := (w - z_k)/\Delta$ is an inward normal of P at z_k (Lemma 2.2(a) and the remark after it). So $\eta := (z_{k+1} - z_k) \cdot \hat{w}_k = \sin(\varepsilon_1 + \varphi) \in [0, \sin(2 \arcsin \beta)]$, and $\sin(2 \arcsin \beta) < 0.0201 < t$.^[III] By Lemma 5.3, $|z_{k+1} w| = \Delta$. In the same way $(z_{k-1} - z_k) \cdot \hat{w}_k = \sin(\varepsilon_2 - \varphi) \in [0, 0.0201)$ and $|z_{k-1} w| = \Delta$.

Now induct outwards from the base, with no wrap-around: going right from z , each vertex reached has a left T -neighbour, so whenever it has a right T -neighbour it is an interior vertex and the step applies; going left from z' is symmetric. So every vertex of Z lies on $C(w, \Delta)$. Two points of $C(w, \Delta)$ at distance 1 are at angle α_1 about w , and by the rotational-sense remark

each step to the right is counterclockwise about w . Points of X are at most Δ apart, and a chord $2\Delta \sin(\theta/2) \leq \Delta$ has $\theta \leq \pi/3$; so Z spans at most $\pi/3$, and monotone steps cannot close up into a cycle. $\square$

Lemma 5.8 (one centre per component). *If A and B are triangles, at v_A and v_B , whose good bases lie in the same component Z , then $w_{v_A} = w_{v_B}$.*

Proof. Write $w_A := w_{v_A}$, $w_B := w_{v_B}$, and let $z'_B z_B$ be the base of B . By Lemma 5.7 for A , $z'_B, z_B \in C(w_A, \Delta)$; by the definition of rungs, $z'_B, z_B \in C(w_B, \Delta)$. Suppose $w_A \neq w_B$; then w_A is the mirror image of w_B in the line $z'_B z_B$. Put $a := (w_B - z_B)/\Delta$ and $e := z'_B - z_B$. From $|z'_B - w_B| = \Delta$ we get $a \cdot e = 1/(2\Delta)$ exactly. The mirror image of a in the line $\mathbb{R}e$ is $a^* = 2(a \cdot e)e - a$, and $w_A - z_B = \Delta a^*$. So $(w_A - z_B) \cdot a = \Delta(2(a \cdot e)^2 - 1) < 0$. But w_B is a diametral partner of z_B and $w_A \in X$, so $(w_A - z_B) \cdot a \geq 0$ by Lemma 2.2(a). Contradiction. $\square$

So all triangles with good base in Z share one centre w . The apex v_A of such a triangle lies on $C(w, \Delta_2)$ and on the perpendicular bisector of its base (both v_A and w are equidistant from the base endpoints), on the same side as the base; so, seen from w , it lies at the angle of the base midpoint. Order the triangles with good base in Z along Z . Let A, B be consecutive, B to the right, with bases (z_i, z_{i+1}) and (z_{i+k}, z_{i+k+1}) . Then the apexes are at angle $k\alpha_1$ about w , with $k\alpha_1 \leq \pi/3 - \alpha_1 < \pi/3$ by Lemma 5.7, and $k\alpha_1 \geq \alpha_2 := 2 \arcsin(1/(2\Delta_2))$ because $|v_A v_B| \geq 1$.

The arc between consecutive apexes. Let γ_{AB} be the shorter arc of $C(w, \Delta_2)$ from v_A to v_B . For $x \in X \setminus \{w\}$, $B(x, \Delta_2) \cap C(w, \Delta_2)$ is the arc $\{\theta : \cos(\theta - \theta_x) \geq |wx|/(2\Delta_2)\}$ of angular width at most π . It contains $v_A, v_B \in K$, which are less than π apart about w , so it contains γ_{AB} . Hence $\gamma_{AB} \subseteq K \subseteq B(w, \Delta_2)$, and therefore $\gamma_{AB} \subseteq \partial K$.

Lemma 5.9 (the K-path is forced along the arc). *Let $v_0 := v_A$ and follow pure right T-edges in the K-row: $v_0, v_1, \dots$. As long as $v_i \in \gamma_{AB} \setminus \{v_B\}$ and v_i has a pure right T-neighbour, that neighbour is the point v^* of γ_{AB} at angle α_2 counterclockwise from v_i about w .*

Proof. Normal fact. $u_{v_i} = \hat{w}_i := (w - v_i)/\Delta_2$. For $i = 0$ this holds because $w = w_{v_A}$. For v_i in the relative interior of γ_{AB} , K contains the arc on both sides of v_i , and $K \subseteq B(w, \Delta_2)$; so every inward normal ν at v_i satisfies $(y - v_i) \cdot \nu \geq 0$ for y on the arc on both sides, which forces ν to be a positive multiple of $\hat{w}_i$. Since u_{v_i} is an inward normal of K at v_i , $u_{v_i} = \hat{w}_i$.

v^* lies on γ_{AB} . Otherwise v_B would be at angle in $(0, \alpha_2)$ from v_i , and $|v_i v_B| < 1$.

The step. Use coordinates with second axis $u := \hat{w}_i$ and first axis $u^\perp$ (counterclockwise tangent). Then $v_{i+1} - v_i = (\cos \psi, \sin \psi)$ with $\psi \in (0, \arcsin \beta)$ (T-step form, with $u_{v_i} = \hat{w}_i$). From $v_{i+1} \in K \subseteq B(w, \Delta_2)$, $|v_{i+1} - w|^2 = \Delta_2^2 + 1 - 2\Delta_2 \sin \psi \leq \Delta_2^2$, so $\psi \geq \psi^* := \arcsin(1/(2\Delta_2))$, the elevation of v^* . Suppose $\psi > \psi^*$. Write $u_{v_{i+1}} = (\sin \varphi, \cos \varphi)$; $|\varphi| \leq \beta$ because the T-edge is good. From $v_i \in K$, $(v_i - v_{i+1}) \cdot u_{v_{i+1}} = -\sin(\psi + \varphi) \geq 0$, so $\varphi \leq -\psi < 0$. From $v^* \in K$ (as $v^* \in \gamma_{AB}$),

$$(v^* - v_{i+1}) \cdot u_{v_{i+1}} = (\cos \psi^* - \cos \psi) \sin \varphi + (\sin \psi^* - \sin \psi) \cos \varphi \geq 0.$$

But both terms are negative. So $\psi = \psi^*$ and $v_{i+1} = v^*$. $\square$

So the path moves along γ_{AB} in steps of exactly α_2 about w , and it cannot pass v_B . It reaches v_B only if $j\alpha_2 = k\alpha_1$ for some integer $j \geq 1$; otherwise it stops at a point $b_{AB} \in \gamma_{AB} \setminus \{v_B\}$ of the K-row that has no pure right T-neighbour. (Its right slot is unused or used by a mixed T-edge.) In particular a pure right T-neighbour of a point of γ_{AB} never lies on another part of ∂K .

The path cannot land on v_B . Since $\alpha_2 > \alpha_1$, $j\alpha_2 = k\alpha_1$ forces $k > j$, so $j(\alpha_2 - \alpha_1) = (k - j)\alpha_1 \geq \alpha_1$ and

$$k\alpha_1 = j\alpha_2 \geq \frac{\alpha_1 \alpha_2}{\alpha_2 - \alpha_1} = \frac{1}{1/\alpha_1 - 1/\alpha_2}.$$

For $\rho \geq 10^4$ put $f(\rho) := 1/(2\arcsin(1/(2\rho)))$. From $x \leq \arcsin x \leq x + x^3/(6(1-x^2))$ (the Taylor coefficients of $\arcsin$ beyond the first are at most $\frac{1}{6}$) we get $\rho - 1/(23\rho) \leq f(\rho) \leq \rho$. Hence $1/\alpha_1 - 1/\alpha_2 = f(\Delta) - f(\Delta_2) \leq t + 1/(23\Delta_2)$. At $\tau = \sqrt{3}/2$ we have $t \leq 1$ and $t - \tau = (1 - t^2)/(2\Delta_2) \leq 1/(2\Delta_2)$, so $t < 0.8661$. Therefore $k\alpha_1 \geq 1/(t + 1/(23\Delta_2)) > 1/0.8662 > 1.154 > \pi/3$,^[III] contradicting $k\alpha_1 < \pi/3$. So every pair (A, B) of consecutive triangles in the same Z yields a point b_{AB} of the K-row in the half-open arc $[v_A, v_B)$ with no pure right T-neighbour, and hence a pair $(b_{AB}, \text{succ } b_{AB})$ counted in B_K .

Counting. Distinct points b give distinct pairs $(b, \text{succ } b)$. The half-open arcs $[v_A, v_B) \subseteq \partial K$ are pairwise disjoint:

- *Same Z :* they are consecutive intervals on one circle.
- *Different components $Z \neq Z'$ with the same centre w :* their angular ranges about w are disjoint, since a point of one strictly inside the range of the other would be at angle less than α_1 , hence at distance less than 1, from one of its points. The apex arcs lie in these ranges.
- *Different centres $w \neq w'$:* let $\gamma \subseteq C(w, \Delta_2)$ and $\gamma' \subseteq C(w', \Delta_2)$ be nondegenerate arcs of ∂K sharing a point x . If x were in the relative interior of γ , then near x the boundary ∂K coincides with γ on both sides, and γ' would contain a nondegenerate piece of ∂K lying on both circles, forcing $w = w'$. So x is an endpoint of both arcs; for the half-open arcs this means $x = v_A = v_{A'}$. The triangle at a K-vertex is unique, so $A = A'$.

Hence $B_K \geq \sum_Z (m_Z - 1)$, where m_Z is the number of triangles with good base in Z and the sum is over components with $m_Z \geq 1$. Each such Z is a path (Lemma 5.7); its right end has no pure right T-neighbour and so gives a pair counted in B_D , and distinct Z give distinct pairs. So $B_D \geq \#\{Z : m_Z \geq 1\}$. Adding, $B_K + B_D \geq \sum_Z m_Z \geq q_2 - C_{\text{bad}}$, and (4) gives

$$e_T \leq s - q_2 + C_{\text{bad}} + 75,$$

which is (3).

Conclusion for Region III. In all three cases, $e(S) \leq s + p + 2C_{\text{bad}} + 81$.

Remark 5.10. In Regions II and III, the argument of Appendix B (ownership and pairing, rhombi, and the weighted bound with $2p$) is no longer needed. What replaces it is the observation that a rung beyond the first-order count is always paid for, either through p on the Δ_2 side or by a missing T-edge.

6. POINTS WITHOUT A Δ_2 -PARTNER

In this section we are in the nondegenerate regime with $\Delta_2 \geq 10^4$; no assumption on τ is made. We bound $\deg y + m_y$ for $y \in R$.

Theorem 6.1. *All but fewer than $7.1 \cdot 10^7$ points $y \in R$ satisfy $m_y \leq 4$, and hence $\deg y + m_y \leq 10$.*

Theorem 6.2. *Let $0 < \varepsilon_0 \leq 1/50$, put $\varepsilon := \tan \varepsilon_0$, and suppose $t > \sqrt{3}/2 + \varepsilon$. Then all but at most $2\pi \cdot 468^2/\varepsilon_0 + 5106 + 465^2$ points $y \in R$ satisfy $m_y \leq 3$, and $m_y \leq 2$ if $\deg y = 6$; hence $\deg y + m_y \leq 8$. In Region II this holds with $\varepsilon_0 = 1/50$, and then the exceptional set has fewer than $7.1 \cdot 10^7$ points.*

In Region II, $t \geq \tau \geq \sqrt{3}/2 + 3\beta$ if $t \leq 1$, and $t > 1$ otherwise; since $\tan(1/50) < 0.0201 < 3\beta$, the hypothesis of Theorem 6.2 holds with $\varepsilon_0 = 1/50$.

6.1. Structure of R .

- (a) *Points of $R \cap D$.* If $y \in R \cap D$, then by Lemma 2.2(a) (which holds for every point of D) all other points of X lie in an open half-plane bounded by a line through y . The δ -neighbours of y are pairwise at least 60° apart as seen from y , so $\deg y \leq 3$ and $\deg y + m_y \leq 6$.
- (b) $\partial K \subseteq \bigcup_{w \in X} C(w, \Delta_2)$. A point $x \in K$ with $|x - w| < \Delta_2$ for all (finitely many) $w \in X$ is an interior point of K . Hence a point of $X \setminus D$ on ∂K has a Δ_2 -partner, i.e. lies in S . With Lemma 2.1(a): $R \setminus D \subseteq \text{int } K$ and $S \setminus D = (X \setminus D) \cap \partial K$. So every δ -neighbour of a point of X lies in $\text{int } K$ (and then in $R \setminus D$), in $S \setminus D \subseteq \partial K$, or in D , which is disjoint from K .
- (c) *Depth.* For $y \in R \setminus D$ put $h := \text{dist}(y, \partial K) > 0$. Every neighbour q of y in S is outside $\text{int } K$: either $q \in S \setminus D \subseteq \partial K$, or $q \in D$ with $\text{dist}(q, K) \geq t$ (Lemma 2.1(c)). The segment $[y, q]$ leaves K at a point $b \in \partial K$, so $h \leq |yb| \leq 1$, and $h \leq 1 - t$ if $q \in D$. Consequently, if $h > 1$ then $m_y = 0$ and $\deg y + m_y \leq 6$.

From now on let $y \in R \setminus D$ with $h \leq 1$. Let $p_0 \in \partial K$ be a nearest boundary point, $n_0 := (p_0 - y)/h$, which is an outer normal of K at p_0 , $H_0 := \{x : (x - p_0) \cdot n_0 \leq 0\} \supseteq K$, and $\ell_0 := \partial H_0$.

Small K . If the perimeter of K is less than 464, then $\text{diam } K < 232$ and, since the discs of radius $\frac{1}{2}$ about points of $X \setminus D \subseteq K$ are disjoint, $|X \setminus D| \leq (2 \cdot 232 + 1)^2 = 465^2$. We may therefore assume that the perimeter of K is at least 464, and we define

$\gamma_y :=$ the closed arc of ∂K of points at arc-length distance at most 232 from p_0 .

6.2. Thin K .

Lemma 6.3 (thin K). *If some point $b \in \partial K \cap B(y, 1)$ does not lie on γ_y , then K lies in a rectangle of length $L < 400$ and width $W < 9$. Hence $|X \setminus D| < 5106$.*

Proof. Let ν_b be an outer normal of K at b , $H_b := \{x : (x - b) \cdot \nu_b \leq 0\} \supseteq K$ and $\ell_b := \partial H_b$. Then $\text{dist}(y, \ell_0) = h \leq 1$ and $\text{dist}(y, \ell_b) \leq |yb| \leq 1$. Both arcs of ∂K from p_0 to b have length greater than 232.

Case $\nu_b = n_0$. Then $\ell_0 = \ell_b$: otherwise one of H_0, H_b is strictly contained in the other, and the point $p_0 \in \ell_0$ or $b \in \ell_b$ lies outside the smaller half-plane, although both lie in K . So $[p_0, b] \subseteq K \cap \ell_0 \subseteq \partial K$ is a boundary arc of length $|p_0 b| \leq 2 < 232$, a contradiction.

Case $\nu_b \neq \pm n_0$. The lines ℓ_0, ℓ_b meet at a point A , and $Z := H_0 \cap H_b$ is a closed cone with apex A and opening $\theta := \pi - \angle(n_0, \nu_b) \in (0, \pi)$. The point $y \in Z$ has distance at most 1 from both bounding lines; the set of such points of Z is a rhombus whose far vertex is on the bisector at distance $1/\sin(\theta/2)$ from A , so $|yA| \leq 1/\sin(\theta/2)$. The point p_0 lies on the boundary ray of Z in ℓ_0 and b on the other boundary ray, and $|Ap_0| \leq |Ay| + |yp_0| \leq 1 + 1/\sin(\theta/2)$, and likewise for $|Ab|$. If $[p_0, b] \subseteq \partial K$ we conclude as before. Otherwise the line $p_0 b$ meets ∂K only in p_0, b ; let H^+ be the closed half-plane bounded by it that contains A . Then $\partial K \cap H^+$ is one of the two arcs from p_0 to b , and $K \cap H^+$ is a convex subset of the triangle $Ap_0 b$. By monotonicity of perimeter, the length of this arc is at most $|Ap_0| + |Ab| \leq 2 + 2/\sin(\theta/2)$. If $\theta \geq 1^\circ$ this is less than $231.2 < 232$,^[N] a contradiction. So $\theta < 1^\circ$.

Fatness step ($\theta < 1^\circ$, or $\nu_b = -n_0$, in which case Z is a strip and we put $\theta := 0$). Let e be the direction of the bisector of Z (the direction of the strip if $\theta = 0$), oriented so that the cone opens towards increasing values of the level $x \cdot e$, and let $e^\perp$ be a unit normal; call $x \cdot e^\perp$ the *height*. The cross-section of Z at level λ has length $w(\lambda) = w(\lambda_y) + 2 \tan(\theta/2)(\lambda - \lambda_y)$, where λ_y is the level of y and $w(\lambda_y) = (d_1 + d_2)/\cos(\theta/2) \leq 2/\cos(\theta/2) =: w_y$, with $d_1, d_2 \leq 1$ the distances from y to the two lines. Let $[E_-, E_+] \ni \lambda_y$ be the range of levels of K , $L := E_+ - E_-$, which is positive since K has interior points, and let W be the range of heights of K . The cross-sections of Z are nested, so $W \leq w_y + 2 \tan(\theta/2)L$; also $L \leq \text{diam } K \leq 2\Delta_2$.

Pick $a, a' \in K$ at levels E_-, E_+ , and let $\lambda_m := (E_- + E_+)/2$. The section of K at level λ_m is a segment of positive length, with top p_{top} and bottom p_{bot} . Let $n_{\text{top}} = \cos \varphi e^\perp + \sin \varphi e$ be an outer normal of K at p_{top} . Writing $a - p_{\text{top}} = -\frac{L}{2}e + v_a e^\perp$ and $a' - p_{\text{top}} = \frac{L}{2}e + v_{a'} e^\perp$ with $|v_a|, |v_{a'}| \leq W$, we have

$$-\frac{L}{2} \sin \varphi + v_a \cos \varphi \leq 0, \quad \frac{L}{2} \sin \varphi + v_{a'} \cos \varphi \leq 0.$$

If $\cos \varphi = 0$, these two inequalities contradict each other, since $L > 0$. If $\cos \varphi < 0$, adding them gives $v_a + v_{a'} \geq 0$; but the midpoint of aa' lies in K at level λ_m , so its height is at most that of p_{top} , i.e. $v_a + v_{a'} \leq 0$. Hence $v_a + v_{a'} = 0$, the midpoint of aa' is p_{top} , and then both inequalities are equalities, so a, a' lie on the supporting line at p_{top} ; now $p_{\text{bot}} = p_{\text{top}} - \mu e^\perp$ with $\mu > 0$ has $n_{\text{top}} \cdot (p_{\text{bot}} - p_{\text{top}}) = -\mu \cos \varphi > 0$, so it violates the supporting half-plane. Therefore $\cos \varphi > 0$, and the inequalities give $|\tan \varphi| \leq 2W/L$. The same holds at p_{bot} with $-e^\perp$ in place of $e^\perp$.

By (b) above, every boundary point x of K lies on some $C(w, \Delta_2)$ with $w \in X$, and then $(x - w)/\Delta_2$ is an outer normal of K at x (as $K \subseteq B(w, \Delta_2)$). Choose such w_{top} for p_{top} and w_{bot} for p_{bot} , and use these normals above. Then, with $\tan \varphi^* := 2W/L$,

$$(w_{\text{bot}} - w_{\text{top}}) \cdot e^\perp = (p_{\text{bot}} - p_{\text{top}}) \cdot e^\perp + \Delta_2(n_{\text{top}} \cdot e^\perp - n_{\text{bot}} \cdot e^\perp) \geq -W + 2\Delta_2 \cos \varphi^*,$$

while $|w_{\text{bot}} - w_{\text{top}}| \leq \Delta \leq 1.94 \Delta_2$. Suppose $L \geq 400$. Then $\tan \varphi^* \leq 2w_y/400 + 4 \tan(\theta/2)$ and $W \leq w_y + 4\Delta_2 \tan(\theta/2)$, so

$$\frac{2\Delta_2 \cos \varphi^* - W}{\Delta_2} \geq 2\left(1 - \frac{\tan^2 \varphi^*}{2}\right) - 4 \tan(\theta/2) - \frac{w_y}{10^4} > 1.96 > 1.94,^{[N]}$$

a contradiction. Hence $L < 400$ and $W \leq w_y + 800 \tan(0.5^\circ) < 9$. So K lies in a rectangle of size less than 400×9 , and the disjoint discs of radius $\frac{1}{2}$ about the points of $X \setminus D$ lie in a rectangle of area less than $401 \cdot 10$; so $|X \setminus D| < 4010/(\pi/4) < 5106$. $\square$

The argument uses only boundary points of K and centres in X , so corners of K and boundary points that are not in X are covered.

6.3. Big turning.

Lemma 6.4. *Let $\varepsilon_0 > 0$. The number of $y \in R \setminus D$ with $h \leq 1$ and $\text{turn}(\gamma_y) > \varepsilon_0$ is less than $2\pi M/\varepsilon_0$, where $M = 468^2$. For $\varepsilon_0 = 1/50$ this is less than $6.89 \cdot 10^7$.*

Proof. If $b \in \gamma_y$ then $|b - p_0| \leq 232$ (a chord is at most the arc), so $|b - y| \leq 233$. The points y are at least 1 apart, so for fixed b at most $(2 \cdot 233.5)^2 \leq 468^2$ of them qualify (disjoint discs of radius $\frac{1}{2}$ in $B(b, 233.5)$). Apply Lemma 2.5 to the arcs γ_y , indexed by y .^[N] $\square$

6.4. The flat case.

Lemma 6.5 (flat case). *Suppose $\partial K \cap B(y, 1) \subseteq \gamma_y$ and $\text{turn}(\gamma_y) \leq \varepsilon_0$ with $0 < \varepsilon_0 \leq 1/50$, and put $\varepsilon := \tan \varepsilon_0$. Use coordinates with origin p_0 , first axis a unit tangent $t_0 \perp n_0$ and second axis n_0 ; so $y = (0, -h)$. Then there is a concave function f on $[-231, 231]$ with $f(0) = 0$, $f \leq 0$ and $|f(x)| \leq \varepsilon|x|$, whose graph is contained in $\gamma_y \subseteq \partial K$, and $U := B(y, 1) \cap \{(x, z) : z < f(x)\} \subseteq \text{int } K$. Write a neighbour q of y as $q = y + (\sin \psi, \cos \psi)$, with ψ measured from n_0 . Then:*

- (F1) *every neighbour $q \notin \text{int } K$ (in particular every neighbour in S , and every neighbour in D) has $\cos \psi \geq h + f(\sin \psi) \geq h - \varepsilon|\sin \psi|$; hence $\cos \psi > -\varepsilon|\sin \psi|$, i.e. $|\psi| < \pi/2 + \varepsilon_0$;*
- (F2) *every neighbour $q \in S \setminus D$ has $\cos \psi = h + f(\sin \psi) \in [h - \varepsilon, h]$;*
- (F3) *every neighbour $q \in D$ (in S or in R) has $\cos \psi \geq h + t + f(\sin \psi) \geq h + t - \varepsilon$;*
- (F4) *every neighbour $q \in \text{int } K$ has $\cos \psi < h$.*

Proof. The normal set $N(\gamma_y)$ is an arc of S^1 of length at most ε_0 containing n_0 . Parametrise γ_y by arc length $\lambda \in [-232, 232]$ with $\gamma(0) = p_0$. Every one-sided tangent is perpendicular to a normal in $N(\gamma_y)$, hence within ε_0 of $\pm t_0$, and after orienting, $\gamma'(\lambda) \cdot t_0 \geq \cos \varepsilon_0 > 0$. So $x(\lambda) := \gamma(\lambda) \cdot t_0$ is strictly increasing with $|x(\pm 232)| \geq 232 \cos \varepsilon_0 > 231$, and γ_y contains the graph of a function f on $[-231, 231]$ with $|f'| \leq \varepsilon$ almost everywhere; as $f(0) = 0$, $|f(x)| \leq \varepsilon|x|$. Also $f \leq 0$ because $K \subseteq H_0$. At each graph point the outer normals have n_0 -component at least $\cos \varepsilon_0 > 0$, and K lies in the corresponding supporting half-plane; so K lies below every supporting line of the graph, and f is concave. The set U is convex (a disc intersected with the strict hypograph of a concave function; the x -range of $B(y, 1)$ is $[-1, 1]$), it contains y (as $-h < 0 = f(0)$), and $U \cap \partial K = \emptyset$, since $\partial K \cap B(y, 1) \subseteq \gamma_y$ and, x being monotone along γ_y , the points of $\gamma_y \cap B(y, 1)$ lie on the graph, where $z = f(x)$. A connected set meeting $\text{int } K$ and missing ∂K lies in $\text{int } K$; so $U \subseteq \text{int } K$.

(F1) If $q \notin \text{int } K$ then $q \notin U$, so $q_z \geq f(q_x)$, i.e. $\cos \psi - h \geq f(\sin \psi)$; and $h > 0$. The inequality $\cos \psi + \tan \varepsilon_0 |\sin \psi| > 0$ is equivalent to $\cos(|\psi| - \varepsilon_0) > 0$, i.e. $|\psi| < \pi/2 + \varepsilon_0$. (F2) A neighbour in $S \setminus D$ lies on $\partial K \cap B(y, 1)$, hence on the graph. (F3) The graph point $g = (\sin \psi, f(\sin \psi))$ lies in K and, by (F1), directly below q ; so $t \leq \text{dist}(q, K) \leq |q - g| = \cos \psi - h - f(\sin \psi)$. (F4) A neighbour in $\text{int } K$ is not on ∂K . If it had $\cos \psi \geq h$, it would satisfy $z \geq 0 \geq f(x)$; points strictly above a graph point are outside K (which lies below the supporting lines), and points on the graph are on ∂K . So $\cos \psi - h < f(\sin \psi) \leq 0$. $\square$

By (b) above, every neighbour of y lies in $\text{int } K$, in $S \setminus D$, or in D .

6.5. Proofs of Theorems 6.1 and 6.2.

Proof of Theorem 6.1. Take $\varepsilon_0 = 1/50$. Points of $R \cap D$ have $\deg y + m_y \leq 6$, and points of $R \setminus D$ with $h > 1$ have $m_y = 0$. Let $y \in R \setminus D$ with $h \leq 1$. If the perimeter of K is less than 464, there are at most 465^2 such points in total. If some point of $\partial K \cap B(y, 1)$ is not on γ_y , then Lemma 6.3 applies, and there are fewer than 5106 such points in total. If $\text{turn}(\gamma_y) > \varepsilon_0$, then y is one of fewer than $6.89 \cdot 10^7$ points (Lemma 6.4). Otherwise Lemma 6.5 applies: by (F1) every neighbour of y in S lies in the open arc $|\psi| < \pi/2 + \varepsilon_0$, of length less than $4\pi/3$. Distinct neighbours are at least 60° apart as seen from y , and an open arc of length less than 240° contains at most 4 such points. So $m_y \leq 4$. The exceptional points number fewer than $6.89 \cdot 10^7 + 5106 + 465^2 < 7.1 \cdot 10^7$. $\square$

Proof of Theorem 6.2. The exceptional points are as in the proof of Theorem 6.1; so let y be in the flat case of Lemma 6.5, with $h \leq 1$. Note $t > \sqrt{3}/2 + \varepsilon > \frac{1}{2} + 2\varepsilon$.

- (i) *At most one neighbour in D .* By (F3) a neighbour in D has $\cos \psi \geq t - \varepsilon > \sqrt{3}/2$, i.e. $|\psi| < 30^\circ$; two neighbours at angular distance at least 60° cannot both lie in this open arc.
- (ii) *At most one neighbour in $S \setminus D$ with $\psi \in [0, \pi)$, and at most one with $\psi \in (-\pi, 0]$.* If $0 \leq \psi_1 < \psi_2$ with $\psi_2 - \psi_1 \geq 60^\circ$ are both in $S \setminus D$, then $\psi_2 < 90^\circ + \varepsilon_0 < 91.2^\circ$ by (F1), so $(\psi_1 + \psi_2)/2 \in [30^\circ, 61.2^\circ)$ and $(\psi_2 - \psi_1)/2 \in [30^\circ, 45.6^\circ)$. Hence $\cos \psi_1 - \cos \psi_2 = 2 \sin \frac{\psi_1 + \psi_2}{2} \sin \frac{\psi_2 - \psi_1}{2} \geq \frac{1}{2} > \varepsilon$, contradicting (F2), by which both cosines lie in $[h - \varepsilon, h]$.

Every neighbour in S lies in $S \setminus D$ or in D , so $m_y \leq 3$; and if $m_y = 3$, the neighbours in S are one point of D at some ψ_D with $|\psi_D| < 30^\circ$ and two points $K_\pm$ of $S \setminus D$.

- (iii) *$\deg y = 6$ and $m_y = 3$ is impossible.* Six neighbours pairwise at least 60° apart are exactly 60° apart, at $\psi_D + 60^\circ k$. So $K_\pm$ occupy two of the five directions $\psi_D \pm 60^\circ$, $\psi_D \pm 120^\circ$, $\psi_D + 180^\circ$, and by (F2) their cosines differ by at most ε . We go through the ten pairs.
 - A pair containing $\psi_D + 180^\circ$ (four pairs): $\cos(\psi_D + 180^\circ) = -\cos \psi_D < -\sqrt{3}/2 < -\varepsilon$, contradicting (F1).

- $\{\psi_D + 60^\circ, \psi_D - 60^\circ\}$: $|\cos(\psi_D + 60^\circ) - \cos(\psi_D - 60^\circ)| = \sqrt{3}|\sin \psi_D| \leq \varepsilon$, so $\cos(\psi_D \pm 60^\circ) \geq \frac{1}{2} \cos \psi_D - \frac{\sqrt{3}}{2}|\sin \psi_D| \geq \frac{1}{2}(1 - \varepsilon^2/3) - \frac{\varepsilon}{2} \geq \frac{1}{2} - \varepsilon$. By (F2), $h \geq \frac{1}{2} - \varepsilon$, and then (F3) gives $\cos \psi_D \geq h + t - \varepsilon \geq \frac{1}{2} + t - 2\varepsilon > 1$, a contradiction.
- $\{\psi_D + 120^\circ, \psi_D - 60^\circ\}$ (and symmetrically $\{\psi_D - 120^\circ, \psi_D + 60^\circ\}$): the two cosines are x and $-x$ with $x = \cos(\psi_D + 120^\circ)$, and both lie in $[h - \varepsilon, h]$, so $|x| \leq \varepsilon/2$ and $h \leq \varepsilon$. As $\psi_D + 120^\circ \in (90^\circ, 150^\circ)$, this forces $\psi_D + 120^\circ \leq 90^\circ + \arcsin(\varepsilon/2)$, so $\psi_D + 60^\circ \in (30^\circ, 31.2^\circ]$. The neighbour at $\psi_D + 60^\circ$ is not in S (as $m_y = 3$), and $\cos(\psi_D + 60^\circ) > \frac{1}{2} > h$, so by (F4) it is not in $\text{int } K$; hence it lies in D , and by (F3) it has $|\psi| < 30^\circ$, a contradiction.
- $\{\psi_D + 120^\circ, \psi_D - 120^\circ\}$: both directions have $|\psi| > 90^\circ$, so their cosines are negative and $h < \varepsilon$ by (F2). The neighbours at $\psi_D \pm 60^\circ$ are not in S , and one of them has $30^\circ < |\psi| \leq 60^\circ$, so $\cos \psi \geq \frac{1}{2} > h$; by (F4) it is not in $\text{int } K$, so it lies in D , and (F3) gives $|\psi| < 30^\circ$, a contradiction.
- The two pairs on one side, $\{\psi_D + 60^\circ, \psi_D + 120^\circ\}$ and $\{\psi_D - 60^\circ, \psi_D - 120^\circ\}$, are excluded by (ii).

Hence $\deg y = 6$ implies $m_y \leq 2$, and $\deg y \leq 5$ implies $\deg y + m_y \leq 8$. For $\varepsilon_0 = 1/50$ the exceptional set has at most $6.89 \cdot 10^7 + 5106 + 465^2 < 7.1 \cdot 10^7$ points. $\square$

Remark 6.6 (regimes). [Theorem 6.1](#) uses nothing about rungs or τ , so it holds in Regions I, II, III, including $\tau = \sqrt{3}/2$, $\tau \leq 0$ and $\tau > 1$. [Theorem 6.2](#) needs $t > \sqrt{3}/2 + \varepsilon$ with $\varepsilon > 0$ fixed; in Region III with $\tau \geq \sqrt{3}/2 + \eta$, $0 < \eta \leq 1/50$, it holds with $\varepsilon_0 = \arctan(\eta/2)$, but then the exceptional count grows like $1/\eta$ and is not uniform as $\eta \rightarrow 0$. For $\tau \leq \sqrt{3}/2$ we only have [Theorem 6.1](#).

Remark 6.7 (local sharpness of [Theorem 6.1](#)). Consider a single disc, $K = B(w, \Delta_2)$ near y , with $|y - w| = \Delta_2 - h$. A neighbour at angle ψ from the outward radial direction satisfies $|q - w|^2 = (\Delta_2 - h)^2 + 1 + 2(\Delta_2 - h) \cos \psi$. So q lies on ∂K if and only if $\cos \psi = c_K(h) := h + (h^2 - 1)/(2(\Delta_2 - h))$, and $\text{dist}(q, K) \geq t$ if and only if $\cos \psi \geq c_D(h) := h + t + ((h + t)^2 - 1)/(2(\Delta_2 - h))$; moreover $c_D(0) = \tau$ and c_D is increasing in h .^[N] Two such D-feasible neighbours 60° apart need $c_D(h) \leq \cos 30^\circ$, so this local picture requires $\tau < \sqrt{3}/2$. It does occur for τ just below $\sqrt{3}/2$: with $w = 0$ and $\Delta_2 = 10^4$, take on $C(w, \Delta_2)$ chords $[p_-, p_+]$ of length 2 (every other chord position), let y be the chord midpoint, $e := (p_+ - p_-)/2$, $\nu := y/|y|$, $z_\pm := y \pm e/2 + (\sqrt{3}/2)\nu$ and $a_\pm := y \pm e/2 - (\sqrt{3}/2)\nu$. In 60-digit decimal arithmetic (script `r4_nb_exact.py`) one checks that $|yp_\pm| = |yz_\pm| = |ya_\pm| = 1$, all pairwise distances are at least 1, the $z_\pm$ lie on $C(w, \Delta_2 + t)$ with $t = 0.86598790\dots$ and $\tau = \sqrt{3}/2 - 5.0 \cdot 10^{-5}$, $\text{dist}(z_\pm, B(w, \Delta_2)) = t$, and $a_\pm \in \text{int } B(w, \Delta_2)$. So every such y has degree 6, two neighbours on ∂K and two D-feasible neighbours, i.e. $\deg y + m_y = 10$ as soon as $z_\pm \in S$. In this one-centre configuration the $z_\pm$ have no Δ_2 -partner; whether $\Theta(n)$ such points y with $z_\pm \in S$ can occur in a global configuration is open. So local arguments of the kind above cannot improve the *pointwise* constant 10. They do improve the average: by [Lemma 9.8](#), this configuration at a good point y (in the sense of [Section 9](#)) whose neighbours $z_\pm$ are diametral endpoints forces $t \in [0.75, 0.95]$, as in the example, and [Theorem 9.1](#) shows that the excess of such points over 8 can be absorbed by their lower neighbours $a_\pm$, which have $\deg + m \leq 7$.

7. THE DEGENERATE REGIME $\Delta > 1.94\Delta_2$

Proposition 7.1. *If $\Delta > 1.94\Delta_2$ and $n \geq 4$, then $\mu(\Delta_2) \leq n + 1$.*

Put $\kappa := (\Delta_2/\Delta)^2 < 1/1.94^2$. Note that this section does not use δ .

Step 1: no two disjoint diametral pairs. Suppose (x, x') and (z, z') are disjoint diametral pairs with midpoints m, m' . The four cross distances satisfy $\sum |ab|^2 = 2\Delta^2 + 4|m - m'|^2 \geq 2\Delta^2$,

and each is either at most Δ_2 or equal to Δ . If at most one equals Δ , the sum is at most $\Delta^2(1 + 3\kappa) < 2\Delta^2$,^[C10] a contradiction. So at least two cross distances equal Δ .

- If two disjoint cross pairs equal Δ , say $|xz| = |x'z'| = \Delta$, then z and x' are the two points of $C(x, \Delta) \cap C(z', \Delta)$, so x, z, z', x' form a rhombus of side Δ whose diagonals satisfy $d_1^2 + d_2^2 = 4\Delta^2$. One diagonal is at least $\sqrt{2}\Delta$, a contradiction.
- If two cross pairs sharing a point equal Δ , say $|xz| = |xz'| = \Delta$, then xzz' is an equilateral triangle of side Δ , and $x' \in C(x, \Delta)$ with $|x'z|, |x'z'| \leq \Delta$ lies on the 60° arc of $C(x, \Delta)$ between z and z' , strictly inside it. One of $|x'z|, |x'z'|$ is at least $2\Delta \sin 15^\circ$, and $(2 \sin 15^\circ)^2 = 2 - \sqrt{3} > \kappa$,^[C10] so that distance lies in (Δ_2, Δ) , which is impossible.

Step 2: the diameter graph is a star. A graph with no two disjoint edges is a star or a triangle. A triangle is impossible for $n \geq 4$: a fourth point is not diametral, so it is within Δ_2 of all three vertices, but $\Delta_2^2 < \Delta^2/3$ is less than the squared circumradius.^[C10] So $D = \{y\} \cup A$ with A the set of diametral partners of y ; put $B := X \setminus D \subseteq K$ and $W := \{y\} \cup A$.

Step 3: where the Δ_2 -pairs are.

- *Within B : none.* For $a \in A$, $K \subseteq B(y, \Delta_2) \cap B(a, \Delta_2)$, a lens whose bounding box has squared diagonal less than $0.24\Delta_2^2$.^[C10] So $\text{diam } K < \Delta_2$.
- *Within A : at most one.* $A \subseteq C(y, \Delta)$, and its pairwise distances are at most Δ_2 (a Δ -pair inside A would create a triangle in the diameter graph). A chord of $C(y, \Delta)$ of length at most $\Delta_2 < \Delta/1.94$ subtends an angle less than 30° at y . So A lies in an arc of angular width less than 60° , on which chord length is increasing in angular separation; a Δ_2 -pair in A must realise the largest pairwise distance, hence be the two extreme points of A .
- *Between y and A : none,* as those distances equal Δ .

Hence $\mu(\Delta_2) \leq 1 + I$, where $I := \#\{(b, w) : b \in B, w \in W, |bw| = \Delta_2\}$.

Step 4: $I \leq |B| + |W|$. We may assume K has nonempty interior: an open chord between two points of K lies in the interior of every disc $B(w, \Delta_2)$, so otherwise $|B| \leq |K \cap X| \leq 1$ and the bound is trivial. For $w \in X$ let $J_w := C(w, \Delta_2) \cap K$.

- *Each J_w is a single closed arc (possibly a point).* For $x \in X \setminus \{w\}$, $C(w, \Delta_2) \cap B(x, \Delta_2)$ is the closed arc $\{\cos(\theta - \theta_x) \geq |wx|/(2\Delta_2)\}$, of angular length at most π . For $p, q \in J_w$ we have $|pq| < \Delta_2 < 2\Delta_2$, so p, q are not antipodal and the minor arc of $C(w, \Delta_2)$ from p to q lies in every $B(x, \Delta_2)$, hence in K . Since $\text{diam } K < \Delta_2$, the angular spread of J_w is less than 60° , so J_w is a single closed arc.
- *No point is interior to two arcs.* Suppose b is a relative-interior point of a nondegenerate arc J_w and $b \in C(w', \Delta_2)$ with $w' \in X \setminus \{w\}$. The function $f(\theta) = |w + \Delta_2 e(\theta) - w'|^2 - \Delta_2^2 = |w - w'|^2 + 2\Delta_2 e(\theta) \cdot (w - w')$ is ≤ 0 on J_w (as $K \subseteq B(w', \Delta_2)$) and vanishes at b , so it has an interior local maximum with value 0. But every local maximum of this shifted cosine is its global maximum $|w - w'|^2 + 2\Delta_2|w - w'| > 0$, a contradiction.
- *Consequences.* Each nondegenerate J_w is a subarc of ∂K , and distinct nondegenerate arcs have disjoint relative interiors. So a point $b \in B$ lies on at most two nondegenerate W -arcs (one on each side); any further incidences of b come from degenerate arcs $J_w = \{b\}$.
- *Counting.* Let I_{nd} be the number of incidences from nondegenerate arcs. Then $I_{\text{nd}} \leq |B| + \#\{b : \text{two } W\text{-arcs meet at } b\}$. Mapping each such b to the W -arc that starts at b in the clockwise direction is injective, so $I_{\text{nd}} \leq |B| + \#\{w : J_w \text{ nondegenerate}\}$. Each degenerate arc contributes at most one incidence. Hence $I \leq |B| + |W|$.

Therefore $\mu(\Delta_2) \leq 1 + |B| + |A| + 1 = n + 1$, proving [Proposition 7.1](#).

8. ASSEMBLY: THE FORMALISED BOUND 15/11

The results of [Sections 3 to 7](#) already give the bound of the previous version of this paper, which is the one formalised in Lean ([Section 14.2](#)). [Theorem 1](#) is proved in [Sections 9 and 10](#) by adding one estimate to this argument.

Theorem 8.1. *There is an absolute constant C_0 ($C_0 = 3N_0 + 2 \cdot 10^8$ suffices) such that every n -point set $X \subset \mathbb{R}^2$ satisfies*

- (a) $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{15}{11}n + C_0$;
- (b) *if moreover $\Delta \leq 1.94\Delta_2$ and $\tau(X) \geq \sqrt{3}/2 + 3/100$, then $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{4}{3}n + C_0$.*

Part (b) is superseded by [Theorem 1\(a\)](#); we keep it because its proof is pointwise ([Theorem 6.2](#)).

regime ($\delta = 1, \beta = 1/100$)	$e(S) \leq$	$\deg y + m_y \leq$	$\min\{\mu(\Delta_2), \mu(\delta)\} \leq$
degenerate: $\Delta > 1.94\Delta_2, n \geq 4$	—	—	$n + 1$ (Proposition 7.1)
nondegenerate, $\Delta_2 < 10^4$	—	—	$3n < 3N_0$ (Section 3)
I: $\tau \leq \sqrt{3}/2 - 2\beta$ (incl. $\tau \leq 0$)	$s + C$	10 (Theorem 6.1)	$\frac{15}{11}n + C$
III: $\sqrt{3}/2 - 2\beta < \tau < \sqrt{3}/2 + 3\beta$	$s + p + C$	10 (Theorem 6.1)	$\frac{15}{11}n + C$
II: $\sqrt{3}/2 + 3\beta \leq \tau \leq 1$	$s + p + C$	8 (Theorem 6.2)	$\frac{4}{3}n + C$
II: $\tau > 1$ ($\Leftrightarrow t > 1$)	$s + C$	8 (Theorem 6.2)	$\frac{4}{3}n + C$

TABLE 1. The regimes in the proof of [Theorem 8.1](#). In the last column $C < 2 \cdot 10^8$; the R -bounds hold for all but fewer than $7.1 \cdot 10^7$ points of R .

The τ -intervals partition $\mathbb{R}$, the split at $\Delta = 1.94\Delta_2$ is exhaustive, and the cases $n \leq 3$ (or X with a single distance) are trivial.

Constants. By [Proposition 4.3](#), $C_{\text{bad}} < 1.83 \cdot 10^7$. By [Section 5](#), in every τ -regime of the nondegenerate regime with $\Delta_2 \geq 10^4$,

$$e(S) \leq s + p + C_e, \quad C_e := 2C_{\text{bad}} + 81 < 3.7 \cdot 10^7$$

(Region I and $\tau > 1$: C_{bad} ; Region II: $C_{\text{bad}} + 5$; Region III: $C_{\text{bad}} + 6$ for $\tau \neq \sqrt{3}/2$ and $2C_{\text{bad}} + 81$ for $\tau = \sqrt{3}/2$). By [Theorem 6.1](#), $\deg y + m_y \leq 10$ for all but $C' < 7.1 \cdot 10^7$ points $y \in R$, and in Region II, by [Theorem 6.2](#), $\deg y + m_y \leq 8$ for all but $C' < 7.1 \cdot 10^7$ points. [Lemma 3.5](#) gives

$$\begin{aligned} \min\{\mu(\Delta_2), \mu(\delta)\} &\leq \frac{15}{11}n + C_e + C' \quad \text{in general, and} \\ \min\{\mu(\Delta_2), \mu(\delta)\} &\leq \frac{4}{3}n + C_e + C' \quad \text{in Region II,} \end{aligned}$$

and $C_e + C' < 1.1 \cdot 10^8 < 2 \cdot 10^8$. The degenerate regime gives $n + 1 \leq \frac{4}{3}n + 1$, and the small cases give at most $3N_0$. So [Theorem 8.1](#) holds with $C_0 = 3N_0 + 2 \cdot 10^8$. Part (b) is the Region II row: its hypotheses are $\Delta \leq 1.94\Delta_2$ and $\tau \geq \sqrt{3}/2 + 3\beta$, and if $\Delta_2 < 10^4$ the small-case bound applies. $\square$

Remark 8.2. (a) *The bottleneck of [Theorem 8.1](#) is the R -term.* The estimate $e(S) \leq s + p + O(1)$ is sharp enough for $\frac{4}{3}$ ([Remark 3.6](#)); the value $\frac{15}{11}$ comes only from the pointwise bound $\deg y + m_y \leq 10$ of [Theorem 6.1](#) in Regions I and III. By [Remark 6.7](#), this pointwise bound cannot be improved by local arguments. [Section 9](#) replaces it by the average bound 8, which removes this bottleneck.

- (b) *What remains for 9/7.* If the average of $\deg y + m_y$ over R were $6 + o(1)$, Region I would give $\frac{9}{7}$, but with the p -term of [Section 5](#) Regions II and III would still give only $\frac{4}{3}$ ([Remark 3.6](#)). So reaching $\frac{9}{7}$ with this method also requires sharpening the rung count; in Region II this is done in [Section 11](#) (the p -term is halved, which allows $\frac{5}{4}$), and Region III is discussed in [Section 15](#); for $\tau > 1$, $\frac{9}{7}$ is reached by a different argument ([Section 12](#)).
- (c) *Consistency with the lower bound.* The construction of `ienjoymath` [[17](#)] for $L \geq 9/7$ lies in Region I ($t/\delta \rightarrow 0$). In it $e(S) = |S|$, every point of S has Δ_2 -degree 2 or 4 (so $Q = \emptyset$ and $\mu(\Delta_2) = \frac{3}{2}|S|$), and every point of R has $m_y = 0$ and $\deg y + m_y \leq 6$; we checked the last statement numerically for $n = 60, 120, 250$ ([Appendix A](#)). This is consistent with [Theorem 6.1](#), which is only an upper bound; in Region I the gap between $\frac{9}{7}$ and the bounds $\frac{15}{11}, \frac{4}{3}, \frac{21}{16}$ is exactly the gap between the average 6 and the averages 10, 8, 7 in the R -term.

9. THE AVERAGE WEIGHT OF R : PROOF OF THEOREM 1(A)

For $y \in R$ we call $v(y) := \deg y + m_y$ the *weight* of y . By [Lemma 3.3](#), $\mu(\delta) = e(S) + \frac{1}{2} \sum_{y \in R} v(y)$, and $v(y) \leq 12$. [Theorem 6.1](#) gives $v(y) \leq 10$ for all but $O(1)$ points. In this section we prove that the average weight is at most 8.

Theorem 9.1. *In the nondegenerate regime with $\Delta_2 \geq 10^4$,*

$$\sum_{y \in R} v(y) \leq 8r + C_3, \quad C_3 := 1.1 \cdot 10^9.$$

No assumption on τ is made. [Theorem 9.1](#) has been through two independent referee passes; both found it correct, and their cosmetic corrections are incorporated below. The proof has two ideas. First, a point of weight at least 9 at which ∂K is flat has one of four local shapes, and in each shape a K-row neighbour and a diametral neighbour have a height difference that [Lemma 2.2\(b\)](#) forces to be within 0.05 of t ([Lemma 9.8](#)); so the shape of the heavy points is determined by t . Second, every such point has one or two neighbours deeper inside K that have weight at most 7 and belong to no other heavy point; each heavy point passes its excess over 8 to them.

9.1. Two lemmas with parameters. The proofs of [Lemmas 6.3](#) to [6.5](#) work with other parameters. For $\Lambda \geq 1$ and $y \in R \setminus D$ with $h \leq 1$, if the perimeter of K is at least 2Λ , let γ_y^Λ be the closed arc of ∂K of points at arc-length distance at most Λ from p_0 ; so $\gamma_y = \gamma_y^{232}$.

Lemma 9.2 (packing in thin K). *Let $1 \leq \ell \leq \Delta_2/100$ and $\Lambda > 115\ell$. Suppose $p, b \in \partial K$ satisfy $|pb| \leq \ell$, and both arcs of ∂K between p and b have length greater than Λ . Then K lies in a rectangle of length less than 250ℓ and width less than 5.4ℓ , and hence $|X \setminus D| < (250\ell + 1)(5.4\ell + 1) \cdot 4/\pi$. For $\ell = 4$ this is less than $3 \cdot 10^4$, and for $\ell = 7$ less than 10^5 .*

Proof. We follow the proof of [Lemma 6.3](#), with p in the role of both p_0 and y . Let ν_p, ν_b be outer normals of K at p, b , with supporting half-planes $H_p, H_b \supseteq K$ and boundary lines ℓ_p, ℓ_b . If $\nu_b = \nu_p$, then $\ell_p = \ell_b$ as in the proof of [Lemma 6.3](#), and $[p, b] \subseteq K \cap \ell_p \subseteq \partial K$ is a boundary arc of length at most $\ell < \Lambda$, a contradiction. If $\nu_b \neq \pm \nu_p$, then $Z := H_p \cap H_b$ is a cone with apex A and opening $\theta \in (0, \pi)$, and p, b lie on its two boundary rays. As in the proof of [Lemma 6.3](#), either $[p, b] \subseteq \partial K$ (a contradiction as before) or one of the two arcs from p to b has length at most $|Ap| + |Ab|$. In the triangle Apb the law of sines gives $|Ap| + |Ab| = |pb|(\sin \angle p + \sin \angle b)/\sin \theta \leq |pb| \cdot 2 \cos(\theta/2)/\sin \theta = |pb|/\sin(\theta/2)$ (and $|Ap| + |Ab| = |pb|$ if $A \in \{p, b\}$). For $\theta \geq 1^\circ$ this is at most $\ell/\sin(0.5^\circ) < 114.6\ell < \Lambda$, a contradiction. So $\theta < 1^\circ$, or $\nu_b = -\nu_p$ (a strip; put $\theta := 0$).

Now apply the fatness step of the proof of [Lemma 6.3](#) with p in place of y . The cross-section of Z at the level of p has length $w_0 = \text{dist}(p, \ell_b)/\cos(\theta/2) \leq \ell/\cos(0.5^\circ) < 1.0001\ell$, since

$p \in \ell_p$. With L and W as there, $W \leq w_0 + 2 \tan(\theta/2)L$ and $L \leq 2\Delta_2$. If $L \geq 250\ell$, then $\tan \varphi^* = 2W/L \leq 2w_0/(250\ell) + 4 \tan(0.5^\circ) < 0.0430$, and

$$\frac{2\Delta_2 \cos \varphi^* - W}{\Delta_2} \geq 2\left(1 - \frac{\tan^2 \varphi^*}{2}\right) - 4 \tan(0.5^\circ) - \frac{w_0}{\Delta_2} > 1.998 - 0.035 - 0.011 > 1.94,$$

which contradicts $|w_{\text{bot}} - w_{\text{top}}| \leq \Delta \leq 1.94\Delta_2$ exactly as in that proof. Hence $L < 250\ell$ and $W < 1.0001\ell + 500\ell \tan(0.5^\circ) < 5.4\ell$. The disjoint discs of radius $\frac{1}{2}$ about the points of $X \setminus D \subseteq K$ lie in a rectangle of area less than $(250\ell + 1)(5.4\ell + 1)$.^[N3] $\square$

The proof of [Lemma 6.4](#), with 232 replaced by Λ , gives the following: *the number of $y \in R \setminus D$ with $h \leq 1$ and $\text{turn}(\gamma_y^\Lambda) > \varepsilon_0$ is less than $2\pi(2\Lambda + 3)^2/\varepsilon_0$. (If $b \in \gamma_y^\Lambda$ then $|b - y| \leq \Lambda + 1$, and at most $(2\Lambda + 3)^2$ points of R lie within that distance of b .)*

Lemma 9.3 (frame). *Let $\Lambda \geq 100$, $1 \leq \rho \leq \Lambda/2$, $0 < \varepsilon_0 \leq 1/50$ with $\Lambda(1 - \cos \varepsilon_0) < 1$, and $\varepsilon := \tan \varepsilon_0$. Let $y \in R \setminus D$ with $h \leq 1$, and suppose that the perimeter of K is at least 2Λ , that $\text{turn}(\gamma_y^\Lambda) \leq \varepsilon_0$, and that $\partial K \cap B(y, \rho) \subseteq \gamma_y^\Lambda$. Use the coordinates of [Lemma 6.5](#) (origin p_0 , first axis t_0 , second axis n_0 , so $y = (0, -h)$), and write $q = (q_x, q_z)$. Then there is a concave function f on $[-(\Lambda - 1), \Lambda - 1]$ with $f(0) = 0$ and $-\varepsilon|x| \leq f(x) \leq 0$, whose graph is contained in γ_y^Λ and contains $\partial K \cap B(y, \rho)$, and $U_\rho := B(y, \rho) \cap \{z < f(x)\} \subseteq \text{int } K$. For every $q \in X \cap B(y, \rho)$:*

- (G1) *if $q \notin \text{int } K$, then $q_z \geq f(q_x)$;*
- (G2) *if $q \in S \setminus D$, then $q_z = f(q_x)$;*
- (G3) *if $q \in D$, then $q_z \geq f(q_x) + t$;*
- (G4) *if $q \in \text{int } K$, then $q_z < f(q_x)$.*

Moreover:

- (G5) *every outer normal of K at a point of γ_y^Λ makes an angle at most ε_0 with n_0 . In particular, if y' also satisfies the hypotheses and $|y - y'| \leq \rho - 1$, then the vectors n_0 of y and y' make an angle at most ε_0 .*

Proof. This is the proof of [Lemma 6.5](#) with 232, 231 and $B(y, 1)$ replaced by Λ , $\Lambda - 1$ and $B(y, \rho)$: the hypothesis $\Lambda(1 - \cos \varepsilon_0) < 1$ gives $\Lambda \cos \varepsilon_0 > \Lambda - 1$, and the x -range $[-\rho, \rho]$ of $B(y, \rho)$ lies inside $[-(\Lambda - 1), \Lambda - 1]$. (G1), (G2) and (G4) are proved as (F1), (F2) and (F4). For (G3): if $q \in D$ had $q_z < f(q_x)$, then $q \in U_\rho \subseteq K$, whereas $D \cap K = \emptyset$ by [Lemma 2.1\(c\)](#); and the graph point below q lies in K , so $t \leq \text{dist}(q, K) \leq q_z - f(q_x)$. (G5): the normal set $N(\gamma_y^\Lambda)$ is an arc of length at most ε_0 containing n_0 . If y' is as stated, its nearest boundary point p'_0 satisfies $|p'_0 - y| \leq 1 + |y - y'| \leq \rho$, so $p'_0 \in \partial K \cap B(y, \rho) \subseteq \gamma_y^\Lambda$, and the vector n_0 of y' is an outer normal of K at p'_0 . $\square$

[Lemma 6.5](#) is essentially the case $\Lambda = 232$, $\rho = 1$. In the situation of [Lemma 9.3](#) we write a δ -neighbour q of y as $q = y + (\sin \psi, \cos \psi)$ and call $c(q) := \cos \psi = q_z + h$ its *height*. By (G1)–(G4) and [Section 6.1\(b\)](#), every neighbour of y is of exactly one of three kinds:

- a *K-neighbour* ($q \in S \setminus D$): $c(q) = h + f(q_x) \in [h - \varepsilon, h]$, and $|\psi| < \pi/2 + \varepsilon_0$ as in (F1);
- a *D-neighbour* ($q \in D$, in S or in R): $c(q) \geq h + t - \varepsilon$;
- an *interior neighbour* ($q \in \text{int } K$, hence $q \in R \setminus D$): $c(q) < h$.

The neighbours of y in S are its K-neighbours and its D-neighbours in S . As in the proof of [Theorem 6.1](#), $m_y \leq 4$.

Lemma 9.4 (diametral points are nearly level). *In the situation of [Lemma 9.3](#), let $z \in D \cap B(y, \rho)$ with $|z_x| \leq \xi$ and $z_z \leq \zeta$, where $\xi, \zeta \leq \rho$. Let z^* be a diametral partner of z and $u := (z^* - z)/\Delta = (a_1, a_2)$. Then $a_2 < 0$ and $|a_1| < \kappa_D |a_2|$ with $\kappa_D := ((\Lambda - 1)\varepsilon + \zeta)/(\Lambda - 1 - \xi)$.*

Consequently, any two points $z, z' \in D \cap B(y, \rho)$ satisfying these bounds have $|z_z - z'_z| < \kappa_D |z_x - z'_x|$.

Proof. Every $k \in K$ has $|k - z^*| \leq \Delta_2 < \Delta = |z - z^*|$, so $(k - z) \cdot u > |k - z|^2 / (2\Delta) \geq 0$. The graph point g directly below z lies in K and strictly below z by (G3), and $(g - z) \cdot u = (f(z_x) - z_z)a_2 > 0$ gives $a_2 < 0$. Put $c := -a_2 > 0$. The graph points $g_\pm := (\pm(\Lambda - 1), f(\pm(\Lambda - 1))) \in K$ give $(g_\pm - z) \cdot u > 0$, i.e. $(\pm(\Lambda - 1) - z_x)a_1 > (f(\pm(\Lambda - 1)) - z_z)c \geq -((\Lambda - 1)\varepsilon + z_z)c$, where $(\Lambda - 1)\varepsilon + z_z > 0$ by (G3). The sign $+$ gives $a_1 > -c((\Lambda - 1)\varepsilon + \zeta)/(\Lambda - 1 - \xi)$ and the sign $-$ gives $a_1 < c((\Lambda - 1)\varepsilon + \zeta)/(\Lambda - 1 - \xi)$. For the consequence, apply Lemma 2.2(a), which holds for every point of D , to z and to z' : $(z' - z) \cdot u_z > 0$ and $(z - z') \cdot u_{z'} > 0$ give $z'_z - z_z < \kappa_D |z'_x - z_x|$ and $z_z - z'_z < \kappa_D |z'_x - z_x|$. $\square$

9.2. Good points. In the rest of this section $\varepsilon_0 = 1/50$, $\varepsilon = \tan \varepsilon_0 < 0.02001$, $\Lambda = 464$ and $\rho = 3$; then $\Lambda(1 - \cos \varepsilon_0) < 0.1$. A point $y \in R \setminus D$ with $h \leq 1$ is *good* if the perimeter of K is at least 928, $\text{turn}(\gamma_y^{464}) \leq \varepsilon_0$, and $\partial K \cap B(y, 3) \subseteq \gamma_y^{464}$. At a good point Lemma 9.3 applies with $\Lambda = 464$, $\rho = 3$, and Lemma 9.4 gives the constant $\kappa_D < (463\varepsilon + 3)/460 < 0.0267$ on $D \cap B(y, 3)$, and $\kappa_D < (463\varepsilon + 1)/462 < 0.0224$ for D-neighbours of y (for which $|z_x| \leq 1$ and $z_z \leq 1 - h \leq 1$).^[N3] By (G5), two good points y, y' with $|y - y'| \leq 2$ have frames whose axes differ by at most ε_0 .

Lemma 9.5. *Fewer than $N_g := 2.74 \cdot 10^8$ points $y \in R$ with $v(y) \geq 7$ are not good.*

Proof. Points of $R \cap D$ and points of $R \setminus D$ with $h > 1$ have $v(y) \leq 6$ (Section 6.1(a),(c)). If the perimeter of K is less than 928, then $\text{diam } K < 464$ and $|X \setminus D| \leq (2 \cdot 464 + 1)^2 < 8.7 \cdot 10^5$. The points with $\text{turn}(\gamma_y^{464}) > \varepsilon_0$ number less than $2\pi \cdot 931^2 \cdot 50 < 2.73 \cdot 10^8$. If some $b \in \partial K \cap B(y, 3)$ is not on γ_y^{464} , then $|p_0 b| \leq 4$ and both arcs from p_0 to b are longer than $464 > 115 \cdot 4$, so Lemma 9.2 with $\ell = 4$ gives $|X \setminus D| < 3 \cdot 10^4$.^[N3] $\square$

9.3. Heavy points. A point $y \in R$ is *heavy* if $v(y) \geq 9$. Neighbours of y are pairwise at least 60° apart as seen from y .

Lemma 9.6 (pairs). *Let y be good. If two K-neighbours of y are at $\psi_1 < \psi_2$, then $|\sin \frac{\psi_1 + \psi_2}{2}| \leq \varepsilon$, so $|\psi_1 + \psi_2| < 2.30^\circ$. If two D-neighbours (in S or not) are at $\psi_1 < \psi_2$, then $|\sin \frac{\psi_1 + \psi_2}{2}| < 0.0448$, so $|\psi_1 + \psi_2| < 5.62^\circ$. Consequently y has at most two K-neighbours and at most two D-neighbours.*

Proof. K- and D-neighbours have $|\psi| < 90^\circ + \varepsilon_0$ (D-neighbours have $c > 0$), so $\psi_2 - \psi_1 \in [60^\circ, 180^\circ + 2\varepsilon_0]$ and $\sin \frac{\psi_2 - \psi_1}{2} \geq \frac{1}{2}$. Since $|c_1 - c_2| = 2|\sin \frac{\psi_1 + \psi_2}{2}| \sin \frac{\psi_2 - \psi_1}{2}$, we get $|\sin \frac{\psi_1 + \psi_2}{2}| \leq |c_1 - c_2|$. For K-neighbours $|c_1 - c_2| \leq \varepsilon < \sin 1.15^\circ$. For D-neighbours, Lemma 9.4 gives $|c_1 - c_2| < 0.0224 \cdot 2 < \sin 2.81^\circ$. Three neighbours of one kind would satisfy $\psi_1 \approx -\psi_2 \approx \psi_3$ up to $2 \cdot 5.62^\circ < 60^\circ$, which is impossible.^[N3] $\square$

Lemma 9.7 (classification). *Let y be good and heavy. Then $(\deg y, m_y) \in \{(6, 4), (5, 4), (6, 3)\}$, and y is of one of the following types (angles are values of ψ).*

- (F6) (6, 4): *K-neighbours within ε_0 of $\pm 90^\circ$, D-neighbours in S within ε_0 of $\pm 30^\circ$, and interior neighbours within ε_0 of $\pm 150^\circ$; $v(y) = 10$.*
- (F5) (5, 4): *the same four neighbours in S , and one interior neighbour with $\psi \in (150^\circ - \varepsilon_0, 210^\circ + \varepsilon_0)$; $v(y) = 9$.*
- (HF) (6, 3): *the six neighbours form a regular hexagon within 2.81° of $\{\pm 30^\circ, \pm 90^\circ, \pm 150^\circ\}$; the neighbours in S are K-neighbours near $\pm 90^\circ$ and one D-neighbour near $+30^\circ$ or -30° (HF1), or D-neighbours near $\pm 30^\circ$ and one K-neighbour near $+90^\circ$ or -90° (HF2); the neighbours near $\pm 150^\circ$ are interior; $v(y) = 9$.*

(cap) (6, 3): the six neighbours form a regular hexagon within 1.15° of $\{0^\circ, \pm 60^\circ, \pm 120^\circ, 180^\circ\}$; K-neighbours near $\pm 60^\circ$, a D-neighbour in S near 0° , interior neighbours near $\pm 120^\circ$ and 180° ; $v(y) = 9$.

We call $F6$, $F5$ and HF fans.

Proof. Since $m_y \leq 4$ and $m_y \leq \deg y \leq 6$, $v \geq 9$ leaves exactly the three pairs listed. A neighbour with $|\psi| > 90^\circ + \varepsilon_0$ is neither a K-neighbour nor (having $c < 0 < h + t - \varepsilon$) a D-neighbour, so it is interior.

$m_y = 4$. The four neighbours in S lie in the open arc $|\psi| < 90^\circ + \varepsilon_0$ and are pairwise at least 60° apart, so they are within ε_0 of $-90^\circ, -30^\circ, 30^\circ, 90^\circ$. By Lemma 9.6 two are K- and two are D-neighbours, and each pair of the same kind is mirror-symmetric up to 5.62° ; so the pairs are $\{\pm 90^\circ\}$ and $\{\pm 30^\circ\}$. If the K-neighbours were near $\pm 30^\circ$, then $h \geq \cos 31.15^\circ > 0.85$, while a D-neighbour near $\pm 90^\circ$ has $c \leq \sin \varepsilon_0 < 0.02 < h + t - \varepsilon$; impossible. This gives the neighbours in S of $F6$ and $F5$. For $\deg y = 6$ the six neighbours are exactly 60° apart, so the remaining two are within ε_0 of $\pm 150^\circ$. For $\deg y = 5$ the fifth neighbour is at least 60° from both K-neighbours, so $\psi \in (150^\circ - \varepsilon_0, 210^\circ + \varepsilon_0)$.

(6, 3). The six neighbours form a regular hexagon $\{\varphi + 60^\circ k\}$. By Lemma 9.6, two of the three neighbours in S are of the same kind and form a mirror pair up to 5.62° (2.30° for a K-pair); so φ is within 2.81° (1.15°) of a multiple of 30° . Neighbours in S have $|\psi| < 90^\circ + \varepsilon_0$, so they lie near $0^\circ, \pm 60^\circ$ in the first case and near $\pm 30^\circ, \pm 90^\circ$ in the second. The mirror pair is $\{\pm 60^\circ\}$, $\{\pm 30^\circ\}$ or $\{\pm 90^\circ\}$, and the third neighbour in S is of the other kind. This leaves six cases.

- K near $\pm 60^\circ$, D near 0° : the cap.
- D near $\pm 60^\circ$, K near 0° : the K-neighbour gives $h \geq \cos 2.81^\circ > 0.998$, so a D-neighbour needs $c \geq h + t - \varepsilon > 0.978$; but the D-neighbours have $c \leq \cos 57.19^\circ < 0.55$. Impossible.
- K near $\pm 30^\circ$, D near $\pm 90^\circ$: $h \geq \cos 31.15^\circ > 0.85$; the D-neighbour has $c \leq \sin 2.81^\circ < 0.05$ but needs $c \geq h + t - \varepsilon > 0.82$. Impossible.
- D near $\pm 30^\circ$, K near $+90^\circ$ or -90° : HF2.
- K near $\pm 90^\circ$, D near $+30^\circ$ or -30° : HF1.
- D near $\pm 90^\circ$, K near $\pm 30^\circ$: the D-neighbours give $h \leq c - t + \varepsilon \leq \sin 2.81^\circ + \varepsilon < 0.07$, the K-neighbours give $h \geq \cos 32.81^\circ > 0.84$. Impossible.

In HF and in the cap, the neighbours near $\pm 150^\circ$, resp. $\pm 120^\circ$ and 180° , have $|\psi| > 90^\circ + \varepsilon_0$ and are interior.^[N3] $\square$

Lemma 9.8 (t is pinned). *If some good point is a fan, then $t \in [0.75, 0.95]$. If some good point is a cap, then $t \in [0.44, 0.56]$. In particular, fans and caps do not both occur in one set X .*

Proof. Let y be a good fan or cap. By Lemma 9.7, y has a K-neighbour k and a D-neighbour $z \in S$ whose heights differ by $\Delta c := c(z) - c(k)$ with

- $\Delta c \in [\cos 32.81^\circ - \sin 2.81^\circ, \cos 27.19^\circ + \varepsilon] \subseteq [0.7914, 0.9096]$ for a fan (using $c(k) \geq h - \varepsilon \geq -\varepsilon$), and
- $\Delta c \in [\cos 1.15^\circ - \cos 58.85^\circ, 1 - \cos 61.15^\circ] \subseteq [0.4824, 0.5175]$ for a cap.

Since $k \in \partial K \cap B(y, 1) \subseteq \gamma_y^{464}$, (G5) shows that $u_k = (w_k - k)/\Delta_2$, an inward normal at k , equals $(\sin \beta', -\cos \beta')$ with $|\beta'| \leq \varepsilon_0$. With $z - k = (\Delta x, \Delta c)$ and $|\Delta x| \leq 2$,

$$(z - k) \cdot u_k = \Delta x \sin \beta' - \Delta c \cos \beta' \leq 2 \sin \varepsilon_0 - 0.48 \cos \varepsilon_0 < 0.$$

So z is not normal for k , and by Lemma 2.2(b), $|z - w_k| = \Delta = \Delta_2 + t$. Expanding $|z - w_k|^2 = |z - k|^2 - 2\Delta_2(z - k) \cdot u_k + \Delta_2^2$ gives exactly

$$(z - k) \cdot u_k = -t + \frac{d^2 - t^2}{2\Delta_2}, \quad d := |z - k| \leq 2.$$

(For $d = 1$ this is [Lemma 2.2\(c\)](#).) Hence $t = \Delta c \cos \beta' - \Delta x \sin \beta' + (d^2 - t^2)/(2\Delta_2)$; this gives $t < 1$, and since $\Delta_2 \geq 10^4$,

$$|t - \Delta c| \leq \Delta c(1 - \cos \varepsilon_0) + 2 \sin \varepsilon_0 + 2 \cdot 10^{-4} < 0.0404.$$

This gives $t \in [0.751, 0.950]$ for a fan and $t \in [0.442, 0.558]$ for a cap.^[N3] $\square$

Remark 9.9. [Lemma 9.8](#) is where the global structure enters: the local configuration of [Remark 6.7](#) is realisable only if t is close to $\frac{\sqrt{3}}{2}$. In 60-digit arithmetic the configuration of [Remark 6.7](#) has $t = 0.86598\dots$, and a chain of caps built on chords of length $\sqrt{3}$ of one circle has $t = 0.49996\dots$ (`r6_n3_examples.py`); both lie in the windows of [Lemma 9.8](#). Outside $[0.44, 0.56] \cup [0.75, 0.95]$, all but fewer than N_g points of R have $v(y) \leq 8$.

9.4. Charging. Each good heavy point y sends one unit of charge to each of its *receivers*: an F6 point to its two interior neighbours near $\pm 150^\circ$; an F5 point to its fifth (interior) neighbour; an HF point to its interior neighbour near $+150^\circ$ (any fixed choice of one of the two works below); a cap to its interior neighbour near 180° . Other points send nothing.

Lemma 9.10 (receivers). *Let a be a receiver of a good heavy point y .*

- (a) $a \in R \setminus D$, and a is not heavy: if y is a fan, then $m_a \leq 1$ and $v(a) \leq 7$; if y is a cap, then $m_a = 0$ and $v(a) \leq 6$.
- (b) a is a receiver of no other good heavy point.

Proof. (a) Receivers are interior neighbours, so they lie in $\text{int } K$ and hence in $R \setminus D$. Let y be a fan. The receiver has $\psi_a \in [147^\circ, 213^\circ]$, so $a_z = \cos \psi_a - h \leq -\cos 33^\circ < -0.838$. A neighbour q of a in S lies in $B(y, 2)$ and outside $\text{int } K$, so $q_z \geq f(q_x) \geq -2\varepsilon > -0.041$ by (G1). Hence $(q - a)_z > 0.797$: as seen from a , q lies within 37.2° of the upward vertical. So does y , since $(y - a)_z = -\cos \psi_a \geq \cos 33^\circ$; and $y \in R$. Three neighbours of a pairwise at least 60° apart do not fit into a window of width 74.4° , so a has at most one neighbour in S , and $v(a) \leq 6 + 1$. Let y be a cap. Then $h \geq c(k) \geq \cos 61.15^\circ > 0.48$ for its K-neighbours k , and the receiver b has ψ_b within 1.15° of 180° , so $b_z < -1.48$. A neighbour $q \in S$ of b would satisfy $q_z \geq -0.041$ and $q_z \leq b_z + 1 < -0.48$; so $m_b = 0$.

(b) Suppose a receives from good heavy points $y \neq y'$. By [Lemma 9.8](#) both are fans or both are caps. Since $|y - y'| \leq 2$, their frames differ by at most ε_0 ; we measure directions from a in the frame of y . If both are caps, then as seen from a , y lies within 1.15° and y' within $1.15^\circ + \varepsilon_0 < 2.3^\circ$ of the upward vertical; but they are distinct neighbours of a , hence at least 60° apart. Impossible. Let both be fans. The directions α, α' of y, y' from a satisfy $|\alpha| \leq 33^\circ$, $|\alpha'| \leq 33^\circ + \varepsilon_0 < 34.2^\circ$ and $|\alpha - \alpha'| \geq 60^\circ$. Hence $|y' - y| = 2 \sin \frac{|\alpha' - \alpha|}{2} \in [1, 1.107]$, and $y' - y$ makes an angle at most $\frac{1}{2}|\alpha + \alpha'| \leq 4.2^\circ$ with the horizontal. Now y has a neighbour q in the direction within 2.81° of $\pm 90^\circ$ on the side of y' (F6 and F5 have K-neighbours near $\pm 90^\circ$, and HF points have six neighbours), and $|y' - q| \leq 0.107 + 2 \sin 3.51^\circ < 0.25 < 1$.^[N3] Distinct points of X are at least 1 apart, so $y' = q$. If y is F6, F5 or HF1, then $q \in S$, whereas $y' \in R$; impossible. So y is HF2 and y' is its neighbour near $+90^\circ$, say (the other case is symmetric), which is not in S ; by the same argument with the roles exchanged, y' is HF2 and y is its neighbour near -90° . The hexagons are regular, so the D-neighbour z of y near $+30^\circ$ and the D-neighbour z' of y' near -30° are both at distance 1 from y and from y' . Both lie above the line yy' (their heights above y , resp. y' , exceed 0.8, while $|y_z - y'_z| < 0.09$), and only one point at distance 1 from both y and y' lies on that side; hence $z = z'$. Then z has four distinct neighbours: y, y' , the D-neighbour of y near -30° , and the D-neighbour of y' near $+30^\circ$. But every point of D has at most three neighbours ([Section 6.1\(a\)](#)): by [Lemma 2.2\(a\)](#), $X \setminus \{z\}$ lies in an open half-plane bounded by a line through z . This contradiction proves (b). $\square$

Proof of Theorem 9.1. Give each $y \in R$ the charge $v(y)$ and apply the rule above; the total charge is unchanged. A good heavy point is not a receiver (receivers have $v \leq 7$), and it ends with charge $10 - 2$ (F6) or $9 - 1$ (F5, HF, cap), i.e. 8. A receiver receives exactly one unit (Lemma 9.10(b)) and ends with at most $7 + 1 = 8$. Every other point of R keeps its weight, which is at most 8 unless the point is heavy and not good; by Lemma 9.5 there are fewer than N_g such points, and each has weight at most 12 (Lemma 3.3). Hence $\sum_{y \in R} v(y) \leq 8r + 4N_g < 8r + 1.1 \cdot 10^9$. $\square$

9.5. Proof of Theorem 1(a). If $n \leq 3$ or X has only one distance there is nothing to prove. In the degenerate regime $\min\{\mu(\Delta_2), \mu(\delta)\} \leq n+1 \leq \frac{4}{3}n+1$ (Proposition 7.1). In the nondegenerate regime with $\Delta_2 < 10^4$ the bound $3n < 3N_0$ of Section 3 applies. In the nondegenerate regime with $\Delta_2 \geq 10^4$, Section 5 gives $e(S) \leq s + p + C_e$ with $C_e < 3.7 \cdot 10^7$ in every τ -regime (see Section 8; the p -term is needed only in Regions II and III, Sections 5.3 and 5.4), and Theorem 9.1 gives $\sum_{y \in R} v(y) \leq 8r + C_3$. Lemma 3.7(a) yields

$$\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{4}{3}n + \frac{1}{3}(C_e + \frac{1}{2}C_3) < \frac{4}{3}n + 2 \cdot 10^8.$$

So Theorem 1(a) holds with $C_0 = 3N_0 + 2 \cdot 10^8$. $\square$

Remark 9.11. (a) The two inputs of the proof are of different status: the bound $e(S) \leq s + p + C$ in Regions II and III (Sections 5.3 and 5.4) is part of the formalised Theorem 8.1, while Theorem 9.1 is not formalised.

- (b) With the rung count of Section 5, the value $\frac{4}{3}$ is the limit of the method in Regions II and III, whatever the average weight (Remark 3.6). In Region I and for $\tau > 1$ there is no p -term, and Section 10 lowers the average weight to 7; in Region II, Section 11 halves the p -term and also lowers the average weight to 7.
- (c) In the construction of ienjoymath [17], $t \rightarrow 0$, so by Lemma 9.8 there are no good heavy points; indeed every point of R has weight at most 6 there (Remark 8.2).

10. REGION I AND $\tau > 1$: PROOF OF THEOREM 1(B)

In Region I and for $\tau > 1$ there is no p -term ($e(S) \leq s + C_{\text{bad}}$, Sections 5.1 and 5.2), so every improvement of the average weight improves the bound (Remark 3.6). We lower the average from 8 to 7. Region II, where a p -term is present, is treated in Section 11.

Theorem 10.1. *Assume the nondegenerate regime with $\Delta_2 \geq 2 \cdot 10^{15}$, and either $\tau \leq \sqrt{3}/2 - 2\beta$ (Region I) or $\tau > 1$. Then*

$$\sum_{y \in R} v(y) \leq 7r + C_4, \quad C_4 := 1.3 \cdot 10^{26}.$$

Status. The argument was checked in two independent referee passes. Both found the conclusion correct and found no counterexample, but found gaps in the text of an earlier write-up: the three-sender case in the band $|t - \frac{1}{2}| < \kappa$ was argued incorrectly (it is replaced by Lemma 10.8 below), the packing step for thin K did not follow as written (it is replaced by Lemma 9.2), several depths were misstated, and the error bookkeeping had not been written out (it was written out in the first referee pass, and is summarised in Lemma 10.3 and Table 2). All these repairs are incorporated here. Computer checks (Appendix A) are floating-point screenings and exact checks of special lattices; they are evidence, not part of the proof.

Proof of Theorem 1(b) for $\tau(X) \leq \sqrt{3}/2 - 1/50$ and for $\tau(X) > 1$, from Theorem 10.1. In the degenerate regime the bound is $n + 1 \leq \frac{21}{16}n + 1$. In the nondegenerate regime with $\Delta_2 < 2 \cdot 10^{15}$ we have $\Delta < 3.9 \cdot 10^{15}$, so $n \leq (2\Delta + 1)^2 < 6.1 \cdot 10^{31}$ and $\min\{\mu(\Delta_2), \mu(\delta)\} \leq 3n < 1.9 \cdot 10^{32}$. Otherwise the hypothesis $\tau(X) \leq \sqrt{3}/2 - 1/50$ of Theorem 1(b) is Region I ($2\beta = 1/50$), where

$e(S) \leq s + C_{\text{bad}}$ by Section 5.1; for $\tau > 1$ there are no rungs and $e(S) \leq s + C_{\text{bad}}$ by Section 5.2. Lemma 3.7(b) and Theorem 10.1 give $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{21}{16}n + \frac{3}{8}(C_{\text{bad}} + \frac{1}{2}C_4) < \frac{21}{16}n + 2.5 \cdot 10^{25}$. So Theorem 1(b) holds in these ranges with $C_1 = 2 \cdot 10^{32}$; the remaining range $\sqrt{3}/2 + 3/100 \leq \tau(X) \leq 1$ is treated in Section 11. $\square$

Throughout this section $\tau \leq \sqrt{3}/2 - 2\beta$ or $\tau > 1$, and $\Delta_2 \geq 2 \cdot 10^{15}$. Since $t = \tau + (1 - t^2)/(2\Delta_2)$, Region I gives $t \leq t_I := \sqrt{3}/2 - 2\beta + 10^{-15} < 0.84604$, and $\tau > 1$ gives $t > 1$.

10.1. Parameters, good points and the flat model. We use Lemmas 9.3 and 9.4 with

$$\varepsilon_0 = 10^{-6}, \quad \varepsilon = \tan \varepsilon_0, \quad \Lambda = 10^9, \quad \rho = 6,$$

so that $\Lambda(1 - \cos \varepsilon_0) < 10^{-3}$, and we put $\delta_0 := 10^{-4}$ and $\kappa := 10^{-2}$. A point $y \in R \setminus D$ with $h \leq 1$ is *good'* if the perimeter of K is at least 2Λ , $\text{turn}(\gamma_y^\Lambda) \leq \varepsilon_0$ and $\partial K \cap B(y, 6) \subseteq \gamma_y^\Lambda$. (Since ∂K consists of arcs of radius Δ_2 , *good'* points exist only if $\Delta_2 \geq 2\Lambda/\varepsilon_0 = 2 \cdot 10^{15}$.)

Lemma 10.2. *Fewer than $N'_g := 2.6 \cdot 10^{25}$ points $y \in R$ with $v(y) \geq 7$ are not *good'*.*

Proof. As in Lemma 9.5: if the perimeter of K is less than 2Λ then $|X \setminus D| \leq (2\Lambda + 1)^2 < 4.1 \cdot 10^{18}$; the turning condition fails for fewer than $2\pi(2\Lambda + 3)^2/\varepsilon_0 < 2.52 \cdot 10^{25}$ points; and if some $b \in \partial K \cap B(y, 6)$ is not on γ_y^Λ , then $|p_0 b| \leq 7$ and Lemma 9.2 with $\ell = 7$ gives $|X \setminus D| < 10^5$. $\square$

Let y be *good'*. We use the frame of Lemma 9.3 and call q_z the *level* of $q \in X \cap B(y, 6)$; for an interior point x we call $-x_z$ its *depth*.

Lemma 10.3 (flat model). *Let y be *good'*. Then:*

- (a) *every point of $(S \setminus D) \cap B(y, 6)$ has level in $[-6\varepsilon, 0] \subseteq [-\delta_0, 0]$, and every interior point has level $< f \leq 0$;*
- (b) *there is a number $H \geq t - \delta_0/2$ such that every point of $D \cap B(y, 6)$ has level in $[H, H + \delta_0/2]$;*
- (c) (pinning) *if $k \in (S \setminus D) \cap B(y, 6)$ and $z \in D \cap B(y, 6)$, then $|(z_z - k_z) - t| \leq 1.3 \cdot 10^{-5}$; so if $B(y, 6)$ contains a point of $S \setminus D$ and a point of D , we may take $H \in [t - \delta_0/2, t + \delta_0/2]$;*
- (d) *if y' is *good'* and $|y - y'| \leq 3$, the levels of a point of $B(y, 6) \cap B(y', 6)$ in the two frames, measured from the respective boundary lines, differ by less than $2 \cdot 10^{-5}$.*

Proof. (a) is (G1), (G2), (G4) with $|f(x)| \leq 6\varepsilon$ for $|x| \leq 6$. (b) By (G3) every level is at least $t - 6\varepsilon$, and by Lemma 9.4 with $\xi = \zeta = 6$ we have $\kappa_D < 2 \cdot 10^{-6}$, so two levels differ by less than $12\kappa_D < 2.4 \cdot 10^{-5}$; let H be the smallest level. (c) As in the proof of Lemma 9.8: put $\Delta c := z_z - k_z \geq t - 6\varepsilon$. The inward normal u_k is within ε_0 of $-n_0$ by (G5), and $|z_x - k_x| \leq 12$. If $\Delta c > 12 \tan \varepsilon_0$, then $(z - k) \cdot u_k < 0$, so z is an exception for k and $(z - k) \cdot u_k = -t + (d^2 - t^2)/(2\Delta_2)$ with $d \leq 12$; this gives $|\Delta c - t| \leq 12\varepsilon_0 + 13\varepsilon_0^2 + 80/\Delta_2 < 1.3 \cdot 10^{-5}$. Otherwise $t - 6\varepsilon \leq \Delta c \leq 12 \tan \varepsilon_0$, and again $|\Delta c - t| \leq 12\varepsilon < 1.3 \cdot 10^{-5}$. With (a) and (b) all levels of $D \cap B(y, 6)$ are within $1.3 \cdot 10^{-5} + 6\varepsilon + 2.4 \cdot 10^{-5} < \delta_0/2$ of t . (d) The frames differ by a rotation of at most ε_0 (G5) and the boundary lines by at most 6ε near these points. $\square$

So, up to errors of at most δ_0 in levels, the neighbourhood of a *good'* point looks like a *flat model*: K-row points at level 0, diametral points at a common level $H \geq t$ (with $H = t$ as soon as a K-row point is within distance 6), interior points below level 0. Directions of unit vectors are determined from levels up to $\delta_0/|\sin \theta|$, i.e. to within 0.007° for directions at least 60° from the vertical and to within 0.9° near the vertical. Every inequality used below holds with a margin larger than these errors; the smallest margins are listed in Table 2.

For a neighbour q of y we use ψ and the height $c(q) = \cos \psi = q_z + h$ as in Section 9. By Lemma 10.3: a K-neighbour has $c \in [h - \delta_0, h]$ and $|\psi| < 90^\circ + \varepsilon_0$; a D-neighbour has $c \in [h + H - \delta_0, h + H + \delta_0]$; an interior neighbour has $c < h$. In particular a *neighbour with*

margin	value	used in
fans excluded: $t > 0.8659$ versus $t \leq t_I < 0.8461$	0.0198	Lemma 10.4
type Dt receivers: $m \leq 1$ (depth ≥ 0.8845 versus $\frac{\sqrt{3}}{2}$)	0.0185	Lemma 10.6
C_φ receivers in band L: depth > 1	0.019	Lemma 10.6
sender directions in bands L and H versus $\pm 60^\circ$	0.65°	Lemma 10.9
band Th: depths forced by a 0° and by a $\pm 60^\circ$ sender	0.42	Lemma 10.8

TABLE 2. The smallest margins in [Section 10](#), against level errors below $5 \cdot 10^{-5}$ and angle errors below 0.01° (away from the vertical).

$c > h$ is a D -neighbour. The proof of [Lemma 9.6](#) now gives: two K -neighbours, or two D -neighbours, of y form a mirror pair up to $|\psi_1 + \psi_2| < 0.012^\circ$, and there are at most two of each kind. For an interior point $x \in B(y, 5)$ at depth d and a neighbour q of x in the direction θ from the upward vertical: if $q \in S \setminus D$ then $\cos \theta \in [d - \delta_0, d]$; if $q \in D$ then $\cos \theta \geq d + H - \delta_0$. In particular, *an interior point of depth greater than $1 + \delta_0$ has no neighbour in S .*

10.2. Neighbours in S and classification.

Lemma 10.4. *Let y be good'.*

- (a) *If $t > 1 + 2\delta_0$, then y has no D -neighbour.*
- (b) *$m_y \leq 3$.*
- (c) *If $m_y = 3$, the neighbours of y in S are two K -neighbours at $\pm\psi_K$ with $60^\circ \leq \psi_K < 90^\circ + \varepsilon_0$ and one D -neighbour at ψ_D with $|\psi_D| \leq \psi_K - 60^\circ$ (up to 0.012°), and $\sin(\psi_K - 30^\circ) - \delta_0 \leq t \leq 1 - \cos \psi_K + \delta_0$. In particular $t \geq \frac{1}{2} - \delta_0$.*

Proof. (a) A D -neighbour would have $c \geq h + H - \delta_0 \geq t - 2\delta_0 > 1$. (b) If $m_y = 4$, then as in the proof of [Lemma 9.7](#) the neighbours in S are K -neighbours within ε_0 of $\pm 90^\circ$ and D -neighbours within ε_0 of $\pm 30^\circ$. By [Lemma 10.3\(c\)](#), t is within δ_0 of their height difference, so $t \geq \cos(30^\circ + \varepsilon_0) - \sin \varepsilon_0 - \delta_0 > 0.8659$. This contradicts $t \leq t_I < 0.8461$ in Region I, and (a) when $t > 1$ (the D -neighbours have $c < 0.87 < 1$). (c) By the pair property, three neighbours in S are of kinds $\{K, K, D\}$ or $\{K, D, D\}$. In the second case the D -pair is at $\pm\psi_D$ with $\psi_D \geq 30^\circ$, and the K -neighbour, at least 60° from both and with $|\psi| < 90^\circ + \varepsilon_0$, forces $\psi_D \leq 30^\circ + \varepsilon_0 + 0.012^\circ$ and a K -neighbour near $\pm 90^\circ$; then $t > 0.8659$ as in (b), impossible. In the first case the K -pair is at $\pm\psi_K$, and the D -neighbour, which has $c > 0$, lies between them at least 60° from both: $|\psi_D| \leq \psi_K - 60^\circ$. By [Lemma 10.3\(c\)](#), $t = \cos \psi_D - \cos \psi_K$ up to δ_0 ; now $\cos \psi_D \geq \cos(\psi_K - 60^\circ)$ and $\cos(\psi_K - 60^\circ) - \cos \psi_K = \sin(\psi_K - 30^\circ)$, and $\cos \psi_D \leq 1$. $\square$

Lemma 10.5 (classification). *Let y be good' with $v(y) \geq 8$. Then, up to the reflection $\psi \mapsto -\psi$ and the errors of the flat model, y is of one of the following types.*

- (A) $(6, 2)$: two K -neighbours at $\pm 30^\circ$; $h = \frac{\sqrt{3}}{2}$; the other four neighbours are interior. (If $t \leq 2\delta_0$, one of the two may be a D -neighbour in S .)
- (B) $(6, 2)$: two D -neighbours in S at $\pm 30^\circ$; $h = \frac{\sqrt{3}}{2} - H > 0$, so $H \in [t - \delta_0, \frac{\sqrt{3}}{2})$; the neighbours at $\pm 90^\circ, \pm 150^\circ$ are interior.
- (C $_\varphi$) $(6, 2)$, $0^\circ \leq \varphi \leq 30^\circ$: a D -neighbour in S at φ and a K -neighbour at $\varphi - 60^\circ$; $h = \cos(60^\circ - \varphi)$ and $t = \sin(30^\circ - \varphi)$; the neighbours at $\varphi + 60^\circ, \varphi \pm 120^\circ, \varphi + 180^\circ$ are not in S , and those at $\varphi + 120^\circ, \varphi + 180^\circ$ are interior.
- (A') $(6, 2)$: K -neighbours at $\pm 60^\circ$, a D -neighbour in R at 0° ; $h = \frac{1}{2}$, $t = \frac{1}{2}$; interior neighbours at $\pm 120^\circ, 180^\circ$.

- (cap) (6, 3): *K-neighbours at $\pm 60^\circ$, a D-neighbour in S at 0° ; $h = \frac{1}{2}$, $t = \frac{1}{2}$; interior neighbours at $\pm 120^\circ, 180^\circ$; $v(y) = 9$.*
- (Dt) (5, 3): *as in Lemma 10.4(c), with $h = \cos \psi_K$ and two interior neighbours in the arc $I_y := [\psi_K + 60^\circ, 300^\circ - \psi_K]$. In Region I, $60^\circ \leq \psi_K \leq 87.8^\circ$; for $\tau > 1$ this type needs $t \leq 1 + 2\delta_0$ and $\psi_K > 89.99^\circ$.*

Every type except the cap has $v(y) = 8$. For $t > 1 + 2\delta_0$ only type A occurs.

Proof. By Lemma 10.4, $v \geq 8$ leaves $(\deg, m) \in \{(6, 2), (5, 3), (6, 3)\}$. For $\deg y = 6$ the neighbours form a regular hexagon.

(5, 3) and (6, 3). Use Lemma 10.4(c). If $\deg y = 6$, the K-pair occupies mirror slots of the hexagon, so $\psi_K \in \{30^\circ, 60^\circ, 90^\circ\}$. For 30° there is no room for the D-neighbour; for 90° the D-neighbour is at $\pm 30^\circ$ and $t > 0.8659$, excluded as in Lemma 10.4; for 60° the D-neighbour is at 0° and $t = 1 - \frac{1}{2} = \frac{1}{2}$: the cap. If $\deg y = 5$, the two other neighbours are not in S . They are not D-neighbours: a second D-neighbour would be the mirror image $-\psi_D$ of the first, at least 60° away, forcing $\psi_D \approx \pm 30^\circ$, $\psi_K \approx 90^\circ$ and $t > 0.8659$. So they are interior, and being at least 60° from the K-neighbours and the D-neighbour, they lie in I_y . The range of ψ_K follows from Lemma 10.4(c): $\sin(\psi_K - 30^\circ) \leq t_I + \delta_0$ gives $\psi_K \leq 87.8^\circ$, and $t > 1$ gives $\cos \psi_K < \delta_0$.

(6, 2), two K-neighbours. They are a mirror pair at $\pm 30^\circ, \pm 60^\circ$ or $\pm 90^\circ$. At $\pm 30^\circ$ this is type A ($h = \cos 30^\circ$; the neighbours at $\pm 90^\circ, \pm 150^\circ$ have $c < h$ and so are interior). At $\pm 60^\circ$ the neighbour at 0° has $c = 1 > h = \frac{1}{2}$, so it is a D-neighbour with $H = \frac{1}{2}$, and $t = \frac{1}{2}$ by pinning; since $m_y = 2$ it lies in R : type A'. At $\pm 90^\circ$ the neighbours at $\pm 30^\circ$ have $c > h$, so they are D-neighbours and $t > 0.8659$; excluded.

(6, 2), two D-neighbours. They are a mirror pair, at $\pm 30^\circ, \pm 60^\circ$ or $\pm 90^\circ$. At $\pm 60^\circ$ the neighbour at 0° has $c = 1 > h$, so it is a D-neighbour at a level about $\frac{1}{2}$ higher than the other two, contradicting Lemma 10.3(b). At $\pm 90^\circ$ we get $h \leq \delta_0$, and the neighbours at $\pm 30^\circ$ are D-neighbours at a level about $\frac{\sqrt{3}}{2}$ higher; again a contradiction. At $\pm 30^\circ$: $c = \frac{\sqrt{3}}{2} = h + H$, and $h > 0$ because $y \in \text{int } K$; the neighbours at $\pm 90^\circ$ are not in S (as $m_y = 2$) and are not D-neighbours ($c < 0.01 < h + H - \delta_0$), so they are interior. This is type B. (Note $H < \frac{\sqrt{3}}{2}$ strictly: $H = \frac{\sqrt{3}}{2}$ would put y on ∂K , but $R \setminus D \subseteq \text{int } K$ by Section 6.1(b).)

(6, 2), a K- and a D-neighbour. If $t \leq 2\delta_0$, both have c within $4\delta_0$ of h , so they form a mirror pair, and as above they are at $\pm 30^\circ$ (type A with one D-neighbour; at $\pm 60^\circ$ and $\pm 90^\circ$ a further D-neighbour at a different level would appear). So let $t > 2\delta_0$; then every D-neighbour has $c > h$. Let $\varphi \in [-30^\circ, 30^\circ]$ be the slot with the largest c ; by symmetry $\varphi \geq 0$. If the D-neighbour in S were not at φ , the neighbour at φ , whose c is at least that of the D-neighbour and hence greater than h , would be a second D-neighbour at the same level, hence its mirror image; then $\varphi = 30^\circ$, the K-neighbour is at -90° and $t > 0.8659$; excluded. So the D-neighbour is at φ . The slot $\varphi - 60^\circ$ has the second largest c ; if the K-neighbour were not there, that slot would have $c > h$ and be a second D-neighbour, which again forces $\varphi = 30^\circ$ and is excluded (for $\varphi = 0$ the slots $\pm 60^\circ$ tie, and we name the K-neighbour's slot $\varphi - 60^\circ$). So the K-neighbour is at $\varphi - 60^\circ$, $h = \cos(60^\circ - \varphi)$, and by pinning $t = \cos \varphi - \cos(60^\circ - \varphi) = \sin(30^\circ - \varphi)$. The slots $\varphi + 120^\circ$ and $\varphi + 180^\circ$ have $c < 0 < h$ and are interior. This is type C_φ ; for $\varphi = 30^\circ$ it is type A with one D-neighbour.

Finally, for $t > 1 + 2\delta_0$ there are no D-neighbours (Lemma 10.4(a)), which leaves type A. $\square$

So in Region I and for $\tau > 1$ every good' point has weight at most 8, except caps, which have weight 9 and need $t = \frac{1}{2}$. We now move the excess over 7.

10.3. Discharging. Every good' point y with $v(y) \geq 8$ sends charge to some of its interior neighbours, its receivers:

- types A and B: $\frac{1}{2}$ to each of the neighbours at $\pm 150^\circ$;
- type C_φ : $\frac{1}{2}$ to each of the neighbours at $\varphi + 120^\circ$ and $\varphi + 180^\circ$;
- type A' : 1 to the neighbour at 180° ;
- cap: 1 to the neighbour at 180° and $\frac{1}{2}$ to each of the neighbours at $\pm 120^\circ$;
- type Dt: $\frac{1}{2}$ to each of its two interior neighbours.

Then every receiver x with $v(x) = 7$ (a *forwarder*) passes all the charge it has received to its neighbour opposite its unique neighbour in S (Lemma 10.6(c) shows that this is well defined). Depths and directions at a receiver are measured in the frame of the good' point from which the charge originates; by Lemma 10.3(d) these frames agree up to the flat-model errors.

Lemma 10.6 (receivers). *Let x be a receiver of a good' point y .*

- The depth of x is: $\sqrt{3}$ for type A; $h + \frac{\sqrt{3}}{2} \geq \frac{\sqrt{3}}{2}$ for type B; $2 \cos(60^\circ - \varphi) \geq 1$ at $\varphi + 120^\circ$ and $\sqrt{3} \cos(30^\circ - \varphi) \geq \frac{3}{2}$ at $\varphi + 180^\circ$ for type C_φ ; $\frac{3}{2}$ at 180° for A' and caps; 1 at $\pm 120^\circ$ for caps; at least $\sin(\psi_K + 30^\circ)$ for type Dt.*
- $m_x \leq 1$, so $v(x) \leq 7$ and x is not a sender. Moreover $m_x = 0$ unless y is of type Dt, or a cap and x is at $\pm 120^\circ$.*
- If $v(x) = 7$, then $\deg x = 6$, the unique neighbour of x in S is a K-row point in the direction β with $|\beta| \leq 30.02^\circ$ from the vertical (with $|\beta| \leq 28^\circ$ in Region I), and the neighbour g of x in the direction $\beta + 180^\circ$ has depth greater than 1.7, so $m_g = 0$ and $v(g) \leq 6$.*

Proof. (a) The depth of the neighbour of y at ψ is $h - \cos \psi$. For type C_φ use $\cos(60^\circ - \varphi) + \cos \varphi = \sqrt{3} \cos(30^\circ - \varphi)$; for type Dt use $\cos \psi_K - \cos(\psi_K + 60^\circ) = \sin(\psi_K + 30^\circ)$ and $\psi_x \in I_y$.

(b) Depth greater than $1 + \delta_0$ gives $m_x = 0$; this covers type A, C_φ at $\varphi + 180^\circ$, and A' and caps at 180° . *Type B*: as seen from x (at $\pm 150^\circ$ from y), y is in the direction $\mp 30^\circ$ and the interior neighbour w of y at $\pm 90^\circ$ is in the direction $\pm 30^\circ$ (since $e(90^\circ) - e(150^\circ) = e(30^\circ)$ for $e(\theta) := (\sin \theta, \cos \theta)$). A neighbour of x in S would be in a direction θ with $\cos \theta \geq \frac{\sqrt{3}}{2} - \delta_0$, i.e. $|\theta| \leq 30.01^\circ$, and at least 60° from both $\pm 30^\circ$; impossible. *Type C_φ at $\varphi + 120^\circ$* : a neighbour in S would be a K-row point (a D-point needs $\cos \theta > 1$) with $\cos \theta \geq 2 \cos(60^\circ - \varphi) - \delta_0$; this forces $\varphi < 0.01^\circ$ and $|\theta| < 0.9^\circ$. But the neighbour w of y at $\varphi + 60^\circ$ is in the direction φ from x (as $e(\varphi + 60^\circ) - e(\varphi + 120^\circ) = e(\varphi)$), so that K-row point would be within 0.02 of w , hence equal to w ; and $w \notin S$. So $m_x = 0$. *Caps at $\pm 120^\circ$* : the depth is 1 up to $2\delta_0$, a D-point would need $\cos \theta > 1$, and a K-row neighbour needs $|\theta| \leq 1.5^\circ$; two such are less than 60° apart, so $m_x \leq 1$. *Type Dt*: a D-neighbour of x would need $\cos \theta \geq \sin(\psi_K + 30^\circ) + H - \delta_0$. In Region I, $\psi_K \leq 87.8^\circ$ gives depth at least $\sin 117.8^\circ > 0.8845$, and $H \geq t - \delta_0 \geq \frac{1}{2} - 2\delta_0$, so this is impossible; K-row neighbours need $\cos \theta \geq 0.8845$, i.e. $|\theta| < 28^\circ$, and two of them would be less than 60° apart. For $\tau > 1$ we have $\psi_K > 89.99^\circ$ and depth at least $\sin 119.99^\circ > 0.8659$; again there is no D-neighbour, and K-row neighbours have $|\theta| \leq 30.02^\circ$. Two of them would have to be within 0.04° of $\pm 30^\circ$, which forces the depth of x to be within 10^{-3} of $\frac{\sqrt{3}}{2}$ and hence ψ_x within 0.2° of 150° or 210° ; then y is within 0.2° of the direction $\mp 30^\circ$ from x , i.e. within 0.01 of one of these K-row points, which is impossible since $y \in R$.

(c) If $v(x) = 7$ then $m_x = 1$ and $\deg x = 6$, so by (b) x is a receiver of type Dt or a cap receiver at $\pm 120^\circ$, and its neighbour in S is a K-row point in the direction β found in (b). The six neighbours of x form a regular hexagon, so g exists, and its depth is the depth d of x plus $\cos \beta \geq d - \delta_0$, so at least $2d - \delta_0 \geq \min\{2 \cdot 0.8845, 2 \cdot 0.8659, 2 - 3\delta_0\} - \delta_0 > 1.7$. $\square$

Lemma 10.7 (cone). *Let x receive charge from s , either directly or by forwarding. The direction θ from x to s and the depth of s are as follows: types A and B: $\theta = \pm 30^\circ$, depth $\frac{\sqrt{3}}{2}$ (A) or at most $\frac{\sqrt{3}}{2} - t + \delta_0$ (B); type C_φ : $\theta = \varphi - 60^\circ$ (from its receiver at $\varphi + 120^\circ$) or $\theta = \varphi$ (from*

its receiver at $\varphi + 180^\circ$), depth $\cos(60^\circ - \varphi)$; A' and caps: $\theta = 0^\circ$ (one unit), and for caps also $\theta = \mp 60^\circ$ (half a unit), depth $\frac{1}{2}$; type Dt: $\theta \in [\psi_K - 120^\circ, 120^\circ - \psi_K]$, depth $\cos \psi_K$; a forwarder: $\theta = \beta$ with $|\beta| \leq 30.02^\circ$. In each case the depth of x is the depth of s plus $\cos \theta$. All these directions lie in $[-60^\circ, 60^\circ]$; so x has at most three senders, and three only in the directions $-60^\circ, 0^\circ, 60^\circ$ up to 1° .

Proof. If x is the neighbour of s at ψ , then s is the neighbour of x in the direction $\psi - 180^\circ$. The list follows from [Lemmas 10.5](#) and [10.6](#). Senders of x are distinct neighbours of x , hence at least 60° apart. $\square$

We split Region I and the range $\tau > 1$ into four bands, which together cover them:

$$L: t \leq \frac{1}{2} - \kappa; \quad Th: |t - \frac{1}{2}| < \kappa; \quad H: \frac{1}{2} + \kappa \leq t \leq t_I \text{ or } 1 < t \leq 1 + 2\delta_0; \quad T: t > 1 + 2\delta_0.$$

Lemma 10.8 (band Th). *Let $|t - \frac{1}{2}| < \kappa$ and let x receive charge. A sender of x in a direction within 3° of $\pm 60^\circ$ forces the depth of x to be at most 1.06; a sender of x in a direction within 3° of 0° forces the depth of x to be at least 1.48. Consequently x never has both kinds of sender.*

Proof. In this band the heavy types are A, B, C_φ with $\varphi \leq 0.67^\circ$ (from $\sin(30^\circ - \varphi) \geq \frac{1}{2} - \kappa - \delta_0$), A' , caps, and Dt with $60^\circ \leq \psi_K \leq 60.67^\circ$ (from $\sin(\psi_K - 30^\circ) \leq \frac{1}{2} + \kappa + \delta_0$). Types A and B send only from $\pm 30^\circ$. A sender within 3° of $\pm 60^\circ$ is a C_φ (direction $\varphi - 60^\circ$), a cap (direction $\mp 60^\circ$) or a Dt with $|\theta| \geq 57^\circ$; its depth is at most $\cos 59.33^\circ + \delta_0 < 0.511$, so the depth of x is at most $0.511 + \cos 57^\circ < 1.06$. A sender within 3° of 0° is a C_φ (direction φ ; depth of x equal to $\sqrt{3} \cos(30^\circ - \varphi) \geq 1.49$), an A' or a cap (depth of x equal to $\frac{3}{2}$), a Dt with $|\theta| \leq 3^\circ$ (depth of x at least $\cos 60.67^\circ + \cos 3^\circ > 1.48$), or a forwarder; a forwarder in this band is a receiver of a Dt or a cap, so its depth is at least $\min\{\sin 90.67^\circ, 1 - 2\delta_0\} > 0.999$, its K-row neighbour is in a direction β with $\cos \beta > 0.998$, i.e. $|\beta| < 3^\circ$, and the depth of x is at least $0.999 + \cos 3^\circ > 1.99$. $\square$

Lemma 10.9 (loads). *Every point of R receives at most one unit of charge in total.*

Proof. Let x receive charge. *Band L.* The heavy types are A, B and C_φ , and $t = \sin(30^\circ - \varphi) \leq \frac{1}{2} - \kappa + \delta_0$ gives $\varphi \geq 0.65^\circ$; types A' , cap and Dt need $t \geq \frac{1}{2} - \delta_0$. By [Lemma 10.6\(b\)](#) all receivers have $m = 0$, so there is no forwarding. The sender directions $\pm 30^\circ, \pm(60^\circ - \varphi), \pm\varphi$ lie in $[-59.35^\circ, 59.35^\circ]$, an interval shorter than 120° , so x has at most two senders, each giving $\frac{1}{2}$.

Band H. The heavy types are A, B and Dt, and $1 - \cos \psi_K + \delta_0 \geq t \geq \frac{1}{2} + \kappa$ gives $\psi_K \geq 60.65^\circ$, so Dt sends from directions in $[-59.35^\circ, 59.35^\circ]$. By [Lemma 10.6](#), the forwarders are receivers of type Dt, and all their senders are of type Dt (receivers of A and B have $m = 0$). A forwarder x' has only one sender: two Dt senders at $\delta_1 < \delta_2$ in $[-59.35^\circ, 59.35^\circ]$ with $\delta_2 - \delta_1 \geq 60^\circ$ have $\delta_1 < -0.65^\circ$ and $\delta_2 > 0.65^\circ$, and the K-row neighbour of x' , at $|\beta| \leq 30.02^\circ$ and at least 60° from both, would need $\delta_1 + 60^\circ \leq \beta \leq \delta_2 - 60^\circ$, i.e. $\delta_2 - \delta_1 \geq 120^\circ$. So every forwarder passes on exactly $\frac{1}{2}$, from a direction $|\beta| \leq 30.02^\circ$. Hence all senders of x give at most $\frac{1}{2}$ and lie in $[-59.35^\circ, 59.35^\circ]$, and there are at most two of them.

Band T. Only type A occurs ([Lemma 10.5](#)); its receivers have $m = 0$, and the senders lie at $\pm 30^\circ$: at most two, each giving $\frac{1}{2}$.

Band Th. By [Lemma 10.7](#), three senders would lie near $-60^\circ, 0^\circ$ and 60° , which [Lemma 10.8](#) forbids; so x has at most two senders. A sender giving more than $\frac{1}{2}$ is an A' or a cap in the direction 0° , or a forwarder passing on 1, in a direction within 3° of 0° . Any other sender would be at least 57° from the vertical, hence (being within 60° of it) within 3° of $\pm 60^\circ$, which [Lemma 10.8](#) forbids. So either x has a single sender, giving at most 1, or at most two senders, each giving $\frac{1}{2}$. (A forwarder receives only half units, from at most two senders, so it passes on $\frac{1}{2}$ or 1; forwarders are not themselves forwarding targets, since targets have $m = 0$.) $\square$

Proof of Theorem 10.1. Give each $y \in R$ the charge $v(y)$ and apply the rules; the total is unchanged. A good' point of weight 8 sends 1 and a cap sends 2; neither receives anything (receivers have $v \leq 7$), so both end at 7. A forwarder has weight 7 and passes on all it receives, so it ends at 7. Every other receiver has weight at most 6 (Lemma 10.6) and receives at most 1 (Lemma 10.9), so it ends at most at 7. Every other good' point has weight at most 7, and so do the points of $R \cap D$ and those with $h > 1$ (weight at most 6). What remains are the points that are not good' and have weight at least 8: there are fewer than N'_g of them (Lemma 10.2); they are not receivers, and they end at most at 12. Hence $\sum_{y \in R} v(y) \leq 7r + 5N'_g < 7r + 1.3 \cdot 10^{26}$. $\square$

Remark 10.10. (a) *Radius-one discharging stops at 7.* In an exact stack of hexagonal rows below a straight K-row and in the exact “cap lattice” at $t = \frac{1}{2}$ (coordinates in $\mathbb{Q}(\sqrt{3})$, script `r6_below_exact_examples.py`), the average weight on R is exactly 6 at every truncation depth, but the weights per period are $8, 6, \dots, 6, 4$ and $9, 7, 6, \dots, 6, 5, 3$ respectively. Any rule that moves charge only between neighbours leaves some point at 7 in these examples. So $\frac{21}{16}$ is the limit of rules of the kind used here.

- (b) *What would give $\frac{9}{7}$ in Region I.* By Remark 3.6, an average weight $6 + o(1)$ on R in Region I would give $\frac{9}{7}$ there, matching the lower bound. No configuration with average weight above 6 was found in a search over clusters of up to 7 points of R (Appendix A); we state this as Question 15.1.

11. REGION II: THEOREM 1(B) FOR $\sqrt{3}/2 + 3\beta \leq \tau \leq 1$

In Region II the bound $e(S) \leq s + p + C$ of Section 5.3 carries the full p -term, and by Remark 3.6 this caps the method at $\frac{4}{3}$ whatever the average weight on R . In this section we show that in Region II with $\tau \leq 1$ the p -term can be halved, $e(S) \leq s + \frac{1}{2}p + C$ (Theorem 11.3), and that the average weight on R is at most 7 there as well (Theorem 11.12). Halving the p -term is exactly what the weights $\frac{5}{8}, \frac{3}{8}$ of Lemma 3.7(b) can absorb (Lemma 11.2). Together with Section 10 this gives Theorem 1(b) on both sides of Region III.

Theorem 11.1 (Theorem 1(b) in full). *Let $X \subset \mathbb{R}^2$ be a set of n points. If $\tau(X) \leq \sqrt{3}/2 - 1/50$ or $\tau(X) \geq \sqrt{3}/2 + 3/100$, then*

$$\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{21}{16}n + C_1, \quad C_1 = 2 \cdot 10^{32}.$$

So the only range of τ in which the bound is still $\frac{4}{3}n + C_0$ is Region III, $\sqrt{3}/2 - 1/50 < \tau(X) < \sqrt{3}/2 + 3/100$.

Status. The two new inputs, Theorem 11.3 and Theorem 11.12, have each been through one independent referee pass. Both passes found the results correct and found no counterexample; they asked for a missing step in the disjointness of the arcs below (Lemma 11.8, the case of two components with the same centre), a corrected justification of the count of shared endpoints (proof of Theorem 11.3), a justification of the crossing argument (Lemma 11.6), explicit constants (Lemma 11.10), a direct proof of $m_y \leq 3$ that does not go through fans (Lemma 11.13), a shorter proof that receivers have $m \leq 1$ (Lemma 11.15), and a correct statement of how the frames of different senders are compared (Lemma 11.16). These corrections are incorporated below. Theorems 11.3 and 11.12 are not formalised. Computer checks (Section 11.3) are exact checks of constants and of special configurations, and floating-point screenings; they are evidence, not part of the proof.

Lemma 11.2 (linear combination with half the p -term). *If $e(S) \leq s + \frac{1}{2}p + C$ and $\sum_{y \in R} v(y) \leq 7r + C''$, then $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{21}{16}n + \frac{3}{8}(C + \frac{1}{2}C'')$.*

Proof. By Lemma 3.3, $\mu(\delta) \leq s + \frac{1}{2}p + \frac{7}{2}r + C + \frac{1}{2}C''$. By Lemma 3.2, $\mu(\Delta_2) \leq \frac{3}{2}s - \frac{1}{2}p$; here Q is, by definition, the set of all points of S of Δ_2 -degree exactly 1, which is what the proof of Lemma 3.2 uses, and $p = |Q|$ is the same quantity as in the bound on $e(S)$. Hence

$$\frac{5}{8}\left(\frac{3}{2}s - \frac{1}{2}p\right) + \frac{3}{8}\left(s + \frac{1}{2}p + \frac{7}{2}r + C + \frac{1}{2}C''\right) = \frac{21}{16}(s + r) - \frac{1}{8}p + \frac{3}{8}(C + \frac{1}{2}C''),$$

and $-\frac{1}{8}p \leq 0$.^[LP] □

(In the terms of Remark 3.6: with $e(S) \leq s + \frac{1}{2}p + O(1)$ the weight on $\mu(\Delta_2)$ must be at least $\frac{1}{2}$ to cancel the p -term, and the value is $\max\{\frac{5}{4}, \frac{3c}{2c+1}\}$ for average weight $2c$. For $2c = 8, 7, 6$ this is $\frac{4}{3}, \frac{21}{16}, \frac{9}{7}$.^[LP])

Proof of Theorem 11.1 from Theorems 11.3 and 11.12. If $n \leq 3$ or X has only one distance there is nothing to prove. In the degenerate regime $\min\{\mu(\Delta_2), \mu(\delta)\} \leq n + 1 \leq \frac{21}{16}n + 1$ (Proposition 7.1). In the nondegenerate regime with $\Delta_2 < 2 \cdot 10^{15}$ the bound $3n < 1.9 \cdot 10^{32}$ of the proof of Theorem 1(b) in Section 10 applies. So let $\Delta_2 \geq 2 \cdot 10^{15}$. If $\tau \leq \sqrt{3}/2 - 1/50$ or $\tau > 1$, Theorem 1(b) is proved in Section 10. Otherwise $\sqrt{3}/2 + 3\beta \leq \tau \leq 1$ (recall $3\beta = 3/100$), and Theorem 11.3, Theorem 11.12 and Lemma 11.2 give

$$\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{21}{16}n + \frac{3}{8}(C_{\text{bad}} + 87 + \frac{1}{2}C_4) < \frac{21}{16}n + 2.5 \cdot 10^{25}.$$

So $C_1 = 2 \cdot 10^{32}$ still suffices. □

11.1. Rungs in Region II: $e(S) \leq s + \frac{1}{2}p + C$.

Theorem 11.3. *In the nondegenerate regime with $\Delta_2 \geq 10^4$ and $\sqrt{3}/2 + 3\beta \leq \tau \leq 1$,*

$$e(S) \leq s + \frac{1}{2}p + C_{\text{bad}} + 87.$$

Throughout this subsection we assume the hypotheses of Theorem 11.3. We use the notation of Sections 5.2 and 5.3: $\theta_0 = \arccos \tau < 26.4^\circ$, $\sigma = \sqrt{1 - \tau^2} \leq \sigma_{\max} < 0.4441$,^[C5] $t \leq 1$ and $t \geq \tau$. All rungs below are good rungs; ρ is their number, and by (R1) of Section 5.3 every vertex is an end of at most one of them. We put

$$\alpha_1 := 2 \arcsin \frac{1}{2\Delta}, \quad \alpha_2 := 2 \arcsin \frac{1}{2\Delta_2}, \quad \varphi_\tau := \arcsin \frac{\sigma}{\Delta} \in [0, \pi/2).$$

So two points of $C(w, \Delta)$ (resp. $C(w, \Delta_2)$) at distance 1 are at angle α_1 (resp. α_2) about w , and $\alpha_2 > \alpha_1$. For a point w we write $\vartheta(x)$ for the polar angle of x about w ; it will only be used on arcs of angular width less than π , on which we fix a continuous branch. “Counterclockwise” refers to this angle. Recall from Section 5.4 (the paragraph on the rotational sense and Lemma 5.6) that in both rows “right” means counterclockwise along the boundary, and that a right T-step between two points of $C(w, \Delta)$ is counterclockwise about w ; the same holds, with the same proof, for a right T-step between two points of $C(w, \Delta_2) \cap \partial K$.

Lemma 11.4 (rung geometry). *Let (v, z) be a rung with $v \in S \setminus D$, and let $w := w_v$, $u := u_v$.*

- (a) $z - w = -(\Delta_2 + \tau)u \pm \sigma u^\perp$ and $|z - w| = \Delta$. Hence $\vartheta(z) = \vartheta(v) + \varsigma \varphi_\tau$ for a sign $\varsigma = \varsigma_{(v,z)} \in \{\pm 1\}$, the sign of the rung (if $\sigma = 0$ we put $\varsigma := +1$).
- (b) $2\varphi_\tau < \alpha_1$.
- (c) Along the segment from v to z the distance to w increases strictly from Δ_2 to Δ , and ϑ is monotone. So the segment is the graph $\{\vartheta = g(\varrho)\}$ of a continuous monotone function g of $\varrho = |x - w| \in [\Delta_2, \Delta]$.

Proof. (a) By Lemma 2.2(c), $z-v = -\tau u \pm \sigma u^\perp$, and $v-w = -\Delta_2 u$. So $|z-w|^2 = (\Delta_2 + \tau)^2 + \sigma^2 = \Delta_2^2 + 2\Delta_2\tau + 1 = \Delta^2$, by the definition $\tau = (\Delta^2 - \Delta_2^2 - 1)/(2\Delta_2)$ (Section 2). The component of $z-w$ along $(v-w)/\Delta_2 = -u$ is $\Delta_2 + \tau > 0$ and the orthogonal component has length σ , so the angle at w between v and z is $\arcsin(\sigma/\Delta) = \varphi_\tau$.^[Ith] (b) $\sin \varphi_\tau = \sigma/\Delta < 1/(2\Delta) = \sin(\alpha_1/2)$, since $\sigma < \frac{1}{2}$ in Region II. (In fact $2\varphi_\tau/\alpha_1 \leq 2\sigma < 0.8881$.)^[Ith] (c) For $x = v + \lambda(z-v)$, $\lambda \in [0, 1]$, we have $x-w = -(\Delta_2 + \lambda\tau)u \pm \lambda\sigma u^\perp$, so $|x-w|^2 = (\Delta_2 + \lambda\tau)^2 + \lambda^2\sigma^2$ is strictly increasing in λ (as $\tau > 0$), and the angle between $x-w$ and $-u$ is $\arctan(\lambda\sigma/(\Delta_2 + \lambda\tau))$, which is nondecreasing in λ . $\square$

Lemma 11.5 (a rung-carrying D-row component lies on one circle). *Let (v, z) be a rung, $w := w_v$, and let Z be the component of z in the graph of pure D-row T-edges.*

- (a) $Z \subseteq C(w, \Delta)$; consecutive points of Z are at angle exactly α_1 about w , all in the counterclockwise sense; Z spans an angle at most $\pi/3$ about w ; and Z is a path.
- (b) If $|Z| \geq 2$ and (v', z') is any rung with $z' \in Z$, then $w_{v'} = w$.

We call w the centre of Z ; by (b) it does not depend on the rung when $|Z| \geq 2$.

Proof. (a) By Lemma 11.4(a), $|zw| = \Delta$, so w is a diametral partner of z and $\hat{w} := (w-z)/\Delta$ is an inward normal of P at z (Lemma 2.2(a) and the remark after it). The angle between $\hat{w}$ and $u_v = (w-v)/\Delta_2$ is the angle at w of the triangle vzw ; it is opposite the side vz of length 1, hence acute, and by the law of sines it is at most $\arcsin(1/\Delta)$. Also $\angle(u_v, u_z) \leq \beta$ because the rung is good. Let z_1 be a T-neighbour of z in Z (on either side). Then $(z_1 - z) \cdot u_z \in (0, \beta)$, so $\angle(z_1 - z, u_z) \geq \pi/2 - \arcsin \beta$ and

$$\eta := (z_1 - z) \cdot \hat{w} \leq \sin(\arcsin \beta + \beta + \arcsin(1/\Delta)) < 0.0201 < t.$$
^[Ith]

By Lemma 5.3 (whose threshold is at least t since $t \leq 1$), $|z_1 w| = \Delta$. From here on the proof of Lemma 5.7 applies verbatim: a vertex $z_k \in Z \cap C(w, \Delta)$ that has both a left and a right T-neighbour has a normal cone within $\arcsin \beta$ of u_{z_k} , and its other T-neighbour is again on $C(w, \Delta)$; inducting outwards from z (every vertex reached after the first step has the vertex it was reached from as a T-neighbour, so it is an interior vertex whenever it has a further T-neighbour) puts every vertex of Z on $C(w, \Delta)$. The induction never starts at an end vertex of Z other than z itself, and at z the direction $\hat{w}$ is controlled by the rung triangle as above. The statements about angles, the counterclockwise sense, the span and the path structure follow as in Lemma 5.7.

(b) As $|Z| \geq 2$ and Z is connected, z' has a T-neighbour $z'_1 \in Z$. By (a) applied to (v', z') , $z', z'_1 \in C(w_{v'}, \Delta)$, and by (a) applied to (v, z) , $z', z'_1 \in C(w, \Delta)$. If $w \neq w_{v'}$, these are two circles of radius Δ whose centres are points of X , at distance $d \leq \Delta$; they meet in two points at distance $2\sqrt{\Delta^2 - d^2/4} \geq \sqrt{3}\Delta > 1$, which contradicts $|z'z'_1| = 1$. So $w_{v'} = w$. $\square$

By (R1) at D-vertices, a component Z with $|Z| = 1$ carries at most one rung. For a component Z let m_Z be the number of rungs whose D-end lies in Z ; so $\sum_Z m_Z = \rho$, and $m_Z \geq 2$ implies $|Z| \geq 2$. For such Z , with centre w , all these rungs have K-ends on $C(w, \Delta_2)$ (Lemma 11.5(b)), and by Lemma 11.4(a) all their endpoints lie in an arc of angular width at most $\pi/3 + 2\varphi_\tau < \pi$ about w .

Lemma 11.6 (order). *Let Z have centre w and let $(v_A, z_A) \neq (v_B, z_B)$ be rungs with $z_A, z_B \in Z$ and $\vartheta(z_A) < \vartheta(z_B)$. Then $\vartheta(v_A) < \vartheta(v_B)$.*

Proof. By (R1) the four points are distinct, and $v_A, v_B \in C(w, \Delta_2)$, so $\vartheta(v_A) \neq \vartheta(v_B)$. Suppose $\vartheta(v_A) > \vartheta(v_B)$. By Lemma 11.4(c), each rung is the graph of a continuous function g_A, g_B of $\varrho \in [\Delta_2, \Delta]$, inside an annular sector of opening less than π . Now $g_A - g_B$ is positive at $\varrho = \Delta_2$ and negative at $\varrho = \Delta$, so it vanishes at some $\varrho^* \in (\Delta_2, \Delta)$: the two segments cross at a

point x that is interior to both. If the four points are not collinear, the triangle inequality gives $|v_A v_B| + |z_A z_B| < |v_A x| + |x v_B| + |z_A x| + |x z_B| = |v_A z_A| + |v_B z_B| = 2$, so one of $|v_A v_B|, |z_A z_B|$ is less than 1, a contradiction. If they are collinear, both segments lie on one line through x , along which $|\cdot - w|$ increases from the v -end to the z -end on both segments (Lemma 11.4(c)); at the interior point x this is the same direction of the line, so both v_A and v_B are the point of that line at distance Δ_2 from w on the same side of x , i.e. $v_A = v_B$, a contradiction. $\square$

Gaps. Let Z be a component with $m_Z \geq 2$ and centre w . Order its rungs $A_1, \dots, A_{m_Z}$ by ϑ of their D-ends (distinct by (R1)). A *gap* is a pair $(A, B) = (A_i, A_{i+1})$ of consecutive rungs; Z has $m_Z - 1$ gaps. Write $A = (v_A, z_A)$, $B = (v_B, z_B)$ and ς_A, ς_B for their signs. By Lemma 11.5(a), $\vartheta(z_B) - \vartheta(z_A) = k\alpha_1$ with an integer $1 \leq k$, $k\alpha_1 \leq \pi/3$; by Lemma 11.4(a) and Lemma 11.6,

$$(5) \quad 0 < \vartheta(v_B) - \vartheta(v_A) = k\alpha_1 - (\varsigma_B - \varsigma_A)\varphi_\tau < \pi.$$

Let γ_{AB} be the arc of $C(w, \Delta_2)$ from v_A counterclockwise to v_B .

Lemma 11.7 (the arc of a gap). *Let (A, B) be a gap.*

- (a) $\gamma_{AB} \subseteq \partial K$.
- (b) *Follow pure right T-edges in the K -row from $v_0 := v_A$: $v_0, v_1, \dots$. As long as $v_i \in \gamma_{AB} \setminus \{v_B\}$ and v_i has a pure right T-neighbour, that neighbour is the point of γ_{AB} at angle α_2 counterclockwise from v_i . Consequently either the walk reaches v_B , which happens only if $\vartheta(v_B) - \vartheta(v_A) = j\alpha_2$ with an integer $j \geq 1$, and then $v_1, \dots, v_{j-1}$ are K -row points in the relative interior of γ_{AB} ; or it stops at a K -row point $b \in \gamma_{AB} \setminus \{v_B\}$ that has no pure right T-neighbour.*

Proof. (a) This is the paragraph “The arc between consecutive apexes” in Section 5.4, which uses only that $v_A, v_B \in K \cap C(w, \Delta_2)$ are less than π apart about w (5). (b) The proof of Lemma 5.9 uses only (a), the fact that $w_{v_A} = w$ (here Lemma 11.5(b)), and $v_B \in X$; it applies verbatim. The walk cannot pass v_B , since a point of γ_{AB} at angle in $(0, \alpha_2)$ from v_B is at distance less than 1 from it. $\square$

We call the gap *unbroken* if the walk of Lemma 11.7(b) reaches v_B , and *broken* otherwise; U denotes the number of unbroken gaps, and U_Z the number in Z . For an unbroken gap, $j \geq 1$ is its number of K -steps and $k \geq 1$ its number of D-steps, and (5) reads

$$(6) \quad j\alpha_2 = k\alpha_1 - (\varsigma_B - \varsigma_A)\varphi_\tau.$$

Lemma 11.8 (disjointness of gap arcs). *Two closed arcs $\gamma_{AB} \neq \gamma_{A'B'}$ of gaps are disjoint, except when the gaps are consecutive gaps $(A, B), (B, B')$ of the same component, which share exactly the point v_B . In particular the half-open arcs $[v_A, v_B) := \gamma_{AB} \setminus \{v_B\}$ of all gaps are pairwise disjoint.*

Proof. Same component Z . By Lemma 11.6 the K -ends of the rungs of Z are in the same order about w as their D-ends, so the arcs of the gaps of Z are consecutive intervals of angles, $[\vartheta(v_{A_i}), \vartheta(v_{A_{i+1}})]$; two of them meet only if they are consecutive, in the common endpoint.

Different components $Z \neq Z'$ with the same centre w . Both lie on $C(w, \Delta)$, and all points of X on this circle lie in an arc of width at most $\pi/3$ (they are pairwise at most Δ apart). A point of Z' at angle less than α_1 from a point of Z would be at distance less than 1 from it, and it cannot coincide with it. Since consecutive points of Z are exactly α_1 apart, every angle in $[\min \vartheta(Z), \max \vartheta(Z)]$ is within $\alpha_1/2$ of a point of Z ; so no point of Z' has its angle in this range, and every point of Z' is at angle at least α_1 from $\min \vartheta(Z)$ and from $\max \vartheta(Z)$. As consecutive points of Z' are also exactly α_1 apart, Z' cannot have points on both sides of the range of Z . Hence the angular ranges $[\min \vartheta(Z), \max \vartheta(Z)]$ and $[\min \vartheta(Z'), \max \vartheta(Z')]$ are disjoint and at least α_1 apart. By Lemma 11.4(a) the K -ends of the rungs of Z lie in

$[\min \vartheta(Z) - \varphi_\tau, \max \vartheta(Z) + \varphi_\tau]$, and likewise for Z' . Since $2\varphi_\tau < \alpha_1$ (Lemma 11.4(b)), these two closed intervals are disjoint. The arcs of the gaps of Z lie in the first, those of Z' in the second.

Different centres $w \neq w'$. Let x be a common point of $\gamma_{AB} \subseteq C(w, \Delta_2)$ and $\gamma_{A'B'} \subseteq C(w', \Delta_2)$, both nondegenerate arcs of ∂K (Lemma 11.7(a)). If x were in the relative interior of γ_{AB} , then near x the boundary ∂K coincides with γ_{AB} on both sides of x , and $\gamma_{A'B'}$ would contain a nondegenerate arc of ∂K lying on both circles, forcing $w = w'$. So x is an endpoint of both arcs, i.e. the K-end of a rung A or B and of a rung A' or B' . By (R1), x is the K-end of only one rung; the D-end of that rung lies in one component, and the centre of that component is w_x (Lemma 11.5(b)). So both gaps have centre w_x , a contradiction.

For the last statement: the only common point of two closed gap arcs is v_B for consecutive gaps (A, B) , (B, B') , and $v_B \notin [v_A, v_B]$. $\square$

Theorem 11.9 (break count). $e(S) \leq s + U + 70 + C_{\text{bad}}$.

Proof. If the K-row or the D-row has at most 2 points, then $\rho \leq 2$ by (R1), and $e(S) \leq e_T + \rho + C_{\text{bad}} \leq s + 2 + C_{\text{bad}}$ by (T1). Otherwise define B_K, B_D as in Section 5.4; by Lemma 5.6 and Lemma 5.1 (fewer than 71 mixed T-edges in Region II), as in (4),

$$e_T \leq s - B_K - B_D + 70,$$

and a row point with no pure right T-neighbour is the left element of a pair counted in B_K or B_D .

K-breaks. By Lemma 11.7(b) every broken gap (A, B) has a K-row point $b_{AB} \in [v_A, v_B]$ with no pure right T-neighbour. By Lemma 11.8 these points are distinct for distinct gaps, so they give distinct pairs in B_K , and $B_K \geq \sum_Z (m_Z - 1 - U_Z)$, the sum over components with $m_Z \geq 1$.

D-breaks. Every component Z with $m_Z \geq 1$ is a path (Lemma 11.5(a); this includes $|Z| = 1$), and its counterclockwise last vertex has no pure right T-neighbour; distinct components give distinct such vertices. So $B_D \geq \#\{Z : m_Z \geq 1\}$.

Adding, $B_K + B_D \geq \sum_Z (1 + m_Z - 1 - U_Z) = \rho - U$, and

$$e(S) \leq e_T + \rho + C_{\text{bad}} \leq s - (\rho - U) + 70 + \rho + C_{\text{bad}} = s + U + 70 + C_{\text{bad}}. \quad \square$$

Lemma 11.10 (signs of unbroken gaps). (a) *At most 6 unbroken gaps have $\varsigma_A = \varsigma_B$.*

(b) *At most 24 rungs B are shared by two unbroken gaps (A, B) and (B, C) .*

Proof. Arithmetic. Suppose $J\alpha_2 = K\alpha_1$ with integers $J \geq 1, K \geq 1$. Since $\alpha_2 > \alpha_1, K > J$, so $J(\alpha_2 - \alpha_1) = (K - J)\alpha_1 \geq \alpha_1$ and $K\alpha_1 \geq 1/(1/\alpha_1 - 1/\alpha_2)$. As in the paragraph “The path cannot land on v_B ” of Section 5.4, $1/\alpha_1 - 1/\alpha_2 \leq t + 1/(23\Delta_2)$, and $t \leq 1, \Delta_2 \geq 10^4$ give $J\alpha_2 = K\alpha_1 > 0.99999$.^[IIIh]

(a) If $\varsigma_A = \varsigma_B$, (6) reads $j\alpha_2 = k\alpha_1$, so γ_{AB} has angle greater than 0.99999 and length greater than 0.99999 Δ_2 . By Lemma 11.8 the half-open arcs are disjoint, and the perimeter of $K \subseteq B(w, \Delta_2)$ is at most $2\pi\Delta_2$; as $2\pi/0.99999 < 6.3$, there are at most 6 such gaps.^[IIIh]

(b) Let (A, B) and (B, C) be unbroken gaps of one component. If one of them has equal signs, B is one of the at most $2 \cdot 6$ rungs of the gaps in (a). Otherwise $\varsigma_A = -\varsigma_B = \varsigma_C$, and adding (6) for the two gaps gives $(j + j')\alpha_2 = (k + k')\alpha_1$; the walk from v_A reaches v_C along $\gamma_{AB} \cup \gamma_{BC}$, and by the arithmetic above this arc has length greater than 0.99999 Δ_2 . Each point of ∂K lies in at most one half-open gap arc, hence in at most two of the half-open arcs $[v_A, v_C]$, so there are at most $4\pi/0.99999 < 12.6$ such rungs B . In total at most $12 + 12 = 24$.^[IIIh] $\square$

(By the same arithmetic, same-sign unbroken gaps are impossible when $t < 3/\pi - 1/(23\Delta_2)$, since then $k\alpha_1 > \pi/3$; we do not need this.)

Proof of Theorem 11.3. We show $2U \leq p + 34$; with Theorem 11.9 this gives $e(S) \leq s + \frac{1}{2}p + 17 + 70 + C_{\text{bad}}$.

Endpoints. Let (A, B) be a gap. Its K-ends v_A, v_B are rung K-ends. A rung K-end of Δ_2 -degree at least 2 is of type (i) or of type (ii) (Lemma 5.2 and Section 5.3). There are at most 5 vertices of type (i), each the K-end of at most one rung, which is an end rung of at most two gaps; so at most 10 pairs (gap, endpoint) involve a vertex of type (i). If v is of type (ii), its rung partner z has no T-neighbour (Lemma 5.4), so its component is $\{z\}$, $m_{\{z\}} = 1$, and the rung is in no gap. All other gap endpoints have Δ_2 -degree 1, i.e. lie in Q .

Interior walk points. Let (A, B) be unbroken, with walk $v_A = v_0, v_1, \dots, v_j = v_B$. For $0 < i < j$, v_i is a K-row point in the relative interior of $\gamma_{AB} \subseteq C(w, \Delta_2) \cap \partial K$, and $|v_i w| = \Delta_2$. Suppose $|v_i w'| = \Delta_2$ for some $w' \in X \setminus \{w\}$. Since $K \subseteq B(w', \Delta_2)$, γ_{AB} is contained in $C(w, \Delta_2) \cap B(w', \Delta_2)$, which is a closed arc whose endpoints are the points of $C(w, \Delta_2) \cap C(w', \Delta_2)$. So v_i is an endpoint of that arc, and cannot be a relative interior point of γ_{AB} . Hence $v_i \in Q$.

So the closed arc γ_{AB} of an unbroken gap contains the $j + 1 \geq 2$ distinct K-row points $v_0, \dots, v_j$, all in Q except for the type-(i) endpoints counted above.

Counting. By Lemma 11.8 two closed arcs of distinct gaps share a point only if they are consecutive gaps $(A, B), (B, C)$, and then they share only v_B . (A shared endpoint $v_B = v_{A'}$ means, by (R1), $B = A'$; equal left ends, or equal right ends, give the same gap.) By Lemma 11.10(b) this happens for at most 24 rungs B with both gaps unbroken. Hence

$$2U - 10 \leq \sum_{(A,B) \text{ unbroken}} \#(Q \cap \gamma_{AB}) \leq p + 24,$$

and $2U \leq p + 34$. □

Remark 11.11. (a) The factor $\frac{1}{2}$ cannot be improved by an argument of this local kind. Take $w = 0$, K-row points on $C(0, \Delta_2)$ and D-row points on $C(0, \Delta)$, with rungs of alternating signs and one K-step and one D-step per gap, at the value of τ (with $1 - \tau = 1/(8\Delta_2^2) + O(\Delta_2^{-4})$) at which (6) holds with $j = k = 1$. In 60-digit arithmetic at $\Delta_2 = 10^4$ (script `r7_A2_configs.py`, configuration A) all distances in a window of 8 gaps are at least 1, the unit pairs are exactly the expected T-edges and rungs, and the window has $e - s = U - 1$ and $2U$ points of Δ_2 -degree 1. This is a local configuration; whether it extends to a set with $\Theta(n)$ unbroken gaps is not known.

(b) Conversely, a short argument (the radial “shadow” of a K-row point contains no point of the D-row) suggests that $U = O(1)$ in Region II except on a single curve $\alpha_2 - \alpha_1 = 2\varphi_\tau$; this has not been refereed and is not used here.

11.2. Average weight at most 7 in Region II.

Theorem 11.12. *Assume the nondegenerate regime with $\Delta_2 \geq 2 \cdot 10^{15}$ and $\sqrt{3}/2 + 3\beta \leq \tau \leq 1$. Then*

$$\sum_{y \in R} v(y) \leq 7r + C_4, \quad C_4 = 1.3 \cdot 10^{26}.$$

Throughout this subsection we assume the hypotheses of Theorem 11.12. Then $\tau \leq 1$ gives $t \leq 1$ and $t \geq \tau \geq t_{\text{II}} := \sqrt{3}/2 + 3\beta > 0.89602$ (Section 2).^[IIh]

What is used from Sections 9 and 10. We use the parameters $\varepsilon_0 = 10^{-6}$, $\Lambda = 10^9$, $\rho = 6$, $\delta_0 = 10^{-4}$ and the notion of a good' point of Section 10. The following statements are proved in Sections 9 and 10 without using any hypothesis on τ (the standing assumption of Section 10 that $\tau \leq \sqrt{3}/2 - 2\beta$ or $\tau > 1$ is not used in their proofs), and we use them verbatim: Section 6.1, Lemma 9.3 (in particular (G5)), Lemma 9.4, Lemma 10.2, Lemma 10.3, and the consequences stated after Table 2: for a good' point y and a neighbour q of y at angle ψ with height $c(q) =$

$\cos \psi$, a K-neighbour has $c \in [h - \delta_0, h]$ and $|\psi| < 90^\circ + \varepsilon_0$, a D-neighbour (in S or in R) has $c \in [h + H - \delta_0, h + H + \delta_0]$ with $H \geq t - \delta_0/2$, an interior neighbour has $c < h$, and so a neighbour with $c > h$ is a D-neighbour; two K-neighbours, or two D-neighbours, form a mirror pair up to 0.012° , and there are at most two of each kind; and for an interior point $x \in B(y, 5)$ at depth d and a neighbour q of x in direction θ from the upward vertical, $\cos \theta \in [d - \delta_0, d]$ if $q \in S \setminus D$ and $\cos \theta \geq d + H - \delta_0$ if $q \in D$; in particular an interior point of depth greater than $1 + \delta_0$ has no neighbour in S . By Section 6.1(b) every neighbour of y is a K-neighbour, a D-neighbour or an interior neighbour, and the neighbours in S are the K-neighbours and the D-neighbours in S ; there are no points of R on ∂K , and $h > 0$. The Region-dependent results Lemmas 10.4 to 10.9 are *not* used; they are replaced by Lemmas 11.13 to 11.16. We do not use pinning (Lemma 10.3(c)) either: only the lower bound $H \geq t - \delta_0/2$ on the level of diametral points enters.

Lemma 11.13. *Let y be good'. Then y has at most one D-neighbour, and every D-neighbour has $|\psi| < 26.38^\circ$. Consequently $m_y \leq 3$, and if $m_y = 3$ the neighbours of y in S are two K-neighbours and one D-neighbour.*

Proof. A D-neighbour has $c \geq h + H - \delta_0 \geq t - \frac{3}{2}\delta_0 \geq t_{II} - \frac{3}{2}\delta_0 > 0.89587 > \cos 26.38^\circ$, since $h > 0$.^[IIh] Two neighbours are at least 60° apart as seen from y , so they cannot both lie in the open cone $|\psi| < 26.38^\circ$. There are at most two K-neighbours, and every neighbour in S is a K-neighbour or a D-neighbour. $\square$

This replaces the fan analysis of Sections 9 and 10: in particular there are no fans (F6, F5, HF) and no caps among good' points in Region II.

Lemma 11.14 (classification). *Let y be good' with $v(y) \geq 8$. Then $v(y) = 8$ and y is of one of the following types.*

- (A) $(\deg y, m_y) = (6, 2)$: the neighbours form a regular hexagon within 0.012° of $\{\pm 30^\circ, \pm 90^\circ, \pm 150^\circ\}$; the two near $\pm 30^\circ$ are K-neighbours and the other four are interior; $h \geq \sqrt{3}/2$; the neighbours near $\pm 150^\circ$ have depth greater than 1.73.
- (Dt_{II}) $(\deg y, m_y) = (5, 3)$: K-neighbours at $\psi_- < 0 < \psi_+$ with $|\psi_\pm| \in [84.02^\circ, 90.006^\circ]$, one D-neighbour in S at ψ_D with $|\psi_D| < 26.38^\circ$, and two interior neighbours with angles in the arc $I_y := [\psi_+ + 60^\circ, \psi_- + 300^\circ]$; moreover $h \leq 1 - t + \frac{3}{2}\delta_0 < 0.10413$.

Proof. By Lemma 11.13, $m_y \leq 3$, and $\deg y \leq 6$; so $v(y) \geq 8$ leaves $(\deg y, m_y) \in \{(6, 2), (6, 3), (5, 3)\}$.

$\deg y = 6$. Six neighbours pairwise at least 60° apart form a regular hexagon $\{\varphi + 60^\circ k\}$ with $\varphi \in [-30^\circ, 30^\circ]$. The slot at φ has $c \geq \cos 30^\circ$, and one of the slots $\varphi \pm 60^\circ$ has $|\psi| \leq 60^\circ$, hence $c \geq \frac{1}{2}$. If $h < \frac{1}{2}$, both of these slots have $c > h$ and are D-neighbours, contradicting Lemma 11.13. So $h \geq \frac{1}{2}$; then a D-neighbour would need $c \geq h + H - \delta_0 \geq \frac{1}{2} + t_{II} - \frac{3}{2}\delta_0 > 1.39$, which is impossible, so y has no D-neighbour and every slot has $c \leq h$. The slot at φ gives $h \geq \sqrt{3}/2$. A K-neighbour then has $c \geq h - \delta_0 \geq \sqrt{3}/2 - \delta_0$, i.e. $|\psi| \leq 30.012^\circ$.^[IIh] Two slots of the hexagon are 60° apart, so at most two slots qualify, and two only if they lie within 0.012° of $\pm 30^\circ$. Hence $m_y \leq 2$, $(6, 3)$ is impossible, and $(6, 2)$ is type A: the other four neighbours are not in S and are not D-neighbours, so they are interior (Section 6.1(b)). The neighbours near $\pm 150^\circ$ have depth $h - \cos \psi \geq \sqrt{3}/2 + \cos 30.012^\circ > 1.7319$.^[IIh]

$(\deg y, m_y) = (5, 3)$. By Lemma 11.13 the neighbours in S are two K-neighbours, a mirror pair at $\psi_- < 0 < \psi_+$, and a D-neighbour in S at $|\psi_D| < 26.38^\circ$. Since $c(\text{D-neighbour}) \leq 1$, we get $h \leq 1 - H + \delta_0 \leq 1 - t + \frac{3}{2}\delta_0 \leq 1 - t_{II} + \frac{3}{2}\delta_0 < 0.10413$. Each K-neighbour has $\cos \psi_\pm \in [h - \delta_0, h] \subseteq [-\delta_0, 0.10413]$, so $|\psi_\pm| \in [84.02^\circ, 90.006^\circ]$.^[IIh] The two remaining neighbours are not in S , and are not D-neighbours by Lemma 11.13; so they are interior. Each is at least 60°

from ψ_- , ψ_D and ψ_+ . It cannot lie in the upper arc (ψ_-, ψ_+) : there it would lie between ψ_D and ψ_+ (or ψ_-) at least 60° from both, which needs $\psi_+ - \psi_D \geq 120^\circ$, i.e. $\psi_+ > 93.6^\circ$. So both lie in I_y . $\square$

So in Region II every good' point has weight at most 8. We move the excess over 7.

Discharging. Every good' point y with $v(y) = 8$ sends charge to two of its interior neighbours, its *receivers*:

- type A: $\frac{1}{2}$ to each of the neighbours near $\pm 150^\circ$;
- type Dt_{II}: $\frac{1}{2}$ to each of its two interior neighbours.

Then every receiver x with $v(x) = 7$ (a *forwarder*) passes all the charge it has received to the neighbour g of x with $g - x = x - k$, where k is the unique neighbour of x in S (Lemma 11.15(c) shows that k is unique and g exists). The point g is the *target* of x .

Lemma 11.15 (receivers). *Let x be a receiver of a good' point y ; depths and directions at x are measured in the frame of y .*

- If y is of type A, then x has depth greater than 1.73, so $m_x = 0$ and $v(x) \leq 6$.*
- If y is of type Dt_{II}, then x has depth $d_x \in [0.86597, 1.10413]$; x has no D-neighbour; every K-neighbour of x is in a direction β with $|\beta| \leq 30.02^\circ$; y is seen from x in a direction θ_y with $|\theta_y| \leq 35.98^\circ$; and $m_x \leq 1$, so $v(x) \leq 7$.*
- If $v(x) = 7$, then y is of type Dt_{II}, $\deg x = 6$, x has exactly one neighbour k in S , a K-row point in a direction β with $|\beta| \leq 30.02^\circ$, and the target $g = 2x - k$ is a neighbour of x of depth greater than 1.73; so $m_g = 0$ and $v(g) \leq 6$.*

In particular every receiver and every target lies in $R \setminus D$ and has $m \leq 1$, so it is not a sender.

Proof. Receivers are interior neighbours of y , so they lie in $\text{int } K$ and hence in $R \setminus D$. The depth of the neighbour of y at angle ψ is $h - \cos \psi$.

(a) By Lemma 11.14. An interior point of depth greater than $1 + \delta_0$ has no neighbour in S .

(b) Let x be at angle $\psi_x \in I_y$. If $\psi_x \leq 180^\circ$, then $\cos \psi_x \leq \cos(\psi_+ + 60^\circ)$ and $h \geq \cos \psi_+$, so $d_x \geq \cos \psi_+ - \cos(\psi_+ + 60^\circ) = \sin(\psi_+ + 30^\circ) \geq \sin 120.006^\circ > 0.86597$, as $\sin$ is decreasing on $[114^\circ, 120.006^\circ]$; the case $\psi_x \geq 180^\circ$ is symmetric with ψ_- .^[IIh] Also $d_x \leq h + 1 < 1.10413$. A D-neighbour of x would need $\cos \theta \geq d_x + H - \delta_0 > 0.86597 + 0.8958 > 1$; so there is none. A K-neighbour has $\cos \beta \geq d_x - \delta_0 > 0.86587$, i.e. $|\beta| \leq 30.02^\circ$.^[IIh] As seen from x , y is in the direction $\theta_y = \psi_x - 180^\circ \in [\psi_+ - 120^\circ, \psi_- + 120^\circ] \subseteq [-35.98^\circ, 35.98^\circ]$.^[IIh]

Suppose x had two K-neighbours, at $\beta_1 < \beta_2$. They are at least 60° apart and $|\beta_i| \leq 30.02^\circ$, so $\beta_1 \leq -29.98^\circ$ and $\beta_2 \geq 29.98^\circ$. The point $y \in R$ is a third neighbour of x , at least 60° from both. If $\theta_y \leq \beta_1$, then $\beta_1 - \theta_y \leq -29.98^\circ + 35.98^\circ < 60^\circ$; if $\theta_y \geq \beta_2$, then likewise $\theta_y - \beta_2 < 60^\circ$. So $\beta_1 < \theta_y < \beta_2$, and then $\beta_2 - \beta_1 \geq 120^\circ > 60.04^\circ$. This contradiction gives $m_x \leq 1$. (This argument does not depend on t ; in particular it covers the case $|\psi_\pm| \approx 90^\circ$, $h \approx 0$, which occurs for every $t \in [t_{II}, 1]$.)

(c) By (a), y is of type Dt_{II}, and by (b) $m_x = 1$ and $\deg x = 6$, with k a K-row point at $|\beta| \leq 30.02^\circ$. Six neighbours pairwise at least 60° apart form a regular hexagon, so $g = x - (k - x)$ is a neighbour of x . Its depth is $d_x + \cos \beta \geq 2d_x - \delta_0 > 1.7318$.^[IIh] As $g \in B(y, 2)$, it has no neighbour in S . $\square$

Lemma 11.16 (loads). *A forwarder receives exactly $\frac{1}{2}$. Every point of R receives at most 1 in total, directly or by forwarding.*

Proof. Frames. Let x receive charge. Its *senders* are good' points of type A or Dt_{II} of which x is a receiver, and forwarders of which x is the target; a forwarder need not be good', and all statements about it are made in the frame of its own (good', type Dt_{II}) sender. Every direct

sender of x is at distance 1 from x , and the sender of a forwarder of x is at distance at most 2 from x . So all good' points involved lie within distance $4 \leq \rho - 1$ of each other, and by (G5) their frames differ by rotations of at most ε_0 . Directions of unit vectors between points of X therefore agree in all these frames up to $\varepsilon_0 < 10^{-4}$ degrees, which is negligible against the margins below (at least 24°). The quantities m and v do not depend on a frame.

Forwarders. A forwarder has $v = 7$, hence $m = 1$; by Lemma 11.15(a),(c) it is not a receiver of a type-A point and not a target. So all its senders are of type Dt_{II}, at directions in $[-35.98^\circ, 35.98^\circ]$, and its neighbour $k \in S$ is at $|\beta| \leq 30.02^\circ$. Suppose it had two senders, at $\delta_1 < \delta_2$. Then $\delta_2 - \delta_1 \geq 60^\circ$ gives $\delta_1 \leq -24.02^\circ$ and $\delta_2 \geq 24.02^\circ$. The neighbour k is at least 60° from both: if β lies between them this needs $\delta_2 - \delta_1 \geq 120^\circ > 71.96^\circ$, and otherwise $|\beta| \geq 84^\circ$. Both are impossible. So a forwarder has exactly one sender and passes on exactly $\frac{1}{2}$.

Loads. As seen from x , a type-A sender is within 0.012° of $\pm 30^\circ$ (its receivers are near $\pm 150^\circ$), a type-Dt_{II} sender is in $[-35.98^\circ, 35.98^\circ]$ (Lemma 11.15(b)), and a forwarder x' with target x is in the direction of its neighbour k as seen from x' , i.e. at $|\beta| \leq 30.02^\circ$ (Lemma 11.15(c)). Senders of x are distinct neighbours of x , pairwise at least 60° apart, in a window of width less than $72^\circ < 120^\circ$; so x has at most two senders. Each gives $\frac{1}{2}$. Hence the load of x is at most 1. $\square$

Proof of Theorem 11.12. Give each $y \in R$ the charge $v(y)$ and apply the rules; charge moves only between points of R , so the total is unchanged. A good' point of weight 8 (type A or Dt_{II}) sends 1 and receives nothing (receivers and targets have $m \leq 1$, hence weight at most 7), so it ends at 7. A forwarder has weight 7, receives $\frac{1}{2}$ and passes it on, so it ends at 7. Every other receiver or target has weight at most 6 (Lemma 11.15) and receives at most 1 (Lemma 11.16), so it ends at most at 7. Points of $R \cap D$ are never receivers (receivers are interior points of K). Every other point keeps its weight; this is at most 7 unless the point has weight at least 8 and is not good' (Lemma 11.14). By Lemma 10.2 there are fewer than $N'_g = 2.6 \cdot 10^{25}$ such points; they never receive, and they end at most at 12 (Lemma 3.3). Hence $\sum_{y \in R} v(y) \leq 7r + 5N'_g < 7r + 1.3 \cdot 10^{26}$. $\square$

Remark 11.17. (a) The fan windows of Lemma 9.8 ($t \in [0.75, 0.95]$) come from the coarse parameter $\varepsilon_0 = 1/50$ of Section 9. With the parameters of Section 10, Lemma 11.13 shows directly that in Region II ($t \geq 0.896$) no good' point has two D-neighbours, so fans and caps do not occur there.

(b) As in Remark 10.10(a), the exact stack of hexagonal rows below a straight K-row has average weight 6 on R but per-period weights $8, 6, \dots, 6, 4$, independently of t ; so radius-one discharging stops at 7 in Region II as well. An average weight $6 + o(1)$ in Region II would give $\frac{9}{7}$ there by Lemma 11.2 and Remark 3.6.

11.3. Computations for this section. The checks marked ^[IIIh] (and ^[LP] in Lemma 11.2) are elementary numerical inequalities; they were recomputed with 40-digit arithmetic. The following scripts (in `scripts/` of the repository) were run with the repository's Python environment; none of them is part of the proof.

- `r7_A2_constants.py`: the identity $|z - w| = \Delta$ of Lemma 11.4(a) (symbolically), the bound $2\varphi_\tau/\alpha_1 < 0.8881$, the counts 6 and 24 of Lemma 11.10, and the values of the linear combination. All checks pass.
- `r7_A2_configs.py`: exact (60-digit) configurations in the single-centre model near $\tau = 1$ and near $\tau = \sqrt{3}/2 + 3\beta$, including configuration A of Remark 11.11(a), and two candidate configurations that turn out not to be realisable. No configuration violates Theorem 11.9, Lemma 11.10 or Theorem 11.3.
- `r7_A2_shadow.py`: the computations behind Remark 11.11(b) (not used).

- `r7_B_margins.py`: 60-digit interval checks of the angular and depth margins of [Section 11.2](#) (with slightly more conservative constants than in the text). All pass.
- `r7_B_lattice_t1.py`: an exact rational model at $t = 1$; in $2 \cdot 10^4$ random dense configurations the heavy points are of types A and Dt_{II} only, and the final charge is at most 7.
- `r7_B_classify.py` and `r7_B_rules.py`: a slot-model scan over $t \in [0.896, 1]$ (only types A and Dt_{II} occur, maximal weight 8), and a floating-point beam search over $1.8 \cdot 10^5$ configurations of the discharging (maximal final charge 7). They import the helper files `r6_below_classify_B1copy.py` and `r6_below_rule_check_B1copy.py`.
- `r7_refB_margins.py`, `r7_refB_classify.py`, `r7_refB_search.py`, `r7_refB_exact.py`: the referee's independent implementation for [Section 11.2](#): 60-digit margins (all pass, including the linear combination), a classification scan (only types A and Dt_{II}), a floating-point search and an exact search (9506 configurations, maximal final charge 7).

12. THE REGIME $\tau > 1$: PROOF OF THEOREM 1(C)

When $\tau > 1$ the diametral endpoints are far from K , and the δ -graph splits into a part on D and a part inside K . We bound the part inside K by a global argument ([Theorem 12.1](#)) that uses no local classification at all. In this section G is the δ -graph of X (scaled so that $\delta = 1$), and for a subgraph H we write

$$\text{Def}(H) := 3|V(H)| - |E(H)|.$$

Since G has maximum degree at most 6, $\text{Def}(H) \geq 0$ for every induced subgraph, and $\mu(\delta) = |E(G)| = 3n - \text{Def}(G)$.

Theorem 12.1. *Let X be a finite planar set with minimum distance 1, $n = |X| \geq 1$ and $\Delta \leq 1.94 \Delta_2$, and let $K = \bigcap_{w \in X} B(w, \Delta_2)$. Let $Y \subseteq X \cap K$ and $A := Y \cap \partial K$. Then*

$$\text{Def}(G[Y]) \geq 2|A| - C_E n^{2/3}, \quad C_E := 3 \cdot 10^4.$$

In particular, in the nondegenerate regime, $\text{Def}(G[X \setminus D]) \geq 2|S \setminus D| - 3 \cdot 10^4 n^{2/3}$, for every τ .

For the last statement take $Y = X \setminus D$: by [Lemma 2.1\(a\)](#) and [Section 6.1\(b\)](#), $X \setminus D \subseteq K$ and $(X \setminus D) \cap \partial K = S \setminus D$. The constants depend only on $\Delta \leq 1.94 \Delta_2$, not on t .

Status. [Theorem 12.1](#) and [Corollary 12.4](#) were checked in three independent referee passes. The first two found the argument correct but found constant-level gaps in an earlier write-up (the window around a corner in Step 6 omitted nearby corners; the laminarity claim of Step 6 was stated for all gap arcs; the case of small n was missing; a factor in Step 3; the winding argument of [Lemma 12.3](#) was not written out). The complete corrected proof below incorporates these repairs, and the third pass, on this corrected proof, found no errors or gaps. The numerical inequalities are checked in exact arithmetic by `r6_Eprime_full_checks.py` (34 assertions).

12.1. Classical facts. We use the following standard facts.

- (P1) *G is a plane straight-line graph.* Its edges are unit segments between points at mutual distance at least 1. A point of X in the relative interior of an edge ab would be within $\frac{1}{2}$ of a or b ; so no vertex lies inside an edge, and no two edges overlap collinearly or touch at an interior point. If two edges ab and cd crossed properly, the diagonals of the convex quadrilateral $acbd$ would give $|ac| + |bd| < |ab| + |cd| = 2$, which is impossible.
- (P2) *Faces and boundary walks.* For a plane graph with $cc \geq 1$ components, $V - E + F = 1 + cc$, and $\sum_f |f| = 2E$, where $|f|$ is the total length of the boundary walks of the face f (see [\[20\]](#)). Hence

$$\text{Def} = 3 + 3cc + \sum_f (|f| - 3).$$

A bounded face of a simple plane graph has a cycle in its boundary, so $|f| \geq 3$. The outer face has $|f_{\text{out}}| = \sum_j |w_j|$, summed over the components C_j incident with it, where w_j is the outer boundary walk of C_j .

- (P3) *Separation and perimeter.* If a compact set $\Gamma \subset \mathbb{R}^2$ separates a convex body Q from ∞ and $\Gamma \cap Q = \emptyset$, then $\mathcal{H}^1(\Gamma) \geq \text{per } Q$. Indeed the nearest-point map P_Q is 1-Lipschitz [23] and maps Γ onto ∂Q : for $b \in \partial Q$ with outer normal ν , the ray $b + \mathbb{R}_+\nu$ meets Γ , and $P_Q = b$ on that ray.
- (P4) Perimeter is monotone under inclusion of convex sets, and $\text{per } Q \leq \pi \text{diam } Q$ (Cauchy's formula) [23].
- (P5) *Steinhagen's inequality* [24]: every planar convex body has minimal width $w \leq 3r_{\text{in}}$, where r_{in} is its inradius.
- (P6) *Non-interlacing* (a consequence of the Jordan curve theorem, see [20]): if Q is a closed topological disc and a, b, a', b' lie on ∂Q in this cyclic order, then every path in Q from a to a' meets every path in Q from b to b' .
- (P7) *Monotone maps of circles.* A continuous surjection between two circles whose fibres are connected is monotone: preimages of arcs are arcs, and the images of the arcs between cyclically consecutive points of a finite set tile the target circle.
- (P8) If $|A| \geq 2$, the boundary arcs of ∂K between cyclically consecutive points of A have length at least the chord, hence at least 1. So $|A| \leq \text{per } K + 1$, and an arc of length ℓ contains at most $\ell + 1$ points of A . Also $\text{per } K \leq 2\pi\Delta_2$, since $K \subseteq B(w, \Delta_2)$.

Facts about K . For $z \in K$ put $\text{depth}(z) := \text{dist}(z, \mathbb{R}^2 \setminus K) = \min_{w \in X} (\Delta_2 - |z - w|)$. Depth is 1-Lipschitz, and $\text{depth}(z) \geq \eta$ if and only if $B(z, \eta) \subseteq K$. Since ∂K is a finite union of arcs of circles $C(w, \Delta_2)$, K is strictly convex: a segment between two points of K meets ∂K at most in its endpoints. If $\text{int } K = \emptyset$ then $|Y| \leq 1$ and there is nothing to prove; from now on $\text{int } K \neq \emptyset$.

Lemma 12.2 (fatness). *If $\Delta \leq 1.94\Delta_2$ and $\text{int } K \neq \emptyset$, then $\text{diam } K \leq 100w(K)$, and hence $r_{\text{in}}(K) \geq \text{diam } K/300$.*

Proof. Let the minimal width w_0 of K be attained in the direction u_0 , let $v_0 \perp u_0$, and let L be the width in the direction v_0 . Then $\text{diam } K \leq \sqrt{w_0^2 + L^2}$, and we may assume $L \geq 3w_0$. Let $p, q \in K$ be extreme in the directions $-v_0$ and v_0 , and $m := (p+q)/2$. The line $m + \mathbb{R}u_0$ meets K in a segment $[y_-, y_+]$ of length at most w_0 ; write $y_+ = m + su_0$ with $0 \leq s \leq w_0$. As $y_+ \in \partial K$, we have $y_+ \in C(x_+, \Delta_2)$ for some $x_+ \in X$; put $n_+ := (y_+ - x_+)/\Delta_2$. Every $k \in K \subseteq B(x_+, \Delta_2)$ satisfies $(k - y_+) \cdot n_+ \leq -|k - y_+|^2/(2\Delta_2)$. Adding this for $k = p$ and $k = q$ gives $-2su_0 \cdot n_+ < 0$, so $u_0 \cdot n_+ > 0$; using the two inequalities separately gives $|(p - m) \cdot n_+| \leq su_0 \cdot n_+ \leq w_0$, so $|(q - p) \cdot n_+| \leq 2w_0$. Writing $q - p = Lv_0 + \sigma u_0$ with $|\sigma| \leq w_0$, we get $|v_0 \cdot n_+| \leq 3w_0/L$ and $u_0 \cdot n_+ \geq \sqrt{1 - 9w_0^2/L^2}$. The same argument at y_- gives x_- with $u_0 \cdot n_- \leq -\sqrt{1 - 9w_0^2/L^2}$. Hence

$$1.94\Delta_2 \geq |x_+ - x_-| \geq (x_- - x_+) \cdot u_0 \geq 2\Delta_2 \sqrt{1 - 9w_0^2/L^2} - w_0.$$

If $w_0 \leq 0.02\Delta_2$, this gives $\sqrt{1 - 9w_0^2/L^2} \leq 0.98$, so $L/w_0 \leq 3/\sqrt{0.0396} < 15.1$ and $\text{diam } K < \sqrt{1 + 15.1^2} w_0 < 100w_0$. Otherwise $\text{diam } K \leq 2\Delta_2 < 100w_0$. The last claim follows from (P5).^[E] $\square$

(Numerically, the largest aspect ratio $\text{diam } K/w(K)$ seems to be about 8.1, attained by the lens with centre distance $1.94\Delta_2$.)

12.2. Proof of Theorem 12.1. *Step 0 (parameters).* Put $\rho_0 := \lfloor \min(n^{1/3}, \Delta_2/400) \rfloor$.

- (F1) If $\rho_0 = 0$, then $\Delta_2 < 400$, so $\Delta < 776$ and $n \leq (2\Delta + 1)^2 < 1553^2 < 135^3$. Then $2|A| \leq 2n \leq 270n^{2/3}$, and Def ≥ 0 proves the claim.

Otherwise $\rho_0 \geq 1$. For $\rho \in \{\rho_0, \dots, 2\rho_0 - 1\}$ the shells $Y_\rho := \{y \in Y : \rho \leq \text{depth}(y) < \rho + 1\}$ are disjoint; fix ρ with $|Y_\rho| \leq n/\rho_0$ and put $\eta := \rho + \frac{1}{2}$. Then $\frac{3}{2} \leq \eta \leq 2\rho_0 - \frac{1}{2} < 2n^{1/3}$ and $\eta < \Delta_2/200$.

(F2) If $r_{\text{in}}(K) \leq \eta$, then by Lemma 12.2, (P4) and (P8), $|A| \leq \text{per } K + 1 \leq 300\pi\eta + 1$, so $2|A| \leq 1200\pi n^{1/3} + 2 \leq 3772 n^{2/3}$.

(F3) If $\text{per } K < 100\eta$, then $2|A| \leq 200\eta + 2 \leq 402 n^{2/3}$.

From now on $r_{\text{in}}(K) > \eta$ and $\text{per } K \geq 100\eta$.

Step 1 (shell removal). Put $Y_{\text{out}} := \{y \in Y : \text{depth } y < \rho\}$, $Y_{\text{in}} := \{y \in Y : \text{depth } y \geq \rho + 1\}$ and $H_{\text{out}} := G[Y_{\text{out}}]$. Depth is 1-Lipschitz, so no edge joins Y_{out} to Y_{in} . At most $6|Y_\rho|$ edges meet Y_ρ , and $\text{Def}(G[Y_{\text{in}}]) \geq 0$; hence $\text{Def}(G[Y]) \geq \text{Def}(H_{\text{out}}) - 3|Y_\rho| \geq \text{Def}(H_{\text{out}}) - 3n/\rho_0$. Also $A \subseteq Y_{\text{out}}$, and we may assume $A \neq \emptyset$.

Step 2 (the core K' and the retraction P). Let $K' := \{z \in K : \text{depth } z \geq \eta\} = \bigcap_w B(w, \Delta_2 - \eta)$; it is compact and convex, with nonempty interior since the incentre of K has depth $r_{\text{in}} > \eta$. Every point of the drawing of H_{out} has depth less than $\rho + \frac{1}{2} = \eta$, since a point on a unit edge is within $\frac{1}{2}$ of an endpoint. So K' misses the drawing and lies in a single face F_0 of H_{out} .

$\partial K'$ consists of finitely many open smooth arcs, on circles $C(w, \Delta_2 - \eta)$, and finitely many corners c . At a corner, the normal cone $N_{K'}(c)$ is the cone between the unit normals $\nu_i = (c - w_i)/(\Delta_2 - \eta)$, $i = 1, 2$, for two centres $w_1, w_2 \in X$; let $\psi_c \in (0, \pi)$ be its angle. Normal cones of distinct corners have disjoint interiors, so $\sum_c \psi_c \leq 2\pi$.

Let $P : \partial K \rightarrow \partial K'$ be the nearest-point map onto K' , restricted to ∂K . If $P(x) = y$ then $x \in y + N_{K'}(y)$.

- *Offset points.* If y lies on a smooth arc of $C(w, \Delta_2 - \eta)$ with normal $\nu = (y - w)/(\Delta_2 - \eta)$, then the open disc $B(y, \eta)$ lies in $\text{int } K$ and $|y + \eta\nu - w| = \Delta_2$, so the ray $y + \mathbb{R}_+\nu$ leaves K exactly at $y + \eta\nu$. Hence $P^{-1}(y) = \{y + \eta\nu\}$; on each smooth arc, P^{-1} is the dilation about w with factor $\Delta_2/(\Delta_2 - \eta)$.
- *Collapse arcs.* At a corner, $Z_c := P^{-1}(c) = \partial K \cap (c + N_{K'}(c))$ is a closed arc of ∂K ; distinct Z_c are disjoint.

P is continuous and surjective (the ray from $y \in \partial K'$ along an outer normal leaves K at a point x with $P(x) = y$), and its fibres are points or arcs Z_c , hence connected. By (P7), P is monotone. Moreover

$$(7) \quad \text{per } K = \frac{\Delta_2}{\Delta_2 - \eta} \text{per } K' + \sum_c \text{len } Z_c,$$

and, writing $\text{col}(\alpha) := \text{len}(\alpha \cap \bigcup_c Z_c)$ for an arc $\alpha \subseteq \partial K$,

$$(8) \quad \text{len } P(\alpha) \geq \left(1 - \frac{\eta}{\Delta_2}\right)(\text{len } \alpha - \text{col}(\alpha)).$$

Step 3 (collapse length). The points $p_i := c + \eta\nu_i$ lie in $B(c, \eta) \subseteq K$ and on $C(w_i, \Delta_2)$, so they lie on ∂K and are the endpoints of Z_c . The convex set $K \cap (c + N_{K'}(c))$ has perimeter $2\eta + \text{len } Z_c$. Since K lies behind the tangent line to $C(w_i, \Delta_2)$ at p_i , this set lies in the kite with vertices c, p_1, p_2 and the intersection of the two tangent lines, which has right angles at p_1, p_2 and perimeter $2\eta + 2\eta \tan(\psi_c/2)$. By (P4), $\text{len } Z_c \leq 2\eta \tan(\psi_c/2)$. Also $|w_1 - w_2| = 2(\Delta_2 - \eta) \sin(\psi_c/2) \leq 1.94 \Delta_2$, so $\sin(\psi_c/2) \leq 0.97/(1 - 1/200) = 194/199$. With $\theta^* := \arcsin(194/199) > 1.343$ and $\tan \theta^* = 194/\sqrt{1965} \leq 3.26 \theta^*$, and since $\tan x/x$ is increasing on $(0, \pi/2)$, we get $\tan(\psi_c/2) \leq 1.63 \psi_c$ and

$$(9) \quad \sum_c \text{len } Z_c \leq 3.26 \eta \sum_c \psi_c \leq 3.26 \cdot 2\pi \eta < 21\eta.$$

By (7) and $\text{per } K \leq 2\pi\Delta_2$, $\text{per } K' \geq \text{per } K - (21 + 2\pi)\eta$; also $\text{per } K' \geq \frac{199}{200} \cdot 79\eta > 2$ by (F3).^[E]

Step 4 (case (i): F_0 is bounded). In (P2) keep only the outer face and F_0 :

$$\text{Def}(H_{\text{out}}) \geq \sum_{j \text{ outer}} (3 + |w_j|) + |F_0| - 3.$$

Every point of A lies on the outer face (the outer normal ray from it leaves K at once, and the drawing lies in K), hence on the outer walk of its component; write $A_j := A \cap C_j$. If $|A_j| \geq 2$, then w_j is a closed curve through the points of A_j , which are in convex position, so $|w_j| \geq \text{per conv } A_j \geq |A_j|$; if $|A_j| \leq 1$ then $3 + |w_j| \geq |A_j|$. The boundary of F_0 separates K' from ∞ , so $|F_0| \geq \text{per } K'$ by (P3). With Step 3 and $\text{per } K \geq |A| - 1$,

$$\text{Def}(H_{\text{out}}) \geq 2|A| - (21 + 2\pi)\eta - 4.$$

Lemma 12.3 (winding). *Suppose F_0 is the outer face of H_{out} . Let C_j be a component with $k := |A_j| \geq 2$, list A_j in cyclic order on ∂K , and let $\alpha_1, \dots, \alpha_k$ be the arcs of ∂K between consecutive points. Then $|w_j| \geq 2 \sum_{i \neq i^*} \text{len } P(\alpha_i)$, where α_{i^*} is an arc with the largest image.*

Proof. Fix $o \in \text{int } K'$, and let $\gamma : [0, L] \rightarrow \mathbb{R}^2$, $L = |w_j|$, be the closed unit-speed polygonal curve tracing w_j . The outer face of H_{out} lies in the outer face of C_j , which is a connected unbounded open set disjoint from the image of γ ; it contains $K' \ni o$. So o is in the unbounded component of $\mathbb{R}^2 \setminus \gamma([0, L])$, and the winding number of γ about o is 0. For $x \notin \text{int } K'$, the projection inequality $(x - Px) \cdot (o - Px) \leq 0$ gives $(x - o) \cdot (Px - o) \geq |Px - o|^2 > 0$; so $x - o$ and $Px - o$ are never opposite, the straight-line homotopy from γ to $P \circ \gamma$ avoids o , and $P \circ \gamma$ also has winding number 0 about o . Radial projection from o is a homeomorphism $\partial K' \rightarrow S^1$, so $P \circ \gamma$ has degree 0 as a map into $\partial K' \cong \mathbb{R}/(\text{per } K')\mathbb{Z}$. Let $E : \mathbb{R} \rightarrow \partial K'$ be the arc-length covering map; degree 0 means that a lift β of $P \circ \gamma$ is a closed curve in $\mathbb{R}$. As E preserves length and P is 1-Lipschitz, $\text{length}(\beta) \leq L$. A closed curve in $\mathbb{R}$ with image $[m, M]$ has length at least $2(M - m)$. Every point of A_j is visited by γ , so $E([m, M])$ is an arc of $\partial K'$ (or all of $\partial K'$) containing all $P(a)$, $a \in A_j$. The arcs $P(\alpha_i)$ tile $\partial K'$ between consecutive points $P(a)$ (P7), so the complementary open arc of $E([m, M])$ lies inside a single $P(\alpha_i)$. Hence $M - m \geq \sum_{i \neq i^*} \text{len } P(\alpha_i)$. $\square$

Step 5 (case (ii): F_0 is the outer face). Every bounded face has $|f| \geq 3$, so $\text{Def}(H_{\text{out}}) \geq \sum_{j \text{ outer}} (3 + |w_j|)$. For a gap arc α put $e(\alpha) := \max\{0, 1 - \text{len } P(\alpha)\}$. If $|A_j| \geq 2$, Lemma 12.3 gives $|w_j| \geq 2 \sum_{i \neq i^*} (1 - e(\alpha_i)) \geq 2(|A_j| - 1) - 2 \sum_i e(\alpha_i)$, so $3 + |w_j| \geq 2|A_j| - 2 \sum_{\alpha \in \text{gaps}(j)} e(\alpha)$; if $|A_j| \leq 1$ then $3 + |w_j| \geq 2|A_j|$. Summing,

$$(10) \quad \text{Def}(H_{\text{out}}) \geq 2|A| - 2 \sum_{\alpha} e(\alpha),$$

over all gap arcs of all components. Since $\text{len } \alpha \geq 1$, (8) gives $e(\alpha) \leq \eta/\Delta_2 + \text{col}(\alpha)$. There are at most $|A|$ gap arcs, and $|A|\eta/\Delta_2 \leq (2\pi\Delta_2 + 1)\eta/\Delta_2 \leq 2\pi\eta + \frac{1}{200}$ by (P8). Hence

$$(11) \quad \sum_{\alpha} e(\alpha) \leq 2\pi\eta + \frac{1}{200} + \sum_{\alpha: e(\alpha) > 0} \text{col}(\alpha).$$

Step 6 (corner multiplicity). Fix a corner c , and let $\mathcal{F}_c$ be the set of gap arcs α , over all components, with $e(\alpha) > 0$ and $\alpha \cap Z_c \neq \emptyset$.

- *Window.* Let B_c be the set of points of $\partial K'$ at arc distance less than 1 from c . Since $\text{per } K' > 2$, B_c is a proper open arc of length at most 2; let $U_c := P^{-1}(B_c)$, a proper open arc of ∂K . If $\alpha \in \mathcal{F}_c$, then $P(\alpha)$ is an arc of length less than 1 containing c , so $P(\alpha) \subseteq B_c$ and $\alpha \subseteq U_c$. The arc U_c consists of the offset preimage of B_c minus its corners, together with the collapse arcs $Z_{c'}$ of all corners $c' \in B_c$; so $\text{len } U_c \leq 2\Delta_2/(\Delta_2 - \eta) + \sum_{c'} \text{len } Z_{c'} \leq 21\eta + 3$, and U_c contains at most $N := \lfloor 21\eta + 4 \rfloor$ points of A (P8).

- *Laminarity.* U_c is a proper arc, hence linearly ordered, and each $\alpha \in \mathcal{F}_c$ is an interval of U_c with endpoints in $A \cap U_c$. Two arcs of $\mathcal{F}_c$ from the same component are distinct gaps of one cyclic list, so they have disjoint interiors. Two arcs from different components C_j, C_k have four distinct endpoints; if they interlaced, then by (P6) a path in C_j joining the endpoints of the first and a path in C_k joining the endpoints of the second would meet, although distinct components are disjoint as point sets (P1). So $\mathcal{F}_c$ is a laminar family of distinct intervals with endpoints in an N -point set, and such a family has at most $2N - 3 \leq 42\eta + 5$ members (map an interval to the set of elementary segments it covers). The restriction to arcs with $e(\alpha) > 0$ matters here: a component nested inside a gap of another has a “back” gap arc that runs around the rest of ∂K , and such arcs, which are not contained in U_c , can interlace with others.

Hence

$$(12) \quad \sum_{\alpha: e(\alpha) > 0} \text{col}(\alpha) = \sum_c \sum_{\alpha \in \mathcal{F}_c} \text{len}(\alpha \cap Z_c) \leq (42\eta + 5) \sum_c \text{len} Z_c \leq 21\eta(42\eta + 5).$$

Step 7 (assembly). In case (ii), (10), (11) and (12) give $\text{Def}(H_{\text{out}}) \geq 2|A| - 1764\eta^2 - (210 + 4\pi)\eta - \frac{1}{100}$. With case (i),

$$\text{Def}(H_{\text{out}}) \geq 2|A| - E(\eta), \quad E(\eta) := 1764\eta^2 + 223\eta + 4 \leq 7506 n^{2/3},$$

using $\eta < 2n^{1/3}$. The shell cost $3n/\rho_0$ is at most $6n^{2/3}$ if $\rho_0 = \lfloor n^{1/3} \rfloor \geq n^{1/3}/2$. If $\rho_0 = \lfloor \Delta_2/400 \rfloor \geq \Delta_2/800$, it is at most $2400n/\Delta_2$; from $n \leq (2\Delta + 1)^2$ and $\Delta \leq 1.94\Delta_2$ we get $\Delta_2 \geq \sqrt{n}/7.76$ for $n \geq 4$, so the cost is at most $18624\sqrt{n} \leq 18624 n^{2/3}$, and for $n < 4$ it is at most 12. Altogether $\text{Def}(G[Y]) \geq 2|A| - (7506 + 18636) n^{2/3} \geq 2|A| - 3 \cdot 10^4 n^{2/3}$, and the cases (F1)–(F3) cost at most $3772 n^{2/3}$.^[E] $\square$

12.3. The bound for $\tau > 1$.

Corollary 12.4. *In the nondegenerate regime with $\tau > 1$,*

$$\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{9}{7}n + \frac{3}{7}(3 \cdot 10^4 n^{2/3} + 3.4 \cdot 10^4).$$

Proof. Recall (Section 3) that in the nondegenerate regime we may assume $\Delta_2 \geq 10^4$, which the count of bad DD edges in Section 4 requires. Since $\tau > 1$ if and only if $t > 1$, Lemma 2.1(c) gives $\text{dist}(D, K) \geq t > 1$, while $X \setminus D \subseteq K$; so no δ -edge joins D to $X \setminus D$, and $\text{Def}(G) = \text{Def}(G[D]) + \text{Def}(G[X \setminus D])$.

The part on D . The definitions of Section 2 for $S \cap D$ extend verbatim to all of D : fix a diametral partner x_z of each $z \in D$ and put $u_z := (x_z - z)/\Delta$; Lemma 2.2(a) holds for every $z \in D$; call a δ -edge inside D good if $\angle(u_z, u_{z'}) \leq \beta$. On a good edge, Lemma 2.2(a) at both ends gives $a, b > 0$, and $|a + b| \leq \beta$ then gives $a, b \in (0, \beta)$, so it is a T-edge; by (T1) there are at most $|D|$ of them. The count of DD bad edges in Section 4 uses only that points of D are vertices of P and the lenses $B(x_z, \Delta) \cap B(x_{z'}, \Delta)$, so it applies to all of D and gives fewer than $3.4 \cdot 10^4$ bad edges inside D (or at most $6 \cdot 1203$ edges inside D if the perimeter of P is below 1202). Hence $\text{Def}(G[D]) \geq 2|D| - 3.4 \cdot 10^4 \geq 2|S \cap D| - 3.4 \cdot 10^4$.

The part in K is Theorem 12.1: $\text{Def}(G[X \setminus D]) \geq 2|S \setminus D| - 3 \cdot 10^4 n^{2/3}$.

So $\mu(\delta) = 3n - \text{Def}(G) \leq 3n - 2s + 3 \cdot 10^4 n^{2/3} + 3.4 \cdot 10^4$, while $\mu(\Delta_2) \leq \frac{3}{2}s$ by Lemma 3.1. The combination with weights $\frac{4}{7}$ and $\frac{3}{7}$ gives the claim, since $\frac{4}{7} \cdot \frac{3}{2}s + \frac{3}{7}(3n - 2s) = \frac{9}{7}n$.^[E] $\square$

Proof of Theorem 1(c). In the degenerate regime the bound is $n + 1 \leq \frac{9}{7}n + n^{2/3}$. In the nondegenerate regime with $\Delta_2 < 10^4$ we have $n < N_0$, and $3n \leq \frac{9}{7}n + \frac{12}{7}N_0^{1/3} n^{2/3} < \frac{9}{7}n + 2000 n^{2/3}$. Otherwise Corollary 12.4 gives at most $\frac{9}{7}n + \frac{3}{7}(3 \cdot 10^4 + 3.4 \cdot 10^4) n^{2/3} < \frac{9}{7}n + 3 \cdot 10^4 n^{2/3}$. So Theorem 1(c) holds with $C_2 = 3 \cdot 10^4$. $\square$

- Remark 12.5.* (a) Theorem 1(c) matches the global lower bound $L \geq \frac{9}{7}$, in the sense that sets with $\tau(X) > 1$ cannot do better than $\frac{9}{7}$. It does not show that $\frac{9}{7}$ is attained by sets with $\tau(X) > 1$: the constructions of [17] have $t < 1$ and lie in Region I.
- (b) The argument is global. The first layer alone does not pay: there are exact configurations (checked in 60-digit arithmetic, scripts `r6_gap_L_family.py` and `r6_gap_M_pairgap.py`) in which the edges near ∂K do not give a deficiency of 2 per point of A . The missing amount is paid by a second closed walk: the boundary of the hole F_0 in case (i), or the return path of Lemma 12.3 in case (ii).
- (c) Theorem 12.1 holds for every τ . What prevents the same conclusion for $\tau \leq 1$ is the edges between D and $X \setminus D$ (the rungs and the D-neighbours of points of R), which are absent when $\tau > 1$.

13. A LAYER BOUND AND A NEW CLASS OF SETS

In this section X is an arbitrary finite planar set with at least two distinct distances, $\Delta > \Delta_2$ are its two largest distances, and we scale so that $\Delta_2 = 1$. Every distance of X is at most 1 or equal to Δ . L_1 is the set of extreme points of X (vertices of $\text{conv } X$), L_2 is the set of extreme points of $X \setminus L_1$, and $m_3 := n - |L_1| - |L_2|$. Let G be the graph on $L_1 \cup L_2$ whose edges are the Δ_2 -pairs, let G' be its 2-core (repeatedly delete vertices of degree less than 2), and write $L'_i := L_i \cap V(G')$ and $\deg' = \deg_{G'}$. For $p \in X$ let $h(p)$ be the number of points of X at distance Δ_2 from p .

Lemma 13.1 (location). *(i) Both endpoints of every Δ -pair lie in L_1 .*

(ii) Every Δ_2 -pair has an endpoint in L_1 and both endpoints in $L_1 \cup L_2$.

In particular G contains every Δ_2 -pair of X .

Proof. (i) If $|pq| = \Delta$ then $X \subseteq B(q, \Delta)$ and p lies on its boundary circle, so p is an extreme point of X . (ii) Let $|pq| = 1$. If $p \notin L_1$, then p has no Δ -partner by (i), so $X \subseteq B(p, 1)$ and q , lying on the boundary circle, is in L_1 . So some endpoint, say p , lies in L_1 . Suppose $q \notin L_1$. Every point x with $|px| > 1$ has $|px| = \Delta$ and so lies in L_1 . Hence $X \setminus L_1 \subseteq B(p, 1)$ with q on the boundary circle, so q is extreme in $X \setminus L_1$, i.e. $q \in L_2$. $\square$

Proposition 13.2 (core degree of L_2). *Every $q \in L_2$ has $\deg'(q) = 2$.*

Proof. Put $q = 0$. Since q is not extreme it has no Δ -partner (Lemma 13.1(i)), so $X \subseteq B(0, 1)$ and the G -neighbours of q lie on the unit circle. Since $q \in V(G')$, $\deg'(q) \geq 2$. Suppose v_1, v_2, v_3 are three distinct G' -neighbours of q . Each $|v_i v_j|$ is either at most 1 or equal to Δ . The argument follows the three cases of Vesztergombi's inner-point analysis [27, pp. 192–194], with one change: we delete no points, and the contradiction in Cases 1 and 2 is with membership in the 2-core, via $\deg_G \geq \deg'$.

Case 3: at most one pair $v_i v_j$ at distance Δ . Label so that $|v_1 v_2|, |v_2 v_3| \leq 1$. Then v_2 lies on the shorter arc between v_1 and v_3 , and in polar angle we may take v_2 at 0, v_1 at a and v_3 at $-b$ with $0 < a, b \leq 60^\circ$. Since q is not extreme, the closed half-plane $\{x : x \cdot v_2 \leq 0\}$ contains a point $t \in X$, $t \neq q$. For $t \neq 0$ with $t \cdot v_2 \leq 0$ we get $|t - v_2|^2 = |t|^2 - 2t \cdot v_2 + 1 > 1$, so $|tv_2| = \Delta$. By symmetry assume the polar angle φ of t lies in $[90^\circ, 180^\circ]$. Then the angle between t and v_3 is $\varphi + b \in (90^\circ, 240^\circ]$, so $t \cdot v_3 < 0$ and $|tv_3| = \Delta$. Hence t lies on the perpendicular bisector of $v_2 v_3$, which passes through 0, so $\varphi = 180^\circ - b/2$. The angle between t and v_1 is then $180^\circ - b/2 - a \geq 90^\circ$, so $|tv_1|^2 \geq |t|^2 + 1 > 1$ and $|tv_1| = \Delta$. Now v_1, v_2, v_3 lie on both $C(0, 1)$ and $C(t, \Delta)$, two distinct circles, which meet in at most two points. Contradiction. (This case uses only that q is not extreme.)

Case 2: exactly two pairs at Δ . Say $|v_1 v_2| = |v_1 v_3| = \Delta$ and $|v_2 v_3| \leq 1$. Then v_1 lies on the perpendicular bisector of $v_2 v_3$, and we may take $v_1 = (-1, 0)$, $v_{2,3} = (\cos a, \pm \sin a)$ with

$0 < a \leq 30^\circ$ and $\Delta = 2 \cos(a/2)$. We claim v_1 has no Δ_2 -neighbour $r \neq q$; then $\deg_G(v_1) = 1$, contradicting $v_1 \in V(G')$. Such an $r = (x, y)$ satisfies $|r - v_1| = 1$ (so $|r|^2 = -2x$), $|r| \leq 1$, and $|r - v_i| \in [0, 1] \cup \{\Delta\}$ for $i = 2, 3$; by symmetry $y \geq 0$. Write $c = \cos a$, $s_a = \sin a$.

- Both $|r - v_2|, |r - v_3| \leq 1$. Impossible: $|r - v_3|^2 - 1 = -2x(1 + c) + 2ys_a \geq 0$, with equality only if $r = q$.
- $|r - v_2| = \Delta$ (the case $|r - v_3| = \Delta$ is symmetric). Using $|r|^2 = -2x$, this condition is the linear equation $-2x(1 + c) - 2s_a y = 1 + 2c$. Intersecting with the circle $|r - v_1| = 1$ gives $x = (\pm s_a \sqrt{8c + 7} - 3(1 + c)) / (4(1 + c))$, so $|r|^2 = (3(1 + c) \mp s_a \sqrt{8c + 7}) / (2(1 + c))$. The root with $+s_a \sqrt{8c + 7}$ in $|r|^2$ has $|r|^2 \geq 3/2$, outside the disc. For the other root, $|r|^2 \leq 1 \iff 1 + c \leq s_a \sqrt{8c + 7} \iff 1 + c \leq (1 - c)(8c + 7) \iff c^2 \leq \frac{3}{4} \iff a \geq 30^\circ$. So only $a = 30^\circ$ remains, and there $|r - v_3|^2 = 3 + c - s_a \sqrt{8c + 7} = 2$, which lies strictly between 1 and $\Delta^2 = 2 + \sqrt{3}$. Forbidden.
- Both equal Δ . Then r lies on the x -axis, so $r \in \{(0, 0), (-2, 0)\}$, and $(-2, 0)$ is outside the disc.

Case 1: all three pairs at Δ . Then v_1, v_2, v_3 form an equilateral triangle with circumcentre q , and $\Delta = \sqrt{3}$. Let $P \neq q$ be a further Δ_2 -neighbour of some v_i , so $|P - v_i| = 1$ and $|P| \leq 1$. It cannot have both other distances at most 1: $|P - v_i| = 1$ gives $|P|^2 = 2P \cdot v_i$, the two conditions give $P \cdot (v_i - v_j) \leq 0$ for $j \neq i$, and adding and using $\sum_j v_j = 0$ gives $3P \cdot v_i \leq 0$, so $P = q$. Hence $|P - v_j| = \sqrt{3}$ for some $j \neq i$. Solving exactly, there are six candidate points, two for each v_i , each with $|P|^2 = (9 - \sqrt{33})/6$ and at squared distance $(7 - \sqrt{33})/2 \approx 0.63$ from the third vertex, so each is a Δ_2 -neighbour of exactly one v_i . Their pairwise squared distances are $(29 - 5\sqrt{33})/6 \approx 0.046$ for three disjoint pairs (each joining candidates of different vertices), and otherwise $4/3$, $(9 - \sqrt{33})/2 \approx 1.628$ or $(17 - \sqrt{33})/6 \approx 1.876$, all strictly inside the forbidden interval $(1, 3)$. So a set of candidates that can coexist in X has at most two elements, at most two of the v_i have a second Δ_2 -neighbour, and some v_i has $\deg_G(v_i) = 1$, contradicting $v_i \in V(G')$.

With four or more neighbours apply the argument to any three. Hence $\deg'(q) \leq 2$, so $\deg'(q) = 2$. $\square$

Lemma 13.3 (opposite-side lemma). *Let $|pa| = |pb| = |br| = 1$ with $r \neq p$. Suppose a lies strictly on one side of the line pb and r does not lie strictly on that side. Then $|ar| > 1$, so $|ar| = \Delta$.*

Proof. Put $p = 0$, $b = (1, 0)$, $a = (\cos A, -\sin A)$ with $0 < A < \pi$, and $r = b + (\cos F, \sin F)$ with $0 \leq F \leq \pi$; $F \neq \pi$ as $r \neq p$. Then $|r - a|^2 - 1 = 2(1 + \cos F)(1 - \cos A) + 2 \sin F \sin A$, where the first term is positive and the second nonnegative. $\square$

Lemma 13.4. *Let p be an extreme point of X with Δ_2 -neighbours $w_1, \dots, w_k$ in angular order. Then $\deg_G(w_i) = 1$ for every $3 \leq i \leq k - 2$.*

Proof. Since p is extreme, some line through p has $X \setminus \{p\}$ strictly on one side. So the directions of the w_j from p are distinct, lie in an open half-plane, and any two differ by less than 180° ; for $j < i < l$, w_j and w_l lie strictly on opposite sides of the line pw_i . Suppose $r \neq p$ and $|rw_i| = 1$. Put $p = 0$, $w_i = (1, 0)$. Then w_{i-2}, w_{i-1} lie strictly below the x -axis and w_{i+1}, w_{i+2} strictly above; by symmetry assume r is not strictly below. Write $w_{i-2} = e(-\alpha_1)$, $w_{i-1} = e(-\alpha_2)$ with $0 < \alpha_2 < \alpha_1 < 180^\circ$. By Lemma 13.3, $|rw_{i-2}| = |rw_{i-1}| = \Delta$, so r lies on the perpendicular bisector of $w_{i-2}w_{i-1}$, the line through 0 in direction $-\gamma$ with $\gamma = (\alpha_1 + \alpha_2)/2 \in (0^\circ, 180^\circ)$. As $r \neq 0$ is not below the axis, $r = \lambda e(180^\circ - \gamma)$ with $\lambda > 0$. From $|r - w_i| = 1$ we get $\lambda(\lambda + 2 \cos \gamma) = 0$, so $\lambda = -2 \cos \gamma$, whence $\gamma > 90^\circ$ and $\alpha_1 > 90^\circ$. Then $|w_{i-2}w_i| = 2 \sin(\alpha_1/2) > \sqrt{2} > 1$, so $|w_{i-2}w_i| = \Delta$. Let $\theta > 0$ be the direction of w_{i+1} ; then $\alpha_1 + \theta < 180^\circ$, and since the chord length $2 \sin(x/2)$ is strictly increasing on $[0^\circ, 180^\circ]$, $|w_{i-2}w_{i+1}| = 2 \sin((\alpha_1 + \theta)/2) > 2 \sin(\alpha_1/2) = \Delta$, contradicting the maximality of Δ . $\square$

Theorem 13.5 (degree at most 4). *Every $p \in L'_1$ has $\deg'(p) \leq 4$.*

Proof. Suppose $p \in L'_1$ has five G' -neighbours $u_1, \dots, u_5$ in angular order. Among the G -neighbours of p , u_3 has at least two on each side, so $u_3 = u_i$ with $3 \leq i \leq k-2$, and $\deg_G(u_3) = 1$ by Lemma 13.4. But $u_3 \in V(G')$ forces $\deg_G(u_3) \geq 2$. $\square$

Theorem 13.6 (layer bound). *For every finite $X \subset \mathbb{R}^2$ with at least two distinct distances,*

$$\mu(\Delta_2) \leq |L_1| + |L_2| + \frac{1}{2} \sum_{p \in L'_1} (\deg'(p) - 2).$$

Proof. By Lemma 13.1, $\mu(\Delta_2) = e(G)$. Each deletion made while pruning G to its 2-core removes at most one edge, so $e(G) \leq e(G') + |V(G) \setminus V(G')|$. By Proposition 13.2, $e(G') = \frac{1}{2} \sum_{v \in V(G')} \deg'(v) = |V(G')| + \frac{1}{2} \sum_{p \in L'_1} (\deg'(p) - 2)$. Add the two estimates and use $|V(G)| = |L_1| + |L_2|$. $\square$

Corollary 13.7. *Assume (H3): no point of L_1 has four or more points of X at distance Δ_2 . (This holds, for instance, if no four points of X are concyclic.) Then*

$$\mu(\Delta_2) \leq |L_1| + |L_2| + \frac{1}{2} \# \{p \in L'_1 : \deg'(p) = 3\} \leq \frac{3}{2} |L_1| + |L_2|.$$

Consequently, under (H3), if $|L_1| \leq 2m_3$ then Δ and Δ_2 are two distinct distances each occurring at least once and at most n times.

Proof. Under (H3), $2 \leq \deg'(p) \leq h(p) \leq 3$ for $p \in L'_1$. The condition $\frac{3}{2}|L_1| + |L_2| \leq n$ is equivalent to $|L_1| \leq 2m_3$, and $\mu(\Delta) \leq n$ by [16]. $\square$

Corollary 13.8 (unconditional form). *For every finite X with at least two distances, $\mu(\Delta_2) \leq |L_1| + |L_2| + \frac{1}{2} \sum_{p \in L_1} (\min\{h(p), 4\} - 2)^+$. Hence $\mu(\Delta_2) \leq n$ whenever $\sum_{p \in L_1} (\min\{h(p), 4\} - 2)^+ \leq 2m_3$.*

Proof. Combine Theorem 13.6 with $2 \leq \deg'(p) \leq \min\{h(p), 4\}$ on L'_1 (Theorem 13.5). $\square$

Remark 13.9. (a) Corollary 13.7 covers sets not covered by (1). For example, if $|L_1| = |L_2| = 0.4n$ and $m_3 = 0.2n$, the three quantities in (1) are $1.2n$, $1.33n$ and $1.2n$, while Corollary 13.7 gives $\mu(\Delta_2) \leq n$ (under (H3)). Of course (1) needs no hypothesis like (H3).

(b) Using $\deg' \leq 4$, Theorem 13.6 recovers the bound $2|L_1| + |L_2|$ of [7, Theorem 1.3]. Proposition 13.2 and Theorem 13.5 are the inputs from [28] (Propositions 3 and 5–8 there, as used in [7]) needed for this; we prove them directly in the 2-core, without deleting points. We have not checked the other inputs used in [7] for the first two bounds in (1).

(c) In Corollary 13.7, the bound $\frac{3}{2}|L_1| + |L_2|$ is attained under (H3) alone by the following example, which we have checked numerically for $m = 8, 12, 16$ (floating point): a regular m -gon with m even, with a point at distance Δ_2 from both endpoints of every other edge (CDL's “servers”). These examples contain concyclic quadruples; for sets with no four concyclic points see Question 15.3 below.

Proposition 13.10 (distinct middle neighbours). *Let $p \neq p'$ be points of X with a common Δ_2 -neighbour w , and suppose that each of p, p' has Δ_2 -neighbours strictly on both sides of its line to w . Then X contains four concyclic points. In particular, if no four points of X are concyclic, then distinct vertices of $\text{conv } X$ with three Δ_2 -neighbours have distinct middle Δ_2 -neighbours.*

Proof. The point p' is not strictly on both sides of the line pw , so choose a Δ_2 -neighbour a of p strictly on the side of pw on which p' does not lie strictly. Lemma 13.3 (with $b = w$, $r = p'$) gives $|ap'| = \Delta$. Symmetrically there is a Δ_2 -neighbour b of p' with $|bp| = \Delta$. Now $|pb| = |ap'| = \Delta$ and $|pa| = |bp'| = \Delta_2$, so p, b, a, p' are concyclic by the observation in Section 15 (the four points are distinct: $|pa| = \Delta_2 \neq \Delta = |pb|$, and $a \neq p'$, $b \neq p$ since those distances are Δ). For the last

sentence: if $p \in L_1$ has exactly three Δ_2 -neighbours, they lie in an open half-plane at p , and the middle one in angular order has neighbours of p strictly on both sides of its line to p . $\square$

14. SETS IN CONVEX POSITION

A set X is in *convex position* if every point of X is a vertex of $\text{conv } X$; in particular no three points of X are collinear. Throughout this section $|X| = n$, $m := \lfloor n/2 \rfloor$, $N := \binom{n}{2}$, $D = D(X)$ is the number of distinct distances, and $k = k(X)$ is the number of rare distances ($1 \leq \mu(d) \leq n$). A distance is *frequent* if $\mu(d) \geq n + 1$. R_M denotes the vertex set of a regular M -gon. By Hopf–Pannwitz, Δ is always rare; Clemen, Dumitrescu and Liu proved $k \geq 2$ for $n \geq 5$ [7, Theorem 1.2]. For $n = 5$ this is best possible: R_5 has $k = 2$.

Theorem 14.1. *Let X be a set of n points in convex position with $6 \leq n \leq 18$. Then at least three distinct distances each occur at least once and at most n times. The same holds for every even $n \geq 6$.*

The cases have different status:

- $n = 6, 7, 8$: a corollary of published results (Proposition 14.3).
- even $n \geq 10$: Theorem 14.4, a parity-sharp version of the proof of [7, Theorem 1.2];
- $n = 9$: Theorem 14.5, from the classification of Erdős and Fishburn [12];
- $n = 11, 13, 15, 17$: Theorems 14.8 and 14.9, new and computer-assisted. The $n = 7$ case is re-proved the same way, without the literature.

In Section 14.3 we show that sets in convex position with all but w points on one circle have at least $(n - 3w + 1)/4$ rare distances once $n + w \geq 91$.

14.1. Counting and the literature.

Lemma 14.2 (parity-sharp counting). *$(D - k)(n + 1) + k \leq N$. Consequently $k \geq D - m + 1$ whenever $D \geq 1$. More precisely, if $n = 2m + 1$ and $D = m + K$ with $1 \leq K \leq m + 1$, then $k \geq K + 1$; if $n = 2m$ and $D = m + K$ with $1 \leq K < 2m$, then $k \geq K + 2$.*

Proof. The first inequality holds because frequent distances contribute at least $n + 1$ pairs each and rare ones at least 1. It gives $k \geq (D(n + 1) - N)/n = D - \frac{n-1}{2} + \frac{D}{n} > D - m$, hence $k \geq D - m + 1$. For $n = 2m + 1$ the right side is $K + (m + K)/(2m + 1)$, with $0 < (m + K)/(2m + 1) \leq 1$. For $n = 2m$ it is $K + 1 + K/(2m)$, with $0 < K/(2m) < 1$. $\square$

We use the following results from the literature.

- (Alt) Altman [1, 2]: $D \geq m$; for odd n , $D = m$ only for R_n (see also [15, Theorem 1]).
- (Fi) Fishburn [15]: for even $n \geq 8$, $D = n/2$ if and only if X is R_n or n vertices of R_{n+1} . For $n = 6$ the hexagons with $D = 3$ are R_6 , R_7 minus a vertex, and one further hexagon [15, Theorem 2], [10], with multiplicity profiles $(6, 6, 3)$, $(5, 5, 5)$ and $(6, 6, 3)$ respectively (as recorded in [7, §2]; we re-checked the profiles in exact arithmetic).
- (EF96) Erdős–Fishburn [12]: a convex nonagon has exactly 5 distances if and only if its vertices are 9 vertices of R_{10} or of R_{11} .
- (r₂) Erdős–Fishburn [10, §3], quoting Fishburn’s technical report [14]: the largest possible second-largest multiplicity in a convex n -gon is $r_2^{(n)} = 1, 3, 5, 6, 7, 8$ for $n = 3, \dots, 8$.

A set on a circle has every distance class of maximum degree at most 2, hence of size at most n ; so for cocircular X all distances are rare and $k = D$.

Proposition 14.3 (from the literature). *For $n = 6, 7, 8$ in convex position, $k \geq 3$.*

Proof. By (r_2) , $r_2^{(n)} \leq n$ for $n \leq 8$, so at most one distance is frequent and $k \geq D - 1$. By (Alt), $D \geq 3$ for $n = 6, 7$ and $D \geq 4$ for $n = 8$. So $k \geq 3$ unless $n \in \{6, 7\}$ and $D = 3$. For $n = 7$, $D = 3$ forces R_7 (profile $(7, 7, 7)$, $k = 3$); for $n = 6$, (Fi) gives the profiles $(6, 6, 3)$ and $(5, 5, 5)$, so $k = 3$. $\square$

The value $r_2^{(7)} = 7$, and the other values of $r_2^{(n)}$, rest on an unrefered technical report [14] quoted in a refereed paper [10]; for $n = 7$ we give an independent proof in Theorem 14.8.

Theorem 14.4 (even n). *Let X be in convex position with $n = 2m \geq 6$. Then $k \geq 3$. If $n \geq 8$ and $k \leq 3$, then $D = m + 1$.*

Proof. By (Alt), $D \geq m$. If $D \geq m + 1$, Lemma 14.2 gives $k \geq 3$, and $k \geq 4$ if $D \geq m + 2$. If $D = m$ and $n = 6$, (Fi) gives $k = 3$. If $D = m$ and $n \geq 8$, then X is R_n , with profile $(n, \dots, n, n/2)$, or n vertices of R_{n+1} , with profile $(n-1, \dots, n-1)$; in both cases $k = m \geq 4$. $\square$

Theorem 14.4 is essentially the argument of [7, Theorem 1.2], with the counting done with parity. It adds to the literature only for even $n \geq 10$.

Theorem 14.5 ($n = 9$). *Every set of 9 points in convex position has $k \geq 3$.*

Proof. Here $m = 4$. If $D = 4$, then $X = R_9$ by (Alt) and $k = 4$. If $D = 5$, then X is cocircular by (EF96) and $k = D = 5$. If $D \geq 6$, Lemma 14.2 gives $k \geq 3$. $\square$

14.2. $n = 7, 11, 13, 15, 17$: an ordinal reduction and SAT certificates. Label $X = (p_0, \dots, p_{n-1})$ in convex cyclic order, let $d_1 < \dots < d_D$ be the distinct distances, and colour each pair $\{i, j\}$ with $c(ij) = t$ where $|p_i p_j| = d_t$.

Lemma 14.6 (quadrilateral inequality). *For $i < j < k < l$ (in cyclic order), $|p_i p_k| + |p_j p_l| > |p_i p_j| + |p_k p_l|$ and $|p_i p_k| + |p_j p_l| > |p_j p_k| + |p_l p_i|$.*

Proof. The diagonals of the convex quadrilateral $p_i p_j p_k p_l$ meet at a point O that lies on neither of the sides $p_i p_j$, $p_k p_l$. So $|p_i p_k| + |p_j p_l| = (|p_i O| + |O p_j|) + (|p_k O| + |O p_l|) > |p_i p_j| + |p_k p_l|$, by the strict triangle inequality. The other inequality is the same. $\square$

Ordinal form. Let $x_1 \leq x_2$ be the two diagonal colours, and $y_1 \leq y_2$ the colours of one pair of opposite sides. Then it is *not* the case that $x_1 \leq y_1$ and $x_2 \leq y_2$, since otherwise $|p_i p_k| + |p_j p_l| \leq$ the sum of the two sides. We call a colouring of the pairs of an n -cycle's vertex set satisfying this for all 4-tuples and both pairs of opposite sides a *Q-colouring*. This ordinal use of the quadrilateral inequality goes back to Altman [1]. Abstracting a convex distance problem to a finite search in this way is the method of Morić and Pritchard [22], whose Fact 3.1 is Lemma 14.6.

Some of our certificates also use a second valid constraint (Lemma T): if $w \neq w'$ lie on the same open boundary arc between p_i and p_j , then $(c(iw), c(jw)) \neq (c(iw'), c(jw'))$, because a point is determined by its distances to p_i, p_j and a side of the line $p_i p_j$.

Lemma 14.7 (reduction to profiles). *If $k \leq 2$, then at least $D - 2$ colour classes have size at least $n + 1$, and $D \leq D_{\max}(n)$, the largest D with $(D - 2)(n + 1) + 2 \leq N$. For $n = 7, 11, 13, 15, 17$, $D_{\max} = 4, 6, 7, 8, 9$.*

Call a Q-colouring using exactly D colours, all of them, with at least F classes of size at least $n + 1$, an (n, D, F) *ordinal counterexample*. Since Lemma 14.6 (and Lemma T) are necessary for geometric realisability, the following suffices to prove $k \geq 3$ for a given n : *for every $2 \leq D \leq D_{\max}(n)$ there is no $(n, D, D - 2)$ ordinal counterexample*. (We check the smaller D even where (Alt) would exclude them, so that no citation is needed; for $D \leq 4$ we check the stronger statement with $F = 0$, i.e. no Q-colouring with D colours at all.) No realisation step and no floating-point arithmetic enters.

Theorem 14.8. *Every set of n points in convex position with $n \in \{7, 11, 13\}$ has at least three rare distances. For these n , the ordinal quadrilateral inequality (Lemma 14.6) alone implies the statement.*

Proof (computer-assisted). By Lemma 14.7 it suffices that the following instances have no ordinal counterexample:

- $n = 7$: $(7, 2, 0)$, $(7, 3, 1)$, $(7, 4, 2)$;
- $n = 11$: $(11, 2, 0)$, $(11, 3, 0)$, $(11, 4, 0)$, $(11, 5, 3)$, $(11, 6, 4)$;
- $n = 13$: $(13, 2, 0)$, $(13, 3, 0)$, $(13, 4, 0)$, $(13, 5, 0)$, $(13, 6, 4)$, $(13, 7, 5)$.

Each is encoded in CNF and shown unsatisfiable, with the refutation checked as described below and in Appendix A. $\square$

Encoding. Variables $x_{e,c}$ say that pair e has colour c ; there are exactly-one constraints per pair and at-least-one constraints per colour. In the *direct* encoding of Lemma 14.6, each dominated 4-tuple of colours is forbidden. In the *order* encoding, auxiliary literals $y_{e,t} \Leftrightarrow c(e) \geq t$ are introduced and “ $c(a) \leq t \leq c(b)$ and $c(a') \leq u \leq c(b')$ ” is forbidden for both diagonal/side matchings; this is equivalent, since sorted domination holds if and only if some matching dominates. Selector variables f_c imply “class c has at least $n + 1$ pairs” through sequential counters, and $\sum_c f_c \geq F$.

Certification.

- *DRAT/LRAT.* For every instance with $n = 11$ and $n = 13$ listed above, and for $(7, 4, 2)$, a DRUP proof produced by Glucose 4 [3] was verified by `drat-trim` [29], which also emitted an LRAT proof that was re-verified by `lrat-check` [8]. The largest certificate, for $(13, 7, 5)$, has 446,686 lemmas (a 37 MB DRUP proof and an 82 MB LRAT proof). For $n = 13$ all instances were certified in the Q-only encoding. For $n = 11$, $(11, 6, 4)$ was certified in the Q-only encoding (35,510 lemmas), and $(11, 2..5, \cdot)$ in the encoding with Q and Lemma T (Q-only is covered by the Lean proof below). A truncated proof is correctly rejected. On this route, the trusted components are the CNF encoder (a short Python program, cross-validated as in Appendix A) and the proof checkers.
- $n = 7$ *without SAT.* A plain backtracking enumeration of all colourings, with Q checked as soon as its pairs are assigned, finds no ordinal counterexample for $D = 2, 3, 4$. For $D = 4$ it does so with Lemma T switched off.
- *Lean 4.* The statements for $n = 7, 11$ and 13 are formalised in Lean 4 [9] with Mathlib [26]; see below.

$n = 15$ and $n = 17$.

Theorem 14.9. *Every set of n points in convex position with $n \in \{15, 17\}$ has at least three rare distances. For these n , too, the ordinal quadrilateral inequality (Lemma 14.6) alone implies the statement.*

Proof (computer-assisted). Here $m = 7, 8$ and $D_{\max} = m + 1$ (Lemma 14.7). It suffices to refute, for $n = 15$, the instances $(15, D, 0)$ for $2 \leq D \leq 6$ and $(15, D, D - 2)$ for $D = 7, 8$; and for $n = 17$, the instances $(17, D, 0)$ for $2 \leq D \leq 7$ and $(17, D, D - 2)$ for $D = 8, 9$. Only Lemma 14.6 is encoded (order encoding, without Lemma T). The instances with $F = D - 2$ are split by the pair $\{r_1, r_2\}$ of colours that are *not* required to be frequent: in the case $\{r_1, r_2\}$ every other colour class is required to have at least $n + 1$ pairs, the colours r_1, r_2 are required to be used, and the implied bound $|r_1| + |r_2| \leq N - (D - 2)(n + 1)$ is added as a redundant constraint. Every ordinal counterexample falls under at least one of the $\binom{D}{2}$ cases. All $5 + 21 + 28$ cases for $n = 15$ and all $6 + 28 + 36$ cases for $n = 17$ are unsatisfiable; each refutation was produced by the solver Kissat [5] as a DRAT proof and verified by `drat-trim` [29]. $\square$

Symmetry breaking. For $D \geq m$ most case encodings contain symmetry-breaking constraints, justified as follows. Q-colourings, and the case conditions, are invariant under the dihedral group acting on $\{0, \dots, n-1\}$: reversing the cyclic order maps diagonals to diagonals and pairs of opposite sides to pairs of opposite sides. (Reversing the order of the colours is not a symmetry, and is not used.) Let L^* be the largest cyclic length $\min(j-i, n-j+i)$ of a pair of the largest colour D . After a rotation we may assume that $\{0, L^*\}$ has colour D ; this is encoded by selector variables s_L with $s_L \Rightarrow c(0, L) = D$ and “no pair of colour D has cyclic length $> L$ ”, and $\bigvee_L s_L$. The reflection $i \mapsto L - i \pmod{n}$ fixes $\{0, L\}$, preserves cyclic lengths and exchanges the pairs $\{0, L+1\}$ and $\{L, n-1\}$, so we may also require $s_L \Rightarrow c(0, L+1) \leq c(L, n-1)$. As a sanity check, all solutions of two satisfiable instances with lowered thresholds were enumerated and grouped into dihedral orbits (392 solutions in 42 orbits for $n = 7$, $D = 4$; 21,348 solutions in 1,280 orbits for $n = 9$, $D = 5$); every orbit meets the symmetry-breaking constraints.

Cost and trust. The hard cases are those whose non-frequent pair contains the diameter colour D : for $n = 17$, $D = 9$ these took 3 to 12 minutes each, and the largest DRAT proof has about 124 MB. The other cases took seconds. Three cases $((15, 8, \{1, 2\}), (15, 7, \{3, 7\}), (17, 9, \{2, 3\}))$ were re-solved with CaDiCaL [5] and re-verified with `drat-trim`. The same pipeline re-certifies the cases $D = m + 1$ for $n = 7, 9, 11, 13$. On this route the trusted components are the CNF encoder (whose Q part is the order encoding validated in [Appendix A](#)), the symmetry-breaking argument above, and `drat-trim`; the solver is not trusted. The proofs are not stored (they are regenerated deterministically by the scripts), and [Theorem 14.9](#) is *not* part of the Lean formalisation.

The Lean formalisation. The file `Erdos/Erdos/E132Convex.lean` proves, without `sorry`,

```
theorem erdos132_convex7 (p : Fin 7 → EuclideanSpace ℝ (Fin 2)) (hp :
  StrictConvexPolygon p) : 3 ≤ (rareDists p).card
```

and the analogous `erdos132_convex11` and `erdos132_convex13` (file `E132Convex13.lean`) for `Fin 11` and `Fin 13`. Here `StrictConvexPolygon p` means that *every* index-increasing triple (p_i, p_j, p_k) , $i < j < k$, is positively oriented; that is, the vertices are given as a strictly convex polygon listed counter-clockwise. `rareDists p` is the set of distances realised by pairs $i < j$ with multiplicity at most n . The proof formalises [Lemma 14.6](#) (via the intersection point of the diagonals and strict convexity of the Euclidean norm), the passage to ranks and Q-colourings, [Lemma 14.7](#), and the correctness of a bit-vector encoding of the finite statement “every Q-colouring with colours below $D_{\max}$ has at least three classes of size in $[1, n]$ ”. This covers all $D \leq D_{\max}$ at once, with unused colours allowed. The finite statement is closed by Lean’s `bv_decide`, which bit-blasts the formula with a verified bit-blaster, calls the bundled SAT solver CaDiCaL, and checks the resulting LRAT certificate with a checker that is proved correct in Lean.

Trust base. `#print axioms` reports, for each theorem, the standard axioms `propext`, `Classical.choice`, `Quot.sound`, plus one axiom generated by `bv_decide (..._native.bv_decide.ax_1_14)`. That axiom asserts that the verified LRAT checker `verifyBVExpr` returns `true` on the concrete formula and certificate. The value `true` is obtained by running compiled code, not by kernel reduction, so the trust base is comparable to that of `native_decide`: it includes the Lean compiler and runtime. Neither our Python encoder, nor any external SAT solver, nor `drat-trim` is trusted on this route. An attempt to check the $n = 7$ axiom statement by kernel reduction ran for more than 20 minutes without finishing and was stopped.

Unordered point sets. The file `E132ConvexFinset.lean` proves (using only the three standard axioms) that a finite set S in convex position, in the sense of Mathlib’s `ConvexIndependent`, can be listed as a `StrictConvexPolygon`. The proof sorts $S \setminus \{p_0\}$ by orientation about a point

$p_0 \in S$; a negatively oriented increasing triple would put a point of S inside a triangle spanned by others. It also proves that the rare distances of S (unordered pairs) coincide with `rareDists p` for any such listing. This gives `erdos132_convex{7,11,13}_finset`: for $S : \text{Finset}$ with $S.\text{card} = n$ and `ConvexIndependent`, $3 \leq (\text{rareDistsS } S).\text{card}$.

The layer bound. The results of [Section 13](#) are formalised in `E132Layer.lean` without `sorry`, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound` (no `bv_decide` or `native_decide`). This covers the localisation of Δ_2 -pairs to $L_1 \cup L_2$, the degree-2 property of L_2 -vertices in the 2-core, the bound $\deg \leq 4$ on hull vertices, Theorem A, Corollary A1 and its unconditional form. There L_1 is the set of extreme points of X , shown equal to the vertex set of $\text{conv } X$, and the 2-core is the largest subset with minimum degree at least 2.

The $\frac{54}{37}$ bound. The $\frac{54}{37}$ bound for Problem 1.6 of [7], proved in [Appendix B](#), is formalised in full in `E132Main*.lean`, `E132ConvexCurve.lean` and `E132Ve87.lean` (about 7,000 lines):

```
theorem erdos132_main :  $\exists C_0 : \mathbb{R}, \forall X : \text{Finset Pt}, 2 \leq X.\text{card} \rightarrow (\min (\text{mult } X (\text{dist2 } X)) (\text{mult } X (\text{minDist } X)) : \mathbb{R}) \leq 54/37 * X.\text{card} + C_0$ 
```

with `mult X d` the number of unordered pairs of distinct points of X at distance d , and `dist2`, `minDist` the second-largest and smallest distance. So the formal theorem `erdos132_main` is the $\frac{54}{37}$ bound; the $\frac{15}{11}$ bound of [Theorem 8.1\(a\)](#) is formalised separately (below). The formal proof follows the argument of [Appendix B: Sections 2, 4 and 7](#) and Region I as in the text, the ownership and pairing argument in Region II, and the weighted [Lemma B.2](#) in Region III. There is no `sorry`, and `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`. The formalisation also proves Vesztergombi's bound $\mu(\Delta_2) \leq \frac{3}{2}n$ and the Hopf–Pannwitz bound, so no result is used as a black box. In three places it takes a different route from the text: (a) Vesztergombi's bound is proved in the 2-core (every degree-4 core vertex has two degree-2 core neighbours, then double counting), with no point deletion; (b) the bad-edge bounds of [Section 4](#) are obtained by counting over directions of outward normals instead of arc length, so [Lemma 4.2](#) (lens-corner arc length) is not needed; (c) the degenerate-regime incidence bound uses a local charging argument, and the mixed-T bound of [Lemma 5.1](#) is proved in a weaker form (at most 1200 instead of 71), which only changes the constant C_0 .

The $\frac{15}{11}$ bound, [Theorem 8.1\(a\)](#). The bound $\frac{15}{11}$ is formalised in the six files `E132Main15*.lean` (about 5,300 lines), which build on the $\frac{54}{37}$ development without changing it:

```
theorem erdos132_main15 :  $\exists C_0 : \mathbb{R}, \forall X : \text{Finset Pt}, 2 \leq X.\text{card} \rightarrow (\min (\text{mult } X (\text{dist2 } X)) (\text{mult } X (\text{minDist } X)) : \mathbb{R}) \leq 15/11 * X.\text{card} + C_0$ 
```

There is no `sorry`; `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`, and the compiled modules pass `leanchecker`, which re-checks every declaration with the kernel. The set-up, the reductions, [Section 4](#), Region I, the corner lemma, (R1), [Lemma 5.1](#) and [Section 7](#) are taken from the $\frac{54}{37}$ development; [Lemmas 3.3](#) and [3.5](#), [Lemmas 5.3](#) and [5.4](#), the count in Region II, [Section 5.4](#) and [Theorem 6.1](#) are new. The formal proof deviates from the text as follows.

- (a) *[Theorem 6.1 without arc length](#).* The arc γ_y and the concave-graph description of [Lemma 6.5](#) are replaced by a Euclidean-local case split at $y \in R \setminus D$, according to the outer normals of K at the points of $\partial K \cap B(y, 1)$: *flat* (all these normals within $\frac{1}{4}$ of one unit vector; then every neighbour q of y in S has $(q - y) \cdot \nu > -\frac{1}{4}$ for a fixed unit ν , and since an open arc of angle $2 \arccos(-\frac{1}{4}) < 240^\circ$ contains at most 4 points 60° apart, $m_y \leq 4$); *thin* (two nearly opposite normals; the fatness step of [Lemma 6.3](#) puts K in a bounded box, so $|X \setminus D|$ is bounded); and *big turning* (otherwise every direction between two such normals is the outer normal of a support point within distance 150 of y , so $\partial K \cap B(y, 150)$ has turning at least

- $\frac{1}{4}$, which happens for $O(1)$ points y by [Lemma 2.5](#)). The exceptional set has at most 10^7 points.
- (b) *Region III by slots and breaks.* For $\tau < \sqrt{3}/2$, (3) is proved by a count of free *slots* (positions next to a point of S where a T-neighbour could sit), using [Lemma 5.5](#); for $\tau = \sqrt{3}/2$, by a count of *breaks* (points of S without a pure right T-neighbour) together with the circle-arc rigidity of [Lemmas 5.7 to 5.9](#). [Lemma 5.6](#) is not used.
 - (c) *[Lemma 5.4](#) by exact reflection.* Instead of the angle estimate in the text, the formal proof decomposes the T-step exactly with respect to the reflection that exchanges the two Δ_2 -partners and applies the threshold of [Lemma 5.3](#) to both partners; two distinct circles of radius Δ meet in at most two points.
 - (d) In Region II the role of Q is played by the set of rung K-ends of Δ_2 -degree 1, and the constants differ from the text; the formal constant is $C_0 = 5 \cdot 10^9$, as for the $\frac{54}{37}$ bound.

Limitations of the formalisation. (i) The `bv_decide` axiom described above remains. (ii) Only the $n = 7$ LRAT certificate (1.6 MB) is stored and replayed with `bv_check`; for $n = 11, 13$ the certificates (23 MB and 97 MB) are not stored, so compilation re-runs CaDiCaL (about 40 s and 4 min respectively). (iii) [Lemma 4.2](#) is not formalised; the formal proof of the $\frac{54}{37}$ bound does not need it (see above). (iv) [Theorem 6.2](#), and hence [Theorem 8.1\(b\)](#) (the bound $\frac{4}{3}$ in Region II), is not formalised; nor are [Lemma 5.6](#) and the arc-length form of [Lemmas 6.3 to 6.5](#), which the formal proof of [Theorem 8.1\(a\)](#) avoids (see above). (v) *Theorem 1 is not formalised.* The formalisation covers the $\frac{15}{11}$ bound (`erdos132_main15`) and the $\frac{54}{37}$ bound (`erdos132_main`). The bound $\frac{4}{3}$ of [Theorem 1\(a\)](#), which needs [Section 9](#) in addition, is not formalised; a Lean formalisation of it is in progress. Parts (b) and (c) of [Theorem 1](#) ([Sections 10 and 12](#)) are not formalised.

- Remark 14.10.* (a) The method is tight at “three”: $(11, 6, 3)$ is satisfiable (e.g. with class sizes $(12, 9, 12, 12, 6, 4)$), as is $(9, 5, 2)$.
- (b) The ordinal statement fails for $n = 5, 6$: $(6, 3, 1)$ has an ordinal solution with profile $(9, 3, 3)$, so at $n = 6$ geometry beyond [Lemma 14.6](#) is needed.
 - (c) Without the case split by the non-frequent pair, the instance $(15, 8, 6)$ was unresolved after more than 20 minutes of CaDiCaL; the split into $\binom{8}{2}$ cases, together with the symmetry breaking, is what makes $n = 15, 17$ feasible. For $n = 19$ the analogous computation looks feasible but expensive (hours per hard case), and it has not been done.
 - (d) [Theorem 14.8](#) for $n = 7$ also follows from the literature ([Proposition 14.3](#)).

Remark 14.11 (related work). Independently, a Lean project on the Lean Pool [19] proves that for point sets with coordinates in certain restricted fields (e.g. $\mathbb{Q}(\sqrt{2})$), $n \leq 2^r$ where r is the number of rare distances. This gives at least three rare distances for $n \geq 8$ in those coordinate fields; it neither implies nor is implied by [Theorem 14.1](#). Other contributions on the forum [13] concern the first question of [Problem #132](#) (at least two rare distances) for small n and general sets.

14.3. Almost cocircular sets. For sets on one circle all distances are rare. We show that a bounded number of points off the circle cannot destroy this, up to a linear loss. The two ingredients are Kneser’s theorem, which recognises cocircular sets with few distances as subsets of regular polygons (Kneser’s addition theorem was applied to distance sets on circles by Momihara and Shinohara [21]; [Lemma 14.12](#) is a variant of this idea), and a uniform bound for the number of vertices of a regular polygon that a point off its circle can see at chord distances, which follows from a theorem of Beukers and Smyth on torsion points of plane curves.

Lemma 14.12 (cocircular sets with few distances). *Let A be a set of $q \geq 3$ points on a circle Γ with $D(A) \leq q/2 + K'$, where $K' \geq 0$ and $q > 4K' + 2$. Then A is contained in the vertex set of a regular M -gon inscribed in Γ , for some M with $q \leq M \leq q + 2K' + 1$.*

Proof. Identify Γ with $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ by the polar angle. The distance between points with angles a, b determines $a - b$ up to sign, so the difference set $A - A \subseteq \mathbb{T}$ has $|A - A| \leq 2D(A) + 1 \leq q + 2K' + 1$. By Kneser's theorem [18] (see also [25]), applied in the abelian group $\mathbb{T}$ to the finite sets A and $-A$, with H the stabiliser of $A - A$ (a finite subgroup, hence cyclic of some order M), we have $|A - A| \geq |A + H| + |-A + H| - |H| = (2t - 1)M$, where $A + H$ is a union of t cosets of H . Since $q \leq tM$: if $t \geq 2$ then $(2t - 1)M \geq \frac{2t-1}{t}q \geq \frac{3}{2}q$, so $\frac{3}{2}q \leq q + 2K' + 1$, i.e. $q \leq 4K' + 2$, a contradiction. Hence $t = 1$, so A lies in one coset of H , i.e. in the vertex set of a regular M -gon on Γ , and $q \leq M \leq |A - A| \leq q + 2K' + 1$. $\square$

Scale Γ to the unit circle in $\mathbb{C}$, let $\zeta := e^{2\pi i/M}$ and $c_j := |1 - \zeta^j|$. We say that a point $u \in \mathbb{C}$ hits the vertex ζ^a if $|u - \zeta^a| = c_j$ for some j .

Lemma 14.13 (hit lemma). *If $u \neq 0$ and $|u| \neq 1$, then u hits at most 23 vertices of the regular M -gon $\{\zeta^a\}$, for every M .*

Proof. Put $\lambda := \bar{u} \neq 0$ and $\tau := |u|^2 - 1 \neq 0$, and consider the Laurent polynomial

$$F(x, y) := \lambda x + \bar{\lambda} x^{-1} - y - y^{-1} - \tau.$$

For $|x| = |y| = 1$ we have $|u - x|^2 - |1 - y|^2 = -F(x, y)$. So if u hits ζ^a with $|u - \zeta^a| = c_j$, then (ζ^a, ζ^j) and (ζ^a, ζ^{-j}) are *torsion points* (both coordinates roots of unity) of the curve $F = 0$.

F is irreducible. Multiply by xy : $G := xyF = -xy^2 + (\lambda x^2 - \tau x + \bar{\lambda})y - x$. As a polynomial in y over $\mathbb{C}[x]$ it is primitive ($\gcd(x, \bar{\lambda}) = 1$), and it is not divisible by x or y . So a nontrivial factorisation would have two factors of degree 1 in y , and the discriminant

$$(\lambda x^2 - \tau x + \bar{\lambda})^2 - 4x^2 = (\lambda x^2 - (\tau + 2)x + \bar{\lambda})(\lambda x^2 - (\tau - 2)x + \bar{\lambda})$$

would be a square in $\mathbb{C}[x]$. The two quadratic factors have no common root (their difference is $4x$, and neither vanishes at 0), so both would be squares, i.e. $(\tau + 2)^2 = 4|\lambda|^2 = (\tau - 2)^2$, forcing $\tau = 0$. Hence G is irreducible.

$F = 0$ contains no torsion coset of positive dimension. On $\{(\zeta_1 t^p, \zeta_2 t^q)\}$ with $(p, q) \neq (0, 0)$, F becomes a Laurent polynomial in t : if $p, q \neq 0$ its constant term is $-\tau \neq 0$; if $p = 0$ its t^q -coefficient is $-\zeta_2 \neq 0$; if $q = 0$ its t^p -coefficient is $\lambda \zeta_1 \neq 0$.

The Newton polygon of G is the square with vertices $(0, 1), (1, 0), (2, 1), (1, 2)$, of area 2. By the theorem of Beukers and Smyth [4], an irreducible plane curve $f = 0$ that is not a torsion coset has at most $22V(f)$ torsion points, where $V(f)$ is the area of the Newton polygon of f . Hence the curve $F = 0$ has at most 44 torsion points. Distinct hit vertices give torsion points with distinct first coordinates, and each hit gives two of them unless $\zeta^j = \zeta^{-j}$. Here $\zeta^j = 1$ is impossible (it would put u on Γ), and $\zeta^j = -1$ means $|u - \zeta^a| = 2$, which happens for at most two vertices, since the circle of radius 2 about u is not Γ . So if $d \leq 2$ hits have $c_j = 2$ and h is the number of hits, then $2(h - d) + d \leq 44$, and $h \leq 22 + d/2 \leq 23$. $\square$

A computer search (Appendix A) found that for $M \leq 120$ the maximum number of hits is 13; we have not tried to optimise the constant 23.

Theorem 14.14 (almost cocircular sets). *Let X be a set of n points in convex position and Γ a circle, and put $w := |X \setminus \Gamma|$. Suppose $n \geq 3w$ and $|X \cap \Gamma| \geq 4$. Then*

$$k(X) \geq \min\left\{\frac{n - 3w + 1}{4}, m - \frac{w}{2} - 22 + \frac{22}{w + 1}\right\}.$$

In particular $k(X) \geq (n - 3w + 1)/4$ whenever $n + w \geq 91$.

Proof. Put $C := X \cap \Gamma$, $W := X \setminus \Gamma$, $q := |C| = n - w \geq 4$ and $K' := (q - 3)/4 \geq 0$.

Case 1: $D(C) > q/2 + K' = (3q - 3)/4$. Then $D(X) \geq D(C)$, and Lemma 14.2 gives $k \geq D(X) - m + 1 > \frac{3(n-w)-3}{4} - \frac{n}{2} + 1 = \frac{n-3w+1}{4}$.

Case 2: $D(C) \leq q/2 + K'$. Since $q > 4K' + 2$, Lemma 14.12 puts C inside the vertex set of a regular M -gon on Γ with $M \leq q + 2K' + 1 = (3q - 1)/2$.

- (i) *The centre O of Γ is not in X .* Otherwise O is a vertex of $\text{conv } X$, so C lies in a closed half-plane bounded by a line through O , i.e. in a closed semicircle, which contains at most $M/2 + 1 \leq (3q + 3)/4 < q$ vertices of the M -gon. So every $u \in W$ satisfies $u \neq O$ and $u \notin \Gamma$, and Lemma 14.13 applies to it.
- (ii) Within C every distance graph has maximum degree at most 2, hence at most q edges. So a frequent distance of X that is a chord length of the M -gon (a *chord class*) has at least $n + 1 - q = w + 1$ pairs that meet W . Each such pair is either a pair $\{u, c\}$ with $u \in W$, $c \in C$, and c hit by u , or a pair inside W .
- (iii) A frequent distance that is not a chord length has all its pairs meeting W (all distances within C are chord lengths). For a given distance, each $u \in W$ has at most 2 points of Γ at that distance, so at least $n + 1 - 2w$ of these pairs lie inside W .
- (iv) By Lemma 14.13 there are at most $23w$ pairs of the first kind in (ii), and there are at most $\binom{w}{2}$ pairs inside W ; every pair belongs to one distance class. Writing F_{ch} and F_{nc} for the numbers of frequent chord classes and other frequent classes,

$$(w + 1)F_{\text{ch}} + (n + 1 - 2w)F_{\text{nc}} \leq 23w + \binom{w}{2}.$$

- (v) Since $n \geq 3w$ we have $n + 1 - 2w \geq w + 1$, so the number F of frequent distances satisfies $F \leq (23w + \binom{w}{2})/(w + 1) = \frac{w}{2} + 22 - \frac{22}{w+1}$.
- (vi) By (Alt), $D(X) \geq m$, so $k = D(X) - F \geq m - \frac{w}{2} - 22 + \frac{22}{w+1}$.

For the last claim, $m \geq (n - 1)/2$ gives $m - \frac{w}{2} - 22 + \frac{22}{w+1} > \frac{n-1}{2} - \frac{w}{2} - 22 \geq \frac{n-3w+1}{4}$ when $n + w \geq 91$. $\square$

Remark 14.15. (a) For $w = o(n)$ Theorem 14.14 gives $k \geq (\frac{1}{4} - o(1))n$. So the second question of Problem #132 has a positive answer, in a strong form, for sets in convex position that are almost cocircular. What is missing for general convex sets is a stability statement of the form “a convex n -set with at most $n/2 + K$ distances has all but $o(n)$ of its points on one circle”; Kneser’s theorem provides this on a circle (Lemma 14.12), and we know no analogue off it.

- (b) The loss of $3w/4$ and the factor $\frac{1}{4}$ (rather than $\frac{1}{2}$) come from the threshold $q > 4K' + 2$ in Lemma 14.12.

15. OPEN PROBLEMS

Problem 1.6 of [7]. We now know $9/7 \leq L \leq 4/3$, i.e. $1.2857\dots \leq L \leq 1.3333\dots$. The lower bound construction [17] is conjectured there to be optimal. By Theorem 1(c) no set with $\tau(X) > 1$ beats it, and by Theorem 1(b) the upper bound is $\frac{21}{16}$ outside Region III. Two obstacles keep the global bound at $\frac{4}{3}$, both in Region III ($\sqrt{3}/2 - 2\beta < \tau < \sqrt{3}/2 + 3\beta$).

- *The p -term in Region III.* With $e(S) \leq s + p + O(1)$, the value $\frac{4}{3}$ is the best the linear combination can give, whatever the average weight on R (Remark 3.6). In Region II Section 11 halves the p -term. Near $\tau = \sqrt{3}/2$ an exact single-centre “hexagonal strip” shows that $e(S) - s = p$ can hold locally, so a purely local count cannot do the same there (but see Remark 15.2).
- *The weights on R .* In Region III only the average bound 8 of Theorem 9.1 is available; the heavy “fan” configurations of Lemma 9.7 can occur only there (with the parameters

of Section 10 they force t within $2 \cdot 10^{-4}$ of $\frac{\sqrt{3}}{2}$. Elsewhere radius-one discharging stops at average weight 7 (Remark 10.10(a)), which gives $\frac{21}{16}$; average weight 6 would give $\frac{9}{7}$ in Region I.

Question 15.1. *In Region I, is $\sum_{y \in R} (\deg y + m_y) \leq 6|R| + o(n)$?*

A positive answer would show that sets in Region I cannot beat $\frac{9}{7}$. An alternative route to $\frac{9}{7}$ in any regime would be to extend Theorem 12.1 by controlling the δ -edges between D and $X \setminus D$ (Remark 12.5(c)), together with a sharper bound on $\mu(\Delta_2)$ in Regions II and III (in Region II, Section 11 halves the p -term).

Remark 15.2. Work in progress, which has had one independent referee pass but is not included here, uses the diametral partners of the D-row to bound the rungs in all of Region III by $e(S) \leq s + \frac{2}{3}p + O(1)$ (and by $s + \frac{1}{2}p + O(1)$ away from a window of width $O(1/\Delta_2)$ around $\tau = \sqrt{3}/2$). With the linear combination of Remark 3.6, this removes the p -term as an obstacle: the bound in Region III then depends only on the average weight on R there, and an average weight below 8 in Region III would give a bound below $\frac{4}{3}$ for all sets. We do not claim a new bound here.

Genericity. One might hope that excluding four concyclic points forces $\mu(\Delta_2) \leq n + O(1)$, which together with Theorem 13.6 would answer the first question of Problem #132 for such sets. This is false. For $n = 4m \geq 12$, start from $n/2$ points on a circle of radius $\frac{1}{2}$ interleaved with $n/2$ points on a circle of radius R , where $R^2 + R \cos(2\pi/n) = \frac{3}{4}$, and perturb generically within the solution set of the distance equations. This gives convex sets with no four concyclic points and $\mu(\Delta_2) = \frac{5}{4}n$. We have verified these examples to 50 digits for $n = 12, 20, 40$ (all equalities to $5 \cdot 10^{-51}$, with concyclicity determinants at least 10^{-8}). A non-convex family with $\mu(\Delta_2) = \frac{5}{4}n - \frac{3}{2}$ was verified numerically for $n \leq 62$. In both families every distance other than Δ and Δ_2 occurs once, so the first question of Problem #132 still has a positive answer for them, but the second rare distance is not Δ_2 . An elementary observation, probably folklore, explains the role of concyclicity in the extremal configurations: if $|ab| = |cd| = \Delta$ and $|ac| = |bd| = \Delta_2$, then a, b, c, d are concyclic. (The triangles abc and dcb are congruent. A half-turn would make $abdc$ a parallelogram with $|ad|^2 + |bc|^2 = 2(\Delta^2 + \Delta_2^2) > 2\Delta^2$, which is impossible, so $abdc$ is an isosceles trapezoid.)

Question 15.3. *If no four points of X are concyclic, is $\mu(\Delta_2) \leq \frac{5}{4}n + O(1)$?*

Proposition 13.10 records one further consequence of the no-four-concyclic hypothesis; the examples above show that such local constraints do not bound $\mu(\Delta_2)$ by $n + O(1)$.

Beyond the two largest distances. In a counterexample to the first question of Problem #132, every distance other than Δ occurs at least $n + 1$ times, and in particular $\mu(\Delta_2) \geq n + 1$. One might hope that then the third-largest distance Δ_3 is rare. This is false: there is an 8-point set, with coordinates in $\mathbb{Q}(\sqrt{3}, \sqrt{5}, \sqrt{10 + 2\sqrt{5}}, \sqrt{10 - 2\sqrt{5}})$, formed by a regular pentagon and three further points, whose distance multiplicities in decreasing order of distance are 2, 9, 9, 5, 2, 1; so $\mu(\Delta_2) = \mu(\Delta_3) = 9 > 8$ (checked in exact arithmetic, Appendix A). In this example the fourth distance occurs 5 times. In computer searches (rings of points, symmetric orbits, lattice and cyclotomic pools; about 3,500 sets with $\mu(\Delta_2) > n$), some Δ_j with $j \leq 4$ was always rare.

Question 15.4. *Let $n \geq 5$ and let X be an n -point planar set with at least four distinct distances $\Delta > \Delta_2 > \Delta_3 > \Delta_4 > \dots$. Is $\min\{\mu(\Delta_2), \mu(\Delta_3), \mu(\Delta_4)\} \leq n$?*

A positive answer would settle the first question of Problem #132 for all sets with at least four distinct distances.

Convex position.

Conjecture 15.5 (Q-ORD). *For every $n \geq 7$, every Q -colouring of the pairs of $\{0, \dots, n-1\}$ has at least three colour classes of size at most n .*

Q-ORD holds for $n \in \{7, 11, 13, 15, 17\}$ (Theorems 14.8 and 14.9), and it would imply that every set of $n \geq 7$ points in convex position has at least three rare distances. (For even n we have not certified it: our solver runs for $n = 8, 10, 12$ also used Lemma T.) Even with the case split, the hardest cases grow from about half a minute at $n = 13$ to about 12 minutes at $n = 17$, so a proof for all n must be human. Deletion-minimal refutations are not local ($(7, 4, 2)$ already needs 25 of the 35 quadrilaterals), so such a proof will have to be global.

For odd $n = 2m + 1 \geq 7$ (no computation needed), Lemma 14.2 and (Alt) show that a convex set with $k \leq 2$ must have $D = m + 1$, $k = 2$ and $m - 1$ frequent distances. Moreover, writing the frequent multiplicities as $2m + 2 + e_d$ and the two rare ones as a, b , the identity $a + b + \sum_d e_d = m + 2$ holds. (Indeed $D = m$ gives R_n with $k = m \geq 3$, $D \geq m + 2$ gives $k \geq 3$, and the identity is $(m - 1)(2m + 2) + \sum e_d + a + b = m(2m + 1)$.) So for odd $n \geq 19$ the question in convex position is about this single kind of configuration; by Theorem 14.14 such a configuration cannot have all but w of its points on one circle if $n \geq \max\{3w + 11, 91 - w\}$. Beyond three, the second question of Problem #132 asks whether $k \rightarrow \infty$; this is open even in convex position.

APPENDIX A. COMPUTATIONAL VERIFICATION

All scripts are in the directory `scripts/` of the accompanying repository <https://github.com/g8r-b8/erdos132-lean>. None of the computations below is used as a proof step, except the SAT/LRAT certificates and the Lean proof for Theorem 14.8 and the DRAT certificates for Theorem 14.9. The rest are independent checks of steps proved in the text. Tags ^[Ck] in the text refer to sections of `writeup_a2_checks.py`; tags ^[II], ^[III], ^[LP] and ^[N] refer to `r4_regionII_checks.py`, `r4_regionIII_checks.py`, `r4_lp_checks.py`, and `r4_nb_exact.py` together with `r4_nb_referee_checks.py`, respectively; tags ^[N3] and ^[E] refer to `r6_n3_checks.py` and `r6_Eprime_full_checks.py`.

Theorem 1: Sections 2 to 8.

- `writeup_a2_checks.py`: 58 assertions in exact rationals and rational interval arithmetic, with one-sided rational bounds for $\sqrt{\cdot}$, π , $\sin$, $\cos$, $\arcsin$. Sections: C1 the LP certificates of Lemmas B.1 and B.2 (now used only in Appendix B); C2 the identity of Lemma 2.2(c) and the relation between τ and t ; C3 small-angle facts for $\Delta_2 \geq 10^4$; C4 the Region I windows; C5 the Region II constants ((R1), the mixed T-edge count, and, for Appendix B only, (R2), (R2-sym), the rhombus threshold and the pairing ratio); C6 the corner-lemma angles in Region III; C7 the lens constants; C8 case (c2); C9 the bad-edge totals; C10 the degenerate-regime constants; C11 the small-case packing bound. All pass.
- `r4_regionII_checks.py`: 17 assertions in exact rational arithmetic for Section 5.3: the constants (including the type-(i) count), the LP with the p -term, and the identity and threshold of Lemma 5.3 (on 500 exact rational instances and 2000 random exact rational instances). It also runs an adversarial floating-point search against Lemma 5.4 ($2 \cdot 10^5$ samples), and asserts in exact arithmetic a sufficient condition for the angle estimate in the proof of Lemma 5.4 on all of Region II (in a formulation slightly different from the one in the text, with minimum margin 0.088 over the samples). (The Lean proof of Lemma 5.4 uses neither condition; it argues by exact reflection, see Section 14.2.)
- `r4_regionIII_checks.py`: 18 assertions in exact rational arithmetic for Section 5.4: the constants of Lemma 5.5 and of the injectivity step, the corner-lemma angles, the mixed

T-edge count < 76 , the threshold used in [Lemma 5.7](#), the bounds on arcsin and f and the margin $t + 1/(23\Delta_2) < 3/\pi$ in the arithmetic step, and the thin-cap estimate of [Lemma 5.6](#). Floating-point searches (reported, not used): B1, the slot-killing configuration, with maximum $|y - z| = 0.0691$; B2, the forced step $v_{i+1} = v^*$ of [Lemma 5.9](#), with 0 violations in $2 \cdot 10^5$ samples; B3, the minimum of $|j\alpha_2 - k\alpha_1| \Delta_2$, which was $1.7 \cdot 10^{-6} > 0$.

- `r4_lp_checks.py`: exact dual certificates for [Lemma 3.5](#) and [Remark 3.6](#), for all values of $2c$ in the remark, with and without the p -term; it also re-checks the certificates of [Lemmas B.1](#) and [B.2](#).
- `r4_nb_exact.py` (exact fractions and decimals, with rigorous rational bounds for sin, cos, tan): (1) the closed forms of c_K, c_D , the identity $c_D(0) = \tau$ and the monotonicity of c_D used in [Remark 6.7](#); (2) the LP values of [Remark 3.6](#); (3) the constants of [Lemmas 6.3](#) and [6.4](#): the apex arc is at most $231.19 < 232$, the thin-cone margin is positive, $M \leq 468^2$, and the count of [Lemma 6.4](#) for $\varepsilon_0 = 1/50$ is at most $6.88 \cdot 10^7$; (4) the 60-digit verification of the configuration in [Remark 6.7](#).
- `r4_nb_referee_checks.py`: an independent set of exact checks, with rigorous Taylor bounds, of the LP steps and the constants of [Theorems 6.1](#) and [6.2](#).
- `r4_nb_search.py`: an adversarial floating-point search over local boundaries of K with random corners near a point y (reported, not used). No configuration violating the bounds of [Theorems 6.1](#) and [6.2](#) in their range of validity was found.
- `r4_nb_ring16_check.py`: consistency with ienjoymath’s construction (floating point). For $n = 60, 120, 250$, every point of R has $m_y = 0$ and $\deg y + m_y \leq 6$.
- `ref2_numeric.py`: an independent set of 40 exact assertions (second referee pass of the $\frac{54}{37}$ argument) covering the inequalities of `writeup_a2_checks.py`, together with an adversarial floating-point search for (R1), (R2), (R2-sym). The search is reported, not asserted. Its maxima were 0.906, 0.897 and 0.914, against the proved bounds 0.908, 0.918 and 0.948. (Only (R1) is used in the proof of Theorem 1; (R2) and (R2-sym) concern [Appendix B](#).)
- `ref2_ownership.py`: an exhaustive exact check of the ownership/pairing bookkeeping of [Appendix B](#) (not used for Theorem 1), in the abstract model the argument uses: rows that are disjoint unions of paths and cycles of length ≥ 3 , rungs forming a matching, (R2), (R2-sym), and no two consecutive rhombi. It covers all row structures up to isomorphism with at most 6 vertices per row and all partial rung matchings. There were 2,365,426 admissible configurations without rhombi, with maximum ratio exactly $5/4$, and 2,741,183 with rhombi allowed, with maximum ratio exactly $4/3$. Mutation tests show that (R2-sym) and the rule excluding two consecutive rhombi are both needed.
- `ref_degen.py`, `degen_check.py`, `ref2_degen.py`: adversarial searches in the degenerate regime, including 5500 configurations built so that three or more circles $C(w, \Delta_2)$ pass through one vertex of ∂K . The maximum of $\mu(\Delta_2) - n$ found was 0, against the proved bound 1.
- `ref_checks.py`, `a2_checks.py`: random tests of the lens lemma (about 7000 lenses, worst ratio 0.66) and of the exact single-centre strip in the near-resonant regime.

The $\frac{54}{37}$ argument of [Appendix B](#), together with the parts of [Sections 2](#) to [5](#) and [7](#) that it shares with the present proof, was checked twice more, independently, in adversarial “referee” passes. The new parts of the proof of Theorem 1 were checked in independent referee passes as follows: Region II ([Section 5.3](#)): one; Region III with $\tau \neq \sqrt{3}/2$: one; Region III with $\tau = \sqrt{3}/2$: two; [Theorems 6.1](#) and [6.2](#): one. Vesztergombi’s theorem [\[27\]](#) was checked in the original: its main theorem is $\mu(\Delta_2) \leq \frac{3}{2}n$ for all n , with no additive constant.

Theorem 1: [Sections 9](#), [10](#) and [12](#).

- `r6_n3_checks.py`: 36 assertions in exact rational arithmetic with rigorous Taylor enclosures, for [Section 9](#): the constants of [Lemma 9.2](#) for $\ell = 4$ (the fatness margin, the width, the packing count), the count of [Lemma 9.5](#), the constants of [Lemma 9.4](#), the tolerances of [Lemma 9.6](#), every excluded case in the proof of [Lemma 9.7](#), the windows of [Lemma 9.8](#) (and that they are disjoint), the receiver windows and the owner-uniqueness distance of [Lemma 9.10](#), and the linear combination of [Lemma 3.7\(a\)](#). `r6_ref_constants.py` recomputes these constants independently (second implementation, exact).
- `r6_n3_examples.py`, `r6_ref_examples.py` (60-digit decimal arithmetic): a chain of fans (the configuration of [Remark 6.7](#), $t = 0.86598\dots$) and a chain of caps ($t = 0.49996\dots$); they check the weights, the identity of [Lemma 9.8](#), the receivers and their uniqueness, and that every diametral point has degree at most 3.
- Floating-point screenings (reported, not used): `r6_n3_search.py` (about $3 \cdot 10^4$ assignments in a relaxed slot model; it builds in [Lemmas 9.4](#) and [9.8](#), so it tests only the enumeration), `r6_ref_hexscan.py`, `r6_ref_hexscan_loose.py`, `r6_ref_hexscan_corner.py`, `r6_ref_search.py` (hexagon and position scans on flat and cornered boundaries with the exception rule of [Lemma 2.2\(b\)](#) enforced; about $1.4 \cdot 10^5$ heavy configurations, none unclassified and none with t outside the windows of [Lemma 9.8](#)), `r6_ref_receivers.py` (receiver windows on random cornered boundaries), `r6_ref_ring.py` (the construction of [\[17\]](#): no heavy points, all weights at most 6).
- For [Section 10](#) (evidence only): `r6_below_classify.py` (a scan of good' points of weight at least 8 in the flat model; only the types of [Lemma 10.5](#) occur), `r6_below_rule_check.py` (a beam search over several hundred thousand flat configurations implementing the discharging rules; maximum final charge 7), `r6_refB1_roles.py` (a relaxed role model of all senders at 22 values of t ; no receiver with three senders, all loads at most 1), `r6_refB1_caplattice.py` (an exact integer model of the cap lattice at $t = \frac{1}{2}$ with diametral points in R and several adjacent heavy points; $3 \cdot 10^4$ configurations, maximum final charge 7), `r6_refB1_example.py` (an exact instance in which a cap and a type-Dt point feed one forwarder), `r6_below_exact_examples.py` (the exact examples of [Remark 10.10\(a\)](#), in $\mathbb{Q}(\sqrt{3})$), `r6_below_cluster_search.py` (the cluster search of [Remark 10.10\(b\)](#)). The flat-model searches do not contain diametral points in R or unpinned diametral points and rarely contain more than two heavy points; the exact cap-lattice model covers these cases in one family.
- `r6_Eprime_full_checks.py`: 34 assertions in exact rational arithmetic for [Section 12](#): the steps of [Lemma 12.2](#), the constants of Steps 0, 3 and 7 and of the fall-back cases, and the linear combination of [Corollary 12.4](#). `r6_refE_parallel.py` (needs `mpmath`) tests Steps 1–3 of the proof of [Theorem 12.1](#) at 40 digits on 897 configurations: the bound $\text{len } Z_c \leq 2\eta \tan(\psi_c/2)$ is attained to 6 digits but never exceeded, and $\sum_c \text{len } Z_c / \eta \leq 16.7$. The topological Steps 4–6 are not machine-tested. `r6_gap_L_family.py` and `r6_gap_M_pairgap.py` verify the configurations of [Remark 12.5\(b\)](#) in 60-digit arithmetic.

[Theorem 9.1](#) was checked in two independent referee passes (no mathematical error; cosmetic corrections incorporated). [Theorem 10.1](#) was checked in two independent referee passes; both found it correct, with gaps in the text of an earlier write-up that are repaired here (see the status paragraph in [Section 10](#)). The two inputs of [Section 11](#) ([Theorems 11.3](#) and [11.12](#)) were each checked in one independent referee pass, whose corrections are incorporated; the text of [Section 11](#) has not been refereed. [Theorem 12.1](#) and [Corollary 12.4](#) were checked in three independent referee passes, the last one on the complete corrected proof given here. The input $e(S) \leq s + p + C$ of [Sections 5.3](#) and [5.4](#), which [Theorem 1\(a\)](#) uses in Regions II and III, had the referee passes listed above; the passes on [Section 9](#) took it as given.

Section 13.

- `va1b_exact.py` (sympy, mpmath interval arithmetic): exact computation of the six Case 1 candidates of [Proposition 13.2](#) and their pairwise squared distances, with an exhaustive check that no compatible set of candidates covers all three v_i . It also gives a closed form and a rigorous interval check (6000 subintervals, 0 failures) of the Case 2 branch $|r - v_2| = \Delta$.
- `va1c_sympy.py`: symbolic checks of the identity in [Lemma 13.3](#), of $\lambda(\lambda + 2 \cos \gamma) = 0$, of the chord formula, and of the bisector direction in [Lemma 13.4](#).
- `va1c_local.py`: exhaustive search for the configuration of [Lemma 13.4](#) with unit vectors from $\mathbb{Z}[\omega]$ (norm ≤ 147 , exact), $\mathbb{Z}[i]$ (norm ≤ 325 , exact) and N -th roots of unity ($5 \leq N \leq 60$): 0 hits in about $3.2 \cdot 10^6$ configurations. A control run with weakened hypotheses does find hits.
- `va1_clique.py`, `va1_cyclo.py`, `va1c_random.py`: searches over maximal compatible sets in lattice and cyclotomic pools, plus greedy floating-point constructions. Every set was tested against [Lemma 13.1](#), [Proposition 13.2](#), [Theorem 13.5](#), [Theorem 13.6](#) and [Corollaries 13.7](#) and [13.8](#): more than 800,000 sets in the first searches and about 81,000 in the second, with 0 violations.

Section 14.

- `ver_classification_k.py`, `ver_hexagons.py`: exact chord-index profiles for the classified extremal sets used in [Theorems 14.4](#) and [14.5](#) and [Proposition 14.3](#), and strict convexity and multiplicity profiles of the three 3-distance hexagons.
- `counting_parity.py`: the closed forms of [Lemma 14.2](#) for $n < 400$.
- `b3_sat.py`: the CNF encoder (direct and order encodings, with or without Lemma T).
`b3_enum.py`: the standalone backtracking enumerator for $n = 7$.
- `b3_validate.py`: encoder validation. Every 7-subset of R_N , $7 \leq N \leq 16$ (11,440 sets), and every 11-subset of R_N , $N \leq 15$, satisfies all constraints under exact chord-index ranks, as do 3000 (for $n = 7$) and 500 (for $n = 11$) random convex polygons. The $(7, 3, 0)$ enumeration returns exactly the R_7 colouring. Every satisfying assignment printed by the SAT route was re-checked against the independent constraint code of `b3_enum.py`.
- Four solvers (CaDiCaL, Glucose 4, MiniSat 2.2, Lingeling) agree on the $n = 7$ and $n = 11$ instances.
- `b3_drup_check.py`: an independent forward RUP checker, used for the $n = 7$, $n = 9$ and $n = 11$ DRUP proofs. Its self-test rejects an empty proof of a satisfiable instance, a proof with one negated lemma, and a truncated proof.
- `b3_drat_trim.py`: regenerates each CNF and Glucose proof deterministically and runs `drat-trim` with LRAT output, followed by `lrat-check`. The CNF, DRUP and LRAT files are not stored in the repository because of their size.

For [Theorem 14.9](#):

- `r1_sat_split.py`: the case-split encoder (Q-only order encoding, as in `b3_sat.py`, plus the case constraints and the symmetry-breaking constraints). `r1_run.py`, `r1_batch.py`, `r1_lowD.py`: build each CNF, solve it with Kissat writing a DRAT proof, check the proof with `drat-trim`, and delete both files. A case is counted as done only when `drat-trim` reports `s VERIFIED`. All $5 + 21 + 28$ cases for $n = 15$ and $6 + 28 + 36$ cases for $n = 17$ were verified; solve times were at most 56 s for $n = 15$ with symmetry breaking (282 s for the case $\{7, 8\}$ without it) and at most 739 s for $n = 17$, and the DRAT proofs were at most 124 MB.
- `r1_sb_test.py`: the orbit test of the symmetry-breaking constraints described after [Theorem 14.9](#).

instance (n, D, F)	encoding	lemmas	drat-trim	lrat-check
(7, 4, 2)	Q	714	verified	verified
(11, 2, 0), (11, 3, 0), (11, 4, 0)	Q+T	3, 8, 56	verified	verified
(11, 5, 3)	Q+T	524	verified	verified
(11, 6, 4)	Q	35,510	verified	verified
(13, 2, 0), (13, 3, 0), (13, 4, 0), (13, 5, 0)	Q	2, 7, 82, 448	verified	verified
(13, 6, 4)	Q	8,463	verified	verified
(13, 7, 5)	Q	446,686	verified	verified

TABLE 3. Certified refutations for [Theorem 14.8](#).

For [Section 14.3](#):

- **r3_cyc_hits.py**: the maximum number of hits ([Lemma 14.13](#)) for $M \leq 120$. Every point with at least two hits is an intersection of two circles centred at vertices with chord radii; candidates are scanned in floating point with tolerance 10^{-7} , and the best ones are confirmed exactly in $\mathbb{Q}(\zeta_M)$ by arithmetic modulo the cyclotomic polynomial. The maximum found is 13 (at $M = 60, 120$).
- **r3_almostcoc_check.py**: floating-point sanity check of [Theorem 14.14](#) on regular polygons with holes plus up to 3 adversarial points off the circle ($M \leq 60$); the slack against the bound was at least 24 in every case.

Section 15.

- **r2_verify.py**: 50-digit verification of the convex examples with $\mu(\Delta_2) = \frac{5}{4}n$ and no four concyclic points, for $n = 12, 20, 40$. **r2_local_lemmas.py**: symbolic checks of identities used in the local analysis around hull vertices with three Δ_2 -neighbours.
- **r5_verify_n8.py**: exact (sympy) verification of the 8-point example with $\mu(\Delta_2) = \mu(\Delta_3) = 9$, including its full multiplicity profile.

Lean. The whole development (all **E132*.lean** files: the convex-position results, the layer bound, Vesztergombi's and Hopf–Pannwitz's bounds, the $\frac{54}{37}$ bound **erdos132_main** and the $\frac{15}{11}$ bound **erdos132_main15**) builds with **lake exe cache get** followed by **lake build**; on a laptop this takes about 7 minutes after the Mathlib cache is downloaded, of which about 4 minutes is **bv_decide** for $n = 13$ and about 1 minute is the six **E132Main15*** files. The toolchain is Lean **v4.35.0-rc2** with Mathlib.

APPENDIX B. THE 54/37 ARGUMENT FORMALISED IN LEAN

This appendix gives the written counterpart of the formal theorem **erdos132_main** ([Section 14.2](#)): the bound $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{54}{37}n + C_0$ with $C_0 = 3N_0 + 10^8$. It uses [Section 2](#), the reductions [Lemma 3.1](#), [Corollary 3.4](#) and the small cases of [Section 3](#), [Section 4](#), Region I of [Section 5.1](#), (R1) and [Lemma 5.1](#) and [Lemma 5.2](#) of [Section 5](#), and [Section 7](#). It does not use [Lemmas 3.3](#) and [3.5](#), [Lemmas 5.3](#) and [5.4](#), the Region III arguments of [Section 5.4](#), or [Section 6](#). In place of the R -term estimate it uses the trivial bound (E), $\mu(\delta) \leq e(S) + 6r$.

Linear combinations.

Lemma B.1. *Suppose $e(S) \leq ks + C$ with $0 \leq k < 3/2$. Then $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{9}{7.5-k}n + C$.*

Proof. Put $\lambda = (6 - k)/(7.5 - k) \in (0, 1)$. By Lemma 3.1 and Corollary 3.4, $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \lambda \cdot \frac{3}{2}s + (1 - \lambda)(ks + C + 6r) \leq \frac{9}{7.5 - k}(s + r) + C$. For $k = 1, \frac{5}{4}, \frac{4}{3}$ the factor $9/(7.5 - k)$ is $\frac{18}{13}, \frac{36}{25}, \frac{54}{37}$.^[C1] $\square$

Lemma B.2. *Suppose $e(S) \leq s + 2p + C$. Then $\min\{\mu(\Delta_2), \mu(\delta)\} \leq \frac{7}{5}n + C$.*

Proof. By Lemma 3.2 and Corollary 3.4, $\min \leq \frac{4}{5}(\frac{3}{2}s - \frac{1}{2}p) + \frac{1}{5}(s + 2p + C + 6r) = \frac{7}{5}s + \frac{6}{5}r + \frac{C}{5} \leq \frac{7}{5}n + C$.^[C1] $\square$

Region I gives $e(S) \leq s + C_{\text{bad}}$, hence $\frac{18}{13}n + C$ by Lemma B.1, and so does Region II with $\tau > 1$.

Region II with $\sqrt{3}/2 + 3\beta \leq \tau \leq 1$. Here $t \leq 1$, $t \geq \tau$ and $t - \arcsin \beta \geq 0.886$; there are fewer than 71 mixed T-edges (Lemma 5.1), and each vertex has at most one rung (R1). Rungs are written $z = v + o_{\sigma'}(u_v)$ with a sign $\sigma' \in \{+, -\}$, and $|o_{\pm}(u) - o_{\pm}(u')| \leq 2\sigma + \angle(u, u')$ for any choice of signs. By the T-step form, for T-steps e_1 at v_0 and e_2 at z_0 in the same direction, $\angle(e_1, e_2) \leq \angle(u_{v_0}, u_{z_0}) + \arcsin \beta$.

Lemma B.3 ((R2), proximity). *Let (v_0, z_0) be a rung, v_1 the right pure T-neighbour of v_0 and z_1 the right pure T-neighbour of z_0 ; put $e_1 = v_1 - v_0$, $e_2 = z_1 - z_0$. If v_1 has a rung (v_1, z') , then $z' = z_1$.*

Proof. $z' - z_1 = (e_1 - e_2) + (o_{\sigma_1}(u_{v_1}) - o_{\sigma_0}(u_{v_0}))$. Since $\angle(u_{v_0}, u_{z_0}) \leq \beta$ (good rung) and $\angle(u_{v_0}, u_{v_1}) \leq \beta$ (good T-edge), $|z' - z_1| \leq (\beta + \arcsin \beta) + \beta + 2\sigma \approx 0.918 < 1$.^[C5] Distinct points of X are at least 1 apart. $\square$

Lemma B.4 ((R2-sym)). *In the setting of Lemma B.3, if z_1 has a rung (v', z_1) , then $v' = v_1$.*

Proof. $v' = z_1 - o_{\sigma'}(u_{v'})$ and $v_1 = z_0 - o_{\sigma_0}(u_{v_0}) + e_1$, so $v' - v_1 = (e_2 - e_1) - (o_{\sigma'}(u_{v'}) - o_{\sigma_0}(u_{v_0}))$. The angle between $u_{v'}$ and u_{v_0} is at most 3β , as the sum of three good-edge angles along $v' \rightarrow z_1 \rightarrow z_0 \rightarrow v_0$. Hence $|v' - v_1| \leq (\beta + \arcsin \beta) + 3\beta + 2\sigma \leq 2\sigma + 6\beta \approx 0.948 < 1$.^[C5] $\square$

(R3) *Parallel rungs and rhombi.* Rungs with the same vector c satisfy $c \cdot u_v = -\tau$, so u_v is one of the two unit solutions, which lie at $\pm\theta_0$ from $-c$. Rungs form a matching by (R1), so their K-ends are distinct and by Lemma 2.3 have distinct normals. Hence *no three rungs are parallel*, and two parallel rungs have normals exactly $2\theta_0$ apart. If $v_0v_1, v_1z_1, z_1z_0, z_0v_0$ are all unit edges (a *rhombus*), then v_1 and z_0 are the two points of $C(v_0, 1) \cap C(z_1, 1)$, so $v_0 + z_1 = v_1 + z_0$ and the rungs (v_0, z_0) and (v_1, z_1) are parallel. Since the T-edge v_0v_1 is good, $2\theta_0 = \angle(u_{v_0}, u_{v_1}) \leq \beta$. So *rhombi exist only if $\tau \geq \cos(\beta/2)$* , and $\cos(\beta/2) > \sqrt{3}/2 + 3\beta$.^[C5]

Ownership. For a rung $r = (v, z)$, walk right from v along pure K-row T-edges until reaching the first vertex after v that is a rung vertex, or until the walk stops. Let $k \geq 1$ be the number of vertices visited before stopping, v included. Define d in the same way in the D-row, starting from z . Then r *owns* these $k + d$ vertices. Every other row vertex is *unowned*: it has no rung vertex to its left on its T-path or T-cycle. Ownership is well defined, since each owned vertex belongs to the nearest rung vertex at or to its left, and that rung vertex lies on exactly one rung.

Charging. Charge to r the rung itself and the right T-edge out of each vertex it owns. Charge every other pure T-edge to its unowned left endpoint. Every rung and pure T-edge is charged exactly once, and every vertex has at most one right T-edge. Hence $\text{edges}(r) \leq 1 + k + d$, with equality only if both walks end at a rung vertex (“both reach”), and otherwise $\text{edges}(r) \leq k + d$; unowned vertices carry at most one edge each.

Both-reach gaps are $(1, 1)$ or $(\geq 2, \geq 2)$. If both walks reach and $k = 1$, then z_0 has a right neighbour z_1 and v_1 is a rung vertex; by Lemma B.3 its rung is (v_1, z_1) , so $d = 1$. If both reach and $d = 1$, then v_0 has a right neighbour v_1 and z_1 is a rung vertex; by Lemma B.4 its rung is

(v_1, z_1) , so $k = 1$. A $(1, 1)$ gap is a rhombus $v_0 v_1 z_1 z_0$ (by (R1), $z_1 \neq z_0$ and $v_1 \neq v_0$). A cycle with a single rung vertex gives a both-reach gap with k equal to the cycle length, which is at least 3.

Ratios. A rhombus gap has ratio edges/vertices = $3/2$; a both-reach gap with $k, d \geq 2$ has ratio at most $5/4$; every other gap has ratio at most 1. If $\tau < \cos(\beta/2)$ there are no rhombi, so $e(S) \leq \frac{5}{4}s + C_{\text{bad}} + 71$ and Lemma B.1 gives $\frac{36}{25}n + C$.

Pairing. If $\tau \geq \cos(\beta/2)$, pair each rhombus rung $r_0 = (v_0, z_0)$ with its successor $r_1 = (v_1, z_1)$. The rung r_1 is not itself a rhombus rung: otherwise r_0, r_1, r_2 would be three parallel rungs (here $v_2 \neq v_0$ because T-cycles have length at least 3). The pairing is injective, because v_0 is the unique left T-neighbour of v_1 . If r_1 owns m vertices, the pair has ratio at most

$$\max\left\{\frac{4+m}{2+m} : m \geq 4\right\} \cup \left\{\frac{3+m}{2+m} : m \geq 2\right\} = \frac{4}{3}.^{[C5]}$$

Hence $e(S) \leq \frac{4}{3}s + C_{\text{bad}} + 71$, and Lemma B.1 gives $\frac{54}{37}n + C$. This is the only place where $\frac{54}{37}$ is attained.

Region III. By Lemma 5.2 and the angle bound $\min\{2\theta_0, \pi/2 - \theta_0\} > 2\pi/7$,^[C6] at most 6 K-vertices with a rung have Δ_2 -degree at least 2; every other K-vertex with a rung lies in Q . A K-vertex has at most 2 rungs, and (T1) bounds all T-edges, mixed ones included, by s . Hence

$$e(S) \leq s + 2p + 12 + C_{\text{bad}},$$

and Lemma B.2 gives $\frac{7}{5}n + C$. (A “hexagonal strip” around a single centre has average δ -degree 3 locally, so this regime genuinely needs a weighted bound such as Lemma B.2 for this argument.)

regime ($\delta = 1, \beta = 1/100$)	bound on $e(S)$	bound on $\min\{\mu(\Delta_2), \mu(\delta)\}$
degenerate: $\Delta > 1.94\Delta_2, n \geq 4$	—	$n + 1$ (Proposition 7.1)
nondegenerate, $\Delta_2 < 10^4$	—	$3n < 3N_0$ (Section 3)
I: $\tau \leq \sqrt{3}/2 - 2\beta$ (incl. $\tau \leq 0$)	$s + C$	$\frac{18}{13}n + C$
III: $\sqrt{3}/2 - 2\beta < \tau < \sqrt{3}/2 + 3\beta$	$s + 2p + C$	$\frac{7}{5}n + C$
II: $\sqrt{3}/2 + 3\beta \leq \tau < \cos(\beta/2)$	$\frac{5}{4}s + C$	$\frac{36}{25}n + C$
II: $\cos(\beta/2) \leq \tau \leq 1$	$\frac{4}{3}s + C$	$\frac{54}{37}n + C$
II: $\tau > 1$ ($\Leftrightarrow t > 1$)	$s + C$	$\frac{18}{13}n + C$

TABLE 4. The regimes of the $\frac{54}{37}$ argument; here $C < 10^8$.

Assembly. Every constant in the last column is at most $\frac{54}{37}$,^[C1] and $n + 1 \leq \frac{54}{37}n + 1$. With $C_{\text{bad}} + 71 + 12 < 10^8$ (Proposition 4.3),^[C9] the bound holds with $C_0 = 3N_0 + 10^8$.

Remark B.5. The two least standard parts of this argument are the ownership/pairing bookkeeping and the incidence count in Step 4 of Section 7. Both were re-derived independently, and both were also checked by computer in abstract models (Appendix A). The bookkeeping check is exhaustive up to $6 + 6$ row vertices, and both (R2-sym) and the “no three parallel rungs” fact turn out to be necessary for it. For this argument the bottleneck is the near-resonant regime $\tau \geq \cos(\beta/2)$, where rhombi occur; Lemma 5.4 removes it in the proof of Theorem 1.

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